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Department of Information Technology

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Jadeer

Prepared by

Walaa Saif Aleslam Mohammed,
Noor Yousef Algumlas,
Jana Khalid Almohsen,
Rawan Munif Alshammari,
Walaa Abdullah Alhajri,

Supervised by
Dr. Mashael AlSaleh & Dr. Manal AlBahlal

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Jadeer

Walaa Mohammed[1], Noor Algumlas[2], Jana Almohsen[3], Rawan Alshammari[4], Walaa Alhajri[5]

[1]Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 444200088@student.ksu.edu.sa

[2]Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 444200811@student.ksu.edu.sa

[3]Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 444204517@student.ksu.edu.sa

[4]Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 444200348@student.ksu.edu.sa

[5]Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 443201299@student.ksu.edu.sa

Abstract (English):

Recruitment has become increasingly complex due to the rapid evolution of job requirements and the growing number of applicants. Organizations, especially in Saudi Arabia, face significant challenges in efficiently screening candidates, conducting fair evaluations, and making timely hiring decisions. At the same time, job seekers struggle with preparing professional CVs, identifying suitable opportunities, and performing well in interviews. This project presents **Jadeer**, an AI-powered comprehensive recruitment ecosystem designed to streamline the hiring process for both companies and job seekers. **Jadeer** provides organizations with intelligent tools for job posting, AI-driven interviews, and automated candidate evaluation reports, enabling faster and more objective hiring decisions. The platform also supports job seekers through job matching, AI-based CV enhancement, and mock interview simulations with personalized feedback. The system integrates Large Language Models (LLMs), Speech-to-Text (STT), Text-to-Speech (TTS), and a Vocal Confidence Scoring Model trained on the First Impressions V2 dataset, achieving a Mean Absolute Error (MAE) of 0.11. User acceptance testing with seven participants confirmed the platform's usability, with unanimous willingness to recommend it. By combining these technologies, **Jadeer** moves beyond traditional Applicant Tracking Systems and evaluates both candidate qualifications and performance. Ultimately, **Jadeer** aims to improve recruitment efficiency, reduce bias, and empower job seekers, aligning with Saudi Vision 2030 goals of enhancing workforce productivity and labor market efficiency.

Abstract (Arabic):

أصبحت عملية التوظيف أكثر تعقيداً في ظل التطور المتسارع لمتطلبات سوق العمل وتزايد أعداد المتقدمين. تواجه المنظمات، ولا سيما في المملكة العربية السعودية، تحديات جوهرية في فرز المرشحين بكفاءة، وإجراء تقييمات عادلة، واتخاذ قرارات التوظيف في الوقت المناسب. في المقابل، يعاني الباحثون عن عمل من صعوبات في إعداد سير ذاتية احترافية، وتحديد الفرص المناسبة، والتحضير الجيد للمقابلات الوظيفية. تقدم هذه الورقة منصة **جدير**، وهي منصة توظيف ذكية ثنائية الجانب تهدف إلى تحسين عملية التوظيف وتبسيطها لكلا الطرفين. توفر **جدير** للشركات أدوات ذكية لنشر الوظائف وإجراء المقابلات بالذكاء الاصطناعي وتوليد تقارير تقييم المرشحين تلقائياً. كما تدعم الباحثين عن عمل من خلال مطابقة الوظائف وتحسين السيرة الذاتية بالذكاء الاصطناعي وتقديم محاكاة واقعية للمقابلات مع ملاحظات شخصية. يدمج النظام تقنيات متقدمة تشمل النماذج اللغوية الكبيرة، وتحويل الكلام إلى نص، وتحويل النص إلى كلام، ونموذج تقييم الثقة الصوتية المدرَّب على مجموعة بيانات First Impressions V2، محققاً متوسط خطأ مطلق بلغ 0.11. وقد أكد اختبار قبول المستخدمين مع سبعة مشاركين سهولة استخدام المنصة، إذ أبدى جميعهم استعداداً تاماً للتوصية بها. تتجاوز **جدير** أنظمة تتبع المتقدمين التقليدية بتقييمها لكل من مؤهلات المرشحين وأدائهم الفعلي. وتهدف المنصة في نهاية المطاف إلى تحسين كفاءة التوظيف والحد من التحيز وتمكين الباحثين عن عمل، بما يتوافق مع أهداف رؤية المملكة 2030 في تعزيز إنتاجية القوى العاملة وكفاءة سوق العمل.

Keywords:

AI recruitment; HR technology; job seeker support; LLMs; CV enhancement; mock interview; vocal confidence scoring;

1 Introduction

Recruitment has always been crucial for organizational success, but it has become increasingly complex in today's labor market. Companies are struggling to keep up with the rapid changes in jobs and skills. According to the World Economic Forum's Future of Jobs Report 2023, nearly a quarter of current jobs may be disrupted in the next five years (World Economic Forum, 2023). This highlights the importance of hiring the right people quickly and fairly.

In Saudi Arabia, the need for better recruitment systems is even stronger. Vision 2030 makes workforce empowerment and labor market efficiency a national priority [1]. At the same time, the *Monsha'at HR Thematic Report 2022* shows that many organizations, especially SMEs, still face difficulties in attracting and retaining talent [2]. These challenges highlight why new, technology-based solutions are necessary.

Our project, **Jadeer**, aims to address these issues. It is designed as a comprehensive recruitment ecosystem that helps companies filter and evaluate applications with AI tools, while also preparing job seekers with resume improvements and interview practice.

By supporting both sides, **Jadeer** aims to accelerate, equalize, and improve the effectiveness of recruitment.

This report details the development of **Jadeer** and its approach to solving recruitment challenges. It begins by identifying the problems in current recruitment practices, followed by an overview of existing solutions and their limitations. The following sections describe **Jadeer** key features, technical design, and development process, showing how the platform addresses these challenges for both job seekers and Companies.

1.1 The Problem

This section describes the real-world problem that **Jadeer** aims to address by focusing on challenges in the recruitment process for both companies and job seekers.

With the rapid development of the Saudi labor market and the growing demand for qualified talent, significant challenges have emerged for both companies and job seekers during the recruitment process [1]. Companies often receive hundreds of applications for a single position, making it difficult to efficiently:

- Filter and evaluate large volumes of resumes.
- Match candidates' qualifications and skills with job requirements.
- Conduct fair and consistent initial interviews.
- Assess candidate performance objectively and make unbiased decisions.

Many companies rely on traditional Applicant Tracking Systems (ATS), which primarily perform keyword-based screening of resumes. While helpful for reducing manual workload, these systems often fail to capture the deeper aspects of a candidate's capabilities such as reasoning ability, problem-solving, communication skills, and potential for growth. As a result, highly qualified candidates may be overlooked simply because their resumes do not contain exact keyword matches — a limitation that contributes to biased or incomplete talent evaluation. [3]

For example, in Saudi Arabia, large organizations and banks often handle upwards of 2,000 to 5,000 applications during major entry-level or graduate hiring campaigns. In such scenarios, traditional ATS tools routinely filter out up to 75% of these applications based purely on strict, missing keyword patterns, forcing HR teams to spend weeks manually auditing the remaining pipeline. Without efficient and holistic recruitment systems, this leads to heavily delayed hiring decisions, high operational costs, and the loss of top-tier local talent who simply did not format their CVs perfectly.

On the other hand, job seekers face challenges in optimizing their resumes for ATS-compatible formats, finding suitable opportunities, and preparing effectively for interviews, which reduces their chances of securing positions that match their qualifications and aspirations. [4]

In our project, we specifically focus on the early and middle stages of the recruitment process, including resume screening, candidate evaluation, and initial interview preparation — stages that are not only time-consuming and prone to human bias, but also heavily influenced by ATS-based filtering.

This inefficiency affects not only companies and individuals but also the labor market, contributing to underemployment and reduced productivity. Addressing these challenges is essential to improving recruitment quality and aligns with the goals of Saudi Vision 2030 to empower the workforce, enhance labor market efficiency, and improve quality of life [1].

1.2 The Solution

This section presents the proposed solution, **Jadeer**, and explains how it tackles the challenges in the recruitment process. It highlights the platform's key features and the ways it benefits both companies and job seekers.

Jadeer is an AI-powered comprehensive recruitment ecosystem. **Jadeer** is designed to automate and enhance the hiring process for companies while also supporting individuals in their job-seeking journey. The platform provides **companies** with efficient tools for managing job postings, conducting AI-assisted interviews, and receiving insightful candidate performance reports. Since the platform supports company registration, an **admin** is included to verify each company by checking the submitted documents and confirming their authenticity before granting access, ensuring that no one can use a company's name without authorization. On the other hand, it empowers **job seekers** to explore job opportunities, enhance their CVs, and prepare for interviews through AI-driven practice sessions. By addressing these perspectives, **Jadeer** bridges the gap between companies and candidates, aiming to make recruitment more efficient, and fair.

- **Company Side:**

1. **Job Posting:** Companies can either create job postings manually or use **Jadeer's** intelligent job-posting generator to automatically craft job descriptions. This ensures consistency and saves time, while still allowing customization. **Jadeer** provides a centralized space for companies to reach a broad pool of job seekers, maximizing visibility and accessibility.
2. **AI-powered Interviews:** **Jadeer** allows companies to conduct preliminary AI-driven interviews with applicants. The interview is conducted as an interactive voice-and-video meeting led by the AI, where candidates are presented with both technical and psychometric questions to assess the applicant's skillset and personality.
3. **Evaluation of candidates:** The system evaluates candidates based on the accuracy of their responses to both technical and psychometric questions, the clarity of their communication, their provided CV, and voice-tone analysis. Each of these evaluation criteria is assigned a specific weight according to its importance, and the weighted scores are combined to calculate an overall score out of 100. This approach ensures consistent and unbiased evaluations.
4. **Candidate Performance Report and Ranking:** After interviews, **Jadeer** generates a detailed report for each candidate, summarizing their strengths, weaknesses, and overall suitability for the role. Candidates are ranked from highest to lowest fit, allowing companies to make fair and data-driven hiring decisions.

- **Job seeker Side:**

1. **Browsing Job Opportunities:** job seekers can browse job postings published by companies on **Jadeer**. The system highlights opportunities that match their skills and qualifications by aligning keywords from CVs with job contents. To help candidates discover the most relevant job opportunities.
2. **AI-powered CV Enhancement:** job seekers can improve their CVs with AI-driven suggestions tailored to their specialization. The platform can enhance the CV either generally or in a job-specific manner, allowing it to highlight the candidate's strengths

that align with a particular role. It refines language for clarity, emphasizes the most relevant skills, and optimizes the overall structure and formatting and ensures the CV is optimized to pass through ATS filters smoothly, increasing its chances of being noticed by recruiters.

3. **Mock AI Interviews:** job seekers can practice interviews by selecting their specialization area. The AI conducts general interview simulations, evaluating the accuracy and clarity of responses as well as voice tone analysis. Afterward, **Jadeer** provides a comprehensive report that highlights strengths, areas for improvement, and practical tips to perform better in real interviews.

Through its comprehensive recruitment ecosystem design, **Jadeer** delivers an integrated solution that supports both companies and job seekers. By automating core recruitment tasks, offering fair and unbiased evaluations, and providing valuable tools for preparation, **Jadeer** enhances efficiency in the hiring process. This solution ultimately helps companies find the right talent faster while empowering individuals to present themselves more effectively in the job market.

1.3 Objectives

In this section we will define the product, project and learning objectives

- **Product (customer focus-value)**:

  These objectives are about the product, and how we will solve different problems and provide good benefits for users:

  1. **Unified Job Platform:** deliver a unified platform that centralizes all job-related processes for both companies and job seekers.
  2. **Automated HR Support:** support Companies by automating key recruitment tasks (screening, matching, initial interviews, evaluation).
  3. **Efficient Job Posting:** enables companies to post job vacancies and manage applications efficiently.
  4. **AI-Powered Virtual Interviews: c**onduct realistic, AI-powered virtual interviews that simulate professional scenarios.

5. **Candidate Skill Analysis**: analyze candidate skills based on their CV and interview performance and generate a detailed report including a suitability score for each applicant.
6. **Job Matching:** match suitable job opportunities to job seeker based on their skills and experiences from their CV.
7. **Mock Interviews for Practice:** provide AI-driven interview simulations to help job seeker improve their performance.
8. **Personalized Feedback Reports:** generate personalized feedback reports highlighting strengths and weaknesses after mock interviews.
9. **CV Enhancement:** enhance CVs using AI to meet professional standards and pass automated company filters smoothly.

- **Project (solution focus-plan):**

  These objectives outline the development stages of **Jadeer** platform and the outcomes to be achieved by the end of the project. They include:

1. **User Needs Identification:** identify user needs (job seekers & HR managers).
2. **Database Schema Design:** build core database schema.
3. **UI/UX Development:** design and develop user interfaces, ensuring intuitive navigation and accessibility.
4. **Interview Question Generation:** use existing AI capabilities (e.g., GPT) to generate and evaluate domain-specific interview questions (e.g., IT, finance, marketing), relying on prompt engineering and curated examples.
5. **Model Integration:** use existing pre-trained models for speech-to-text (Whisper) and text-to-speech (Google TTS) interview functionalities.
6. **Custom Model Development:** Build and fine-tune our own Speech Emotion Recognition (SER) model to recognize and evaluate candidate performance.
7. **AI-Generated Candidate Report:** AI-generated report combining interview transcript, answer evaluation, voice analysis, CV-job fit and candidate ranking.
8. **Usability Testing & Iteration:** conduct usability testing and iterate based on feedback from target users.

- **Learning (student focus):**

These objectives are about the team and the outcomes for the team while doing this project:

1. **Flutter Development:** learn how to use Flutter framework for building cross-platform mobile applications.
2. **Model Fine-Tuning:** learn how to fine-tune pre-trained AI models for domain-specific tasks.
3. **AI Model Integration:** learn how to integrate multiple AI models (e.g. Whisper, OpenAI GPT, Google TTS) into one system.
4. **AI-Driven UX Design:** learn how to design and implement AI-driven user experiences (e.g. interview simulation, CV enhancement).
5. **System Testing & Validation:** learn how to test and validate AI-powered systems with real users.

1.4 Scope

In this section we will define our boundaries and what is outside the scope of our platform.

The **Jadeer** platform is designed to enhance the recruitment process for both companies and job seekers. It provides companies with tools to generate job postings and conduct AI-powered interviews that evaluate candidates both technically and psychometrically, moving beyond the limitations of traditional Applicant Tracking Systems (ATS) keyword filtering [3] For job seekers, **Jadeer** offers CV refinement and realistic mock interview simulations, helping them prepare more effectively and receive meaningful feedback.

The application will be developed as a mobile application accessible on Android devices. The initial version will support English only, ensuring accessibility for a wide audience. The scope is limited to recruitment-related functions and does not extend to other HR modules such as payroll or attendance management.

The development of **Jadeer** will take place over two semesters, during which the core features will be implemented and tested to deliver a reliable and practical solution.

1.5 Product Vision

This section defines the vision of **Jadeer** by highlighting its purpose, target audience, and unique value. It provides a clear direction for development and explains how the product stands out compared to existing solutions.

For companies and job seekers **Who** struggle with time-consuming, inefficient, and biased recruitment processes. **The Jadeer is an** AI-powered intelligent recruitment platform **That** streamlines the hiring journey by managing resume screening, candidate evaluation, interview preparation, and CV optimization **Unlike** HireVue [5], which focuses mainly on conducting and analyzing AI-driven video interviews, **Our product** provides a unified, end-to-end solution that covers the entire recruitment process from CV optimization and job matching to AI-powered interviews and comprehensive candidate evaluations.

1.6 Methods

To build the **Jadeer** platform systematically, the development lifecycle was structured around the Scrum Framework, executing software creation over multiple iterative sprints. The engineering and research workflow followed five comprehensive stages:

1. **Requirements Elicitation and Analysis:** Understanding user needs by conducting structured interviews and distributing questionnaires to HR specialists and job seekers to define core system functionalities.
2. **System and Architectural Design:** Designing the structural blueprint of the platform, including the client-server system architecture, unified class diagrams, detailed database collections, and procedural user interaction flowcharts.
3. **AI Model Development and Prompt Engineering:** Constructing specialized prompt engineering templates for resume auditing (using GPT-4o), integrating high-fidelity

conversational audio pipelines (via Whisper STT and Google TTS), and configuring the foundational speech feature extractions for our custom vocal analysis.

4. **System Implementation and Integration:** Developing the responsive front-end user interfaces using the Flutter framework and Dart language, scripting serverless backend services within Firebase, and binding third-party cloud AI APIs into unified application workflows.
5. **Experiments and System Evaluation:** Executing functional software verification, conducting model accuracy and performance validation checks, assessing non-functional quality attributes, and performing qualitative User Acceptance Testing (UAT) with real target users.

1.7 Main Contribution

The main contribution of **Jadeer** is delivering a balanced, automated system that serves both recruiters and job seekers simultaneously:

- **For the Recruiters side:** It streamlines early-stage hiring by combining multiple assessment steps into an objective ranking engine, calculating a score out of 100 based on CV matching, technical questions, psychometric fit, and voice tone clarity.
- **For the Job Seeker side:** It actively improves candidate readiness through automated AI CV enhancement tailored to specific roles, and interactive mock interview practice with instant feedback.
- **Market Differentiation and Novelty**

Unlike existing tools that either work as an enterprise-only screening filter or just a standalone resume template builder, **Jadeer** creates a comprehensive recruitment ecosystem. It provides companies with verified, multi-criteria data to make fast decisions, while at the same time giving candidates the professional simulation and improvement tools they need to pass actual company filters and real interviews.

- **Local and Global Impact**
- **Local Impact:** **Jadeer** supports the goals of Saudi Vision 2030 by increasing recruitment efficiency in the local labor market. It especially helps local Small and Medium Enterprises (SMEs) by reducing the high operational costs and time spent on manual hiring workflows.
- **Global Impact:** On a broader scale, the project demonstrates how integrating different AI technologies (Speech-to-Text, Large Language Models, and vocal feature analysis) can reduce personal bias in initial assessments, making the early hiring phase faster, fairer, and more accessible.

The remainder of this document is structurally organized to detail the complete lifecycle of the project. It begins by establishing the background, explaining the core theoretical concepts and technical foundations used in the system. Following this, a literature review evaluates existing market solutions, compares their features, and identifies the gaps that our project addresses. The report then covers system design and development, detailing the development methodology, user requirements, structural diagrams, database models, and application interfaces. Next, the system evaluation showcases the testing procedures and validation results for both model performance and user feedback. Finally, the report concludes by summarizing the overall achievements, highlighting the technical challenges faced, and outlining future enhancements for the platform.

2 Background

In this section, we provide the necessary theoretical and domain background for **Jadeer**. We begin by outlining the current state of recruitment technology, focusing on the use of Applicant Tracking Systems (ATS) and related e-recruitment practices. Following this, we introduce additional technologies and methodologies that **Jadeer** relies on to enhance recruitment processes, including Natural Language Processing (NLP), Large Language Models (LLMs), Speech-to-Text (STT), Text-to-Speech (TTS), Vocal Confidence Scoring, and Psychometric

Evaluations. Finally, we discuss the role of APIs such as Firebase, OpenAI, Whisper, Google Cloud TTS, Hugging Face, and Google ML Kit in enabling the system's intelligent functionality. Together, these concepts form the foundation upon which **Jadeer** is designed and developed.

2.1 Overview of Modern Recruitment Practices

The integration of technology into recruitment has transformed how organizations attract, assess, and hire talent. Traditional paper-based processes have largely been replaced by e-recruitment systems, which automate job postings, streamline CV submissions, and enable faster candidate screening [6]. Video recruitment tools now allow candidates to record or join interviews remotely, saving time and eliminating geographical barriers. While these systems increase efficiency, they also introduce challenges such as reduced personal connection and data security concerns [6].

Building on this landscape, **Jadeer** aims to move beyond administrative automation toward deeper candidate evaluation, combining CV checks with AI-driven simulations to address the limitations of current tools.

2.2 Applicant Tracking Systems (ATS)

As organizations face increasingly high volumes of applications for each job opening, managing candidate information efficiently has become a critical challenge. To address this, recruitment technology has evolved beyond manual review and email-based submissions toward specialized software designed to streamline the process.

An Applicant Tracking System (ATS) is recruitment software designed to help employers manage the large number of applications they receive for job openings. These systems automate core tasks such as storing applicant information (resumes, cover letters, and contact details), tracking candidates throughout the hiring process, and scanning resumes to identify relevant qualifications [7]. By automating these functions, ATS platforms enable HR professionals to reduce manual effort, accelerate hiring cycles, and maintain an organized pool of candidates.

Today, nearly all large organizations, including the majority of Fortune 500 companies, rely on ATS as part of their recruitment infrastructure [7].

2.2.1 ATS Limitations

Despite their widespread adoption, ATS systems face several notable limitations:

- **Keyword Dependence**: Most ATS platforms rely on keyword-based filtering, where recruiters configure the system to search for specific terms related to skills, education, certifications, or experience. As a result, qualified candidates may be overlooked simply because their resumes use different phrasing than the job description. This creates a bias toward applicants who tailor their resumes to match ATS requirements.
- **Limited Contextual Understanding**: Keyword matching ignores semantic meaning and relationships between terms. For instance, an applicant with experience in "software engineering" may not be flagged for a role seeking "programming," even though the skills overlap. Research shows that such keyword-focused methods produce low precision and miss latent semantic connections [8].
- **Incomplete Knowledge Resources**: Even more advanced ATS that integrate semantic ontologies, or occupational classifications remain constrained by limited domain coverage. When a skill or concept is missing from the knowledge base, the system may fail to recognize it, leading to false negatives [8].
- **Lack of Feedback:** While ATS platforms help manage applications on a scale, they rarely provide recruiters with actionable insights into why a candidate is not a good fit. Studies highlight that "common issues such as resume rejection, lack of feedback, and technical problems" remain prevalent in ATS implementations [9].

These constraints highlight that ATS platforms, while useful for reducing the administrative workload of CV handling, focus almost entirely on document level screening. This automation saves time, but it does not reflect the real-world decision-making process of recruiters, which depends heavily on how candidates perform during interviews. Automating CV filtering may help shortlist applicants, but automating candidate evaluation provides far greater value; it captures communication skills and the ability to articulate expertise qualities that cannot be inferred from a resume alone.

Jadeer takes this approach by shifting automation from simple keyword-based screening to more meaningful evaluation. Through features such as AI enhanced CV refinement and interview simulations, the system enables recruiters to assess both the qualifications and the performance of candidates. This not only aligns more closely with actual recruitment practices but also ensures a more informative process for recruiters.

2.3 Natural Language Processing (NLP)

Natural Language Processing (NLP) is a branch of artificial intelligence concerned with enabling computers to analyze, interpret, and generate human language. At its foundation, NLP involves transforming raw text into a structured form that machines can process. Basic preprocessing techniques include tokenization, which breaks text into words or subword units, and the removal of stop-words such as "the" or "and" More advanced approaches use Byte-Pair Encoding (BPE) to represent sub word units, allowing systems to handle out-of-vocabulary words effectively [10]. Additional methods such as stemming, lemmatization, and similarity metrics like minimum edit distance are commonly applied to normalize text and measure the closeness of two-word sequences.

A closely related area is Information Retrieval (IR), which focuses on finding relevant documents in response to a user's information need. IR techniques are designed to handle unstructured text, by indexing terms and ranking documents according to their similarity to a query Manning [11].

While these methods provided foundational capabilities for text processing, they suffer from critical limitations: keyword-based matching cannot capture semantic relationships (e.g., "software engineering" vs. "programming expertise"), bag-of-words models ignore context and word order, and rule-based systems require extensive manual engineering for each edge case.

The emergence of Large Language Models (LLMs) addresses the fundamental limitations of traditional NLP approaches [12]. Rather than relying on rigid keyword matching and handcrafted rules, LLMs learn rich contextual representations from vast text corpora, enabling them to understand semantic similarity, generate contextual content, adapt without retraining by responding to diverse prompts and tasks through prompt engineering [13], eliminating the need for task-specific models, and handling ambiguity by interpreting nuanced language and context that rule-based systems cannot process.

2.3.1 Use of NLP in Jadeer

In **Jadeer**, natural language processing is used in a lightweight and practical way to support features such as extracting keywords from job postings and from the CVs that users upload to their profiles. These keywords are then compared to recommend relevant job postings on the job seeker's "For You" page. This process relies on simple text preprocessing and list-based matching rather than advanced NLP models. More complex language understanding and text generation tasks such as creating job descriptions, enhancing CVs, and generating interview questions are handled separately through Large Language Models (LLMs). This distinction allows **Jadeer** to remain efficient while still benefiting from the richer capabilities of modern generative AI.

2.4 Large Language Models (LLMs)

Language models are systems that assign probabilities to sequences of words, allowing them to predict the likelihood of a word or phrase appearing in context. One of the earliest approaches, the n-gram model, estimated the probability of a word based on a fixed window of preceding words in a training corpus [10]. While n-gram models were effective for tasks such as spelling correction and speech recognition, they suffered from data sparsity and were limited by their short context windows. To address these issues, researchers introduced evaluation metrics such as perplexity and smoothing techniques to improve performance on unseen data.

In contrast to n-gram models, modern LLMs differ fundamentally in both their underlying architecture and their linguistic capabilities. Traditional n-gram models rely on fixed-size context windows and discrete probability counts, which limits their ability to capture long-range dependencies and makes them susceptible to data sparsity issues when encountering unseen word combinations [10]. LLMs, however, are built on transformer architectures that use self-attention mechanisms to model relationships across an entire sequence, regardless of distance [12]. This enables them to learn rich semantic representations and contextual meanings that extend far beyond the shallow, frequency-based statistics of n-grams. Additionally, unlike n-gram models that must be manually engineered for specific tasks, LLMs are trained on large-scale datasets and can generalize across multiple tasks such as reasoning, summarization, question answering, and long-form text generation with significantly higher fluency and coherence [10]. These differences mark a major shift from probabilistic language modeling toward deep contextual understanding.

2.4.1 Use of LLMs in Jadeer

In **Jadeer**, Large Language Models (LLMs) are directly embedded into the platform through the OpenAI API, enabling several core features within the application. Companies can use LLMs to automatically generate clear and professional job-posting descriptions while creating job posts, while job seekers benefit from CV enhancement features where the LLM restructures their CV text, improves the clarity and organization of the content, identifies missing or unclear sections, and produces personalized improvement feedback that is stored and displayed in the application. **Jadeer** also employs LLMs to generate interview questions for both real interviews and mock interviews; mock interview questions are tailored according to the specialty selected by the job seeker before entering the mock interview, whereas real interview questions are generated based on the specific content of each job posting after a company creates a job posting. These LLM-driven functionalities are fully integrated into **Jadeer's** workflows, allowing the system to interpret user inputs, understand job contexts, and produce high-quality outputs in real time. By leveraging the natural-language comprehension capabilities of LLMs, **Jadeer** automates processes that would otherwise require manual review, enabling the platform to scale efficiently while maintaining consistent and context-aware recruitment assistance.

2.5 Speech-to-Text (STT)

Speech recognition, also known as Automatic Speech Recognition (ASR), is the technology that enables computers to convert spoken words and phrases into machine-readable text. Unlike traditional input methods such as keyboards or touchscreens, ASR allows users to interact with devices naturally through voice commands [14].

An ASR system operates by capturing audio signals, splitting them into sound waves, extracting acoustic features (such as pitch, intensity, and duration), and applying algorithms that map these features to the most likely words in a language. This process involves multiple models, including acoustic models, language models, and natural language processing (NLP) techniques [14].

Modern speech recognition is widely applied in domains such as banking, healthcare, marketing, and education. Consumer systems such as Siri, Google Assistant, Alexa, Cortana, and Bixby highlight the everyday role of ASR in enhancing accessibility and convenience [14].

2.5.1 Use of STT in Jadeer

In **Jadeer**, a speech to text (STT) API is used to transcribe candidate interview audio responses after both mock and job interviews. This transcription is then analyzed by the system to provide feedback, enabling a realistic interview experience.

2.6 Text-to-Speech (TTS)

Text-to-Speech (TTS) systems are computer-based applications designed to automatically convert written text into spoken audio. Unlike simple voice response systems that concatenate pre-recorded words for limited domains (e.g., train announcements), a TTS synthesizer is capable of producing novel sentences through processes such as grapheme-to-phoneme transcription. This makes TTS a more flexible technology, enabling the generation of natural sounding speech for any input text [15].

A TTS system does not replicate the exact biological processes of human speech production, which involve complex interactions of airflow, vocal tract dynamics, and cortical control. Instead, most systems adopt simplified engineering architecture composed of two main modules. The Natural Language Processing (NLP) module analyzes input text, producing a phonetic transcription along with prosodic features such as rhythm and intonation. The Digital Signal Processing (DSP) module then transforms this symbolic representation into a speech waveform. This architecture balances linguistic accuracy with computational efficiency, enabling real-time synthesis with manageable resource requirements [15].

2.6.1 Use of TTS in Jadeer

In **Jadeer**, Text-to-Speech (TTS) technology is employed through Google text-to-speech (TTS) API to enhance the realism of both mock and actual interview features. By converting written interview questions into natural sounding spoken output, the system simulates the experience of a live interviewer. This functionality contributes to a more engaging and authentic interview environment.

2.7 Psychometric Evaluations

Psychometric evaluations are standardized assessments designed to measure a candidate's cognitive abilities, personality traits, and behavioral tendencies in a structured and objective way. Recruiters often use these evaluations to gain deeper insights into whether a candidate's skills, motivations, and interpersonal style align with the requirements of a role and the culture of the organization. Common examples include aptitude tests (e.g., numerical, verbal, or logical reasoning), personality inventories, and situational judgment tests. These tools complement CV screening and interviews by providing quantifiable data on attributes that are not easily observable in traditional recruitment stages. As a result, psychometric evaluations have become a common component of modern recruitment processes, helping organizations make more informed and fair hiring decisions [16].

2.7.1 Use of Psychometric Evaluations in Jadeer

In **Jadeer**, psychometric evaluation is integrated into the interview simulation rather than presented as a separate personality test. The system generates behavioral questions based on the Big Five framework, with each question targeting a specific trait such as conscientiousness, openness, agreeableness, extraversion, or emotional stability [17]. These questions are tailored to the job context and specialty, allowing candidates to demonstrate behavioral tendencies through scenario-based interview responses.

After the interview, the candidate's spoken answers are transcribed using Speech-to-Text technology and analyzed through an LLM-based evaluation process. The system considers technical relevance, clarity, communication quality, and behavioral indicators related to the Big Five traits. The results are included in the interview report as supportive feedback, including an overall score, strengths, weaknesses, and improvement advice. This psychometric component is not intended to replace human judgment, but to provide additional behavioral insight alongside the technical evaluation.

2.8 vocal confidence scoring Assessment

Speech contains not only linguistic information, but also paralinguistic cues that reflect how an utterance is delivered. In interview settings, aspects such as vocal fluency, tone stability, speaking rhythm, and hesitation patterns can influence how a candidate's communication style is perceived. Therefore, speech-based analysis can provide additional behavioral information that is not available from written responses alone. In **Jadeer**, this concept is used to estimate an interview-related confidence score as supportive feedback within the interview simulation process.

Psycholinguistic studies have shown that pauses and speech disfluencies are meaningful indicators of speech planning, cognitive processing, and hesitation [18]. In spoken communication, longer silent gaps may suggest uncertainty or reduced fluency, especially when they occur frequently during an answer. Based on this idea, pause analysis can be used as an additional behavioral signal when assessing spoken interview responses. Rather than treating pauses as errors, **Jadeer** considers them as temporal cues that may affect perceived vocal confidence.

In addition to hesitation patterns, speech also contains measurable acoustic characteristics related to vocal behavior. The Geneva Minimalistic Acoustic Parameter Set, known as GeMAPS, was proposed as a standardized acoustic feature set for voice research and affective computing [19]. Its extended version, eGeMAPS, includes acoustic parameters related to pitch, loudness, spectral properties, jitter, and shimmer, which can represent aspects of vocal stability and expressiveness [19]. These handcrafted acoustic features are useful because they provide interpretable measurements of the speech signal.

However, handcrafted acoustic features alone may not capture all high-level contextual and temporal patterns in speech. Recent self-supervised speech models, such as WavLM, learn rich speech representations from large-scale audio data and are designed to support a wide range of speech-processing tasks [20]. WavLM is particularly relevant because it is pretrained to model speech content while also improving performance on non-ASR speech tasks through denoising and contextual representation learning [20]. This makes it suitable for extracting deep speech embeddings that may capture prosodic and paralinguistic patterns beyond traditional acoustic measurements.

For this reason, **Jadeer** adopts a hybrid speech modeling approach. Instead of relying on a single type of feature, the model combines deep speech embeddings extracted using a pretrained WavLM model with interpretable acoustic features extracted using eGeMAPS. This allows the system to benefit from both contextual speech representations and explicit acoustic measurements.

2.8.1 Modeling Approach for vocal confidence scoring

In **Jadeer**, speech-based confidence assessment is formulated as a regression task rather than a classification task. The goal is to predict a continuous score representing interview-related confidence or performance. This formulation aligns with the First Impressions V2 dataset, which provides human-annotated apparent personality traits as well as a job-interview variable represented as a continuous value within the range [0,1] [21]. Therefore, the model predicts a continuous score instead of assigning the candidate to fixed categories.

The dataset annotations are first processed to isolate the interview-related score, which is used as the target variable. The audio is extracted from the original video files, converted into a standardized 16 kHz mono WAV format, and divided into training, validation, and test sets. To reduce identity leakage, the split is performed using speaker-independent grouping, ensuring that the same speaker does not appear across different splits. This helps the model learn general speech patterns related to interview performance rather than memorizing speaker-specific vocal characteristics.

The feature extraction pipeline consists of two parallel components. First, WavLM is used as a pretrained speech representation model to extract 768-dimensional deep embeddings from each audio sample [20]. Second, openSMILE is used to extract eGeMAPS acoustic features from the same audio files [22]. openSMILE is a widely used open-source toolkit for extracting speech and audio features, and it supports standardized feature sets for speech analysis [22]. The eGeMAPS feature set provides interpretable acoustic measurements, while WavLM provides high-level learned representations.

The final model, JadeerNet, follows a late-fusion neural architecture. In this architecture, the WavLM embeddings and eGeMAPS features are processed through separate branches before being concatenated into a unified feature representation. The deep feature branch captures contextual speech patterns, while the acoustic feature branch focuses on explicit vocal characteristics. The combined representation is then passed through a regression head that outputs a score between 0 and 1. The model is trained using Mean Squared Error (MSE), which is appropriate for learning continuous target values.

2.8.2 Use of vocal confidence scoring model in Jadeer

In **Jadeer**, the trained confidence prediction model is integrated into the interview simulation workflow. When a candidate submits a spoken answer, the system processes the audio and extracts both WavLM embeddings and eGeMAPS acoustic features. These features are then passed into the trained regression model to generate an interview-related confidence estimate.

To improve reliability during inference, **Jadeer** also uses Voice Activity Detection (VAD). Specifically, Silero VAD is used to detect whether the uploaded audio contains human speech and to identify speech segments [23]. This step helps prevent the system from generating predictions for invalid inputs, such as silence or non-speech audio. By validating the presence of speech before feature extraction, the system reduces the likelihood of unreliable confidence estimates.

In addition to the model prediction, **Jadeer** incorporates a hesitation-based adjustment using long-pause detection. Silent gaps between speech segments are detected and pauses around or above 0.6 seconds are treated as hesitation cues based on psycholinguistic pause analysis [18]. These pauses are used to adjust the final score because frequent long pauses may indicate reduced fluency or uncertainty during spoken responses.

The resulting score is used as supportive feedback on the candidate's communication style, fluency, and vocal confidence. It is not intended to be used as a standalone hiring decision. Instead, it complements other components in **Jadeer**, such as LLM-based answer evaluation and psychometric-style interview questions. By combining speech-based confidence assessment with content-based and behavioral evaluation, **Jadeer** provides a more comprehensive view of candidate performance during interview simulations.

2.9 APIs used in Jadeer:

2.9.1 Firebase API

Firebase is a cloud-based platform by Google that provides a comprehensive suite of backend services and tools for mobile and web applications. It offers Flutter plugins that allow seamless integration between applications and Firebase's infrastructure, enabling developers to accelerate time-to-market and improve app quality with minimal effort. Firebase services cover core areas such as authentication, real-time databases, analytics, and cloud storage [24].

2.9.1.1 Use of Firebase in Jadeer

In **Jadeer**, Firebase is primarily used to manage user authentication and data storage. By relying on Firebase's secure and scalable infrastructure, the system ensures that candidate and recruiter data remain accessible, synchronized, and protected.

2.9.1.2 Suitability of Firebase for Jadeer

A study [25] shows that Firebase performs well for unstructured data and real-time sync. For **Jadeer**, this means resumes, job postings, and user data can be stored and updated seamlessly, reducing latency and improving user experience while avoiding the overhead of building a custom backend.

2.9.2 OpenAI API

The OpenAI API is a cloud-based service that provides access to advanced artificial intelligence models through a general-purpose "text-in, text-out" interface [26] . By submitting natural language prompts, developers can leverage models for tasks such as summarization, text generation, classification, and reasoning without the need to build or host the models locally. In addition to these capabilities, the API also supports fine-tuning, where developers can train base models on domain-specific datasets to improve performance for specialized tasks [26].

The API offers several model families optimized for different use cases, ranging from lightweight models designed for fast responses to larger models capable of handling complex reasoning and extended conversations [26].

2.9.2.1 Use of OpenAI in Jadeer

In **Jadeer**, the OpenAI API is used to generate tailored job posting description, interview questions, provide CV improvement features, and generate interviews feedback.

2.9.2.2 Suitability of the OpenAI API for Jadeer

The GPT-4 Technical Report [27] shows state-of-the-art results on benchmarks like MMLU and HumanEval, while independent studies confirm the high coherence and fluency of GPT outputs [28] [29]. In **Jadeer**, quality is further enhanced by prompt engineering, which allows the API to generate specific recruitment outputs such as tailored interview questions, job postings and CV improvements.

2.9.3 Whisper API (Speech-to-Text)

Whisper is an automatic speech recognition (ASR) system developed by OpenAI and trained on 680,000 hours of multilingual and multitask supervised data collected from the web. This large and diverse training dataset makes Whisper highly robust against variations in accents, background noise, and technical vocabulary. The system supports both transcription (converting speech into text across multiple languages) and translation (rendering non-English speech into English text), offering broad applicability in multilingual contexts [30].

2.9.3.1 Use of Whisper API in Jadeer

For **Jadeer**, the Whisper API provides the backbone for speech-to-text processing in interview simulations. ensuring that the spoken content is accurately captured for further analysis. By leveraging Whisper, **Jadeer** ensures reliable, accessible transcription capabilities that enhance the realism of the simulated interview process.

2.9.3.2 Suitability of Whisper for Jadeer

OpenAI's Whisper has demonstrated state-of-the-art performance across multiple public datasets, showing robustness to noisy environments, diverse accents, and domain-specific terminology. Its ability to generalize to previously unseen conditions reduces transcription errors compared to traditional ASR models [31] .

2.9.4 Google Cloud Text-to-Speech (TTS) API

Google Cloud Text-to-Speech (TTS) is an API that converts written text or Speech Synthesis Markup Language (SSML) into natural-sounding, synthetic human speech. The service supports multiple audio formats, making it adaptable for integration into applications and media systems. TTS enables developers to provide audible feedback to users, generate voice responses, or augment content with human-like narration [32].

2.9.4.1 Use of Google Cloud (TTS) in Jadeer

In **Jadeer**, the TTS API is used to generate realistic interviewer voices during both mock and job interviews, enhancing the realism of the candidate experience. By converting generated questions into spoken audio, the system offers candidates a more immersive environment that mirrors real-world interview conditions.

2.9.4.2 Suitability of Google Cloud TTS for Jadeer

Studies have shown that Google Cloud TTS achieves high MOS ratings, often approaching the quality of human speech. Tests on US English WaveNet voices, for example, achieved an average MOS of 4.1 out of 5, more than 20% better than standard voices, thereby reducing the gap with human speech by over 70% [33].

2.9.5 Hugging Face API

Hugging Face is an open-source artificial intelligence platform that provides access to machine learning models, datasets, and deployment tools. One of its services, Hugging Face Spaces, allows developers to host machine learning applications and expose them through web interfaces or API endpoints. Gradio is a Python-based framework commonly used to build interactive machine learning demos, and Gradio applications hosted on Hugging Face Spaces can be accessed programmatically through API endpoints [34].

2.9.5.1 Use of Hugging Face in Jadeer

In **Jadeer**, the Hugging Face Gradio API is used to connect the application backend with the deployed vocal confidence scoring model. The custom **Jadeer** model is hosted as a Hugging Face Space and is called from Firebase Cloud Functions using the @gradio/client package. During interview processing, the candidate's recorded audio is submitted to the deployed model, which returns a vocal confidence scoring assessment. This score is then used as part of the interview report, alongside the LLM-based evaluation of answer content and psychometric-style behavioral analysis [35].

2.9.5.2 Suitability of Hugging Face for Jadeer

Hugging Face Gradio API is suitable for **Jadeer** because it allows deployed machine learning models to be accessed through API endpoints without needing a dedicated model serving server. This enables **Jadeer** to call the vocal confidence scoring model from Firebase Cloud Functions during interview processing, making the integration scalable, and suitable for the project's infrastructure constraints.

2.9.6 Google ML Kit API

Google ML Kit is a mobile software development kit that brings Google's machine learning capabilities to Android and iOS applications. It provides ready-to-use APIs for common machine learning tasks, including face detection, text recognition, barcode scanning, image labeling, and pose detection. The Face Detection API can detect faces in images and video, identify facial landmarks and classifications, and process frames in real time directly on the device [36].

2.9.6.1 Use of Google ML Kit in Jadeer

In **Jadeer**, Google ML Kit's Face Detection API is used during mock interviews and job interviews to verify that the candidate remains visible in front of the camera. The camera feed is analyzed during the interview session to detect whether a face is present. If no face is detected, the system displays a warning to the candidate and prevents the interviewee from recording their answers until the candidate's presence is restored. This helps maintain the integrity of the interview environment and discourages candidates from leaving the camera view or relying on external assistance during the session.

2.9.6.2 Suitability of Google ML Kit for **Jadeer**

Google ML Kit is suitable for **Jadeer** because its Face Detection API supports real-time processing on mobile devices. Since face detection is performed on-device, it reduces network dependency and avoids transmitting camera frames to external servers, which is important for privacy in an interview context [36]. Its Android and iOS support also aligns with **Jadeer's** Flutter-based mobile application.

2.10 Integration of Technologies in **Jadeer**

In **Jadeer**, the technologies discussed in the background section work together to create a recruitment system that goes beyond the limitations of traditional Applicant Tracking Systems (ATS). Instead of focusing only on keyword-based CV filtering, **Jadeer** emphasizes a more holistic evaluation of candidates.

Large Language Models (LLMs) are used to generate job descriptions, enhance CVs, create tailored interview questions, and produce interview feedback. Speech-to-Text (STT) technology transcribes candidates' spoken interview answers for analysis. Text-to-Speech (TTS) technology converts written questions into spoken audio, creating a more interactive candidate experience. **Jadeer** also incorporates a vocal confidence assessment model that analyzes acoustic and prosodic features to estimate interview-related confidence, alongside Psychometric-style behavioral questions based on the Big Five framework. Google ML Kit's Face Detection API monitors candidate visibility during interviews to ensure integrity.

APIs such as OpenAI, Whisper, Firebase, Google Cloud TTS, Hugging Face, and Google ML Kit provide the backbone for scalable, secure integration. Together, these technologies enable **Jadeer** to automate candidate assessment in a way that mirrors real-world recruitment more closely than simple CV screening, offering recruiters richer insights and candidates more meaningful feedback.

3 Literature Review

For **Jadeer**, the literature review plays a key role in shaping the system's direction by clarifying what is required and why certain approaches are most suitable. This section looks at existing recruitment platforms and highlights how they function, where they succeed, and where they fall short. Understanding these existing efforts provides context, but the greater value lies in examining the difficulties experienced by recruiters and job seekers, difficulties that many current tools only partially resolve.

The review therefore focuses on competitor analysis, outlining the main capabilities of each platform and the limitations that restrict their effectiveness. By comparing these systems, **Jadeer** can draw meaningful lessons and identify opportunities to design features that directly address persistent gaps, such as lack of candidate feedback, limited fairness, and weak personalization. In this way, the literature review provides a well-grounded basis for **Jadeer's** development, ensuring that the platform grows out of both user needs and an informed view of the competitive landscape.

3.1 Competitive Product Analysis

To better understand the current landscape, the following analysis examines leading recruitment platforms, beginning with HireVue, a widely adopted AI-powered interviewing tool.

hirevue **HireVue**

HireVue is a widely used AI-powered hiring platform that enables organizations to conduct video interviews, assessments, and skill-based evaluations to streamline the recruitment process [37].

Features:

1. **Video Interviewing**: Offers both asynchronous (on-demand) and live video interviews with structured evaluations.
2. **AI-Driven Assessments**: Provides game-based, technical, and language assessments to measure candidate skills and potential.
3. **Find My Fit™ Application**: Matches candidates to roles based on skills, interests, and future potential, supporting diversity and inclusivity.
4. **Human Potential Intelligence**: Uses data science and industrial/organizational psychology to focus on aptitude and attitude, rather than traditional CV-based hiring.

Limitations:

1. **Employer-Centered Evaluation:** HireVue mainly supports hiring teams through interviews and assessments, with greater emphasis on employer decision-making than candidate development or improvement.
2. **Limited Candidate Preparation Support:** The platform is designed for completing recruitment assessments rather than helping candidates practice interviews, improve performance, or receive detailed developmental feedback.

harver **Harver**

Pymetrics (by Harver) is a talent assessment solution originally founded as a neuroscience-based platform using game-like exercises to measure cognitive, emotional, and social traits. After being acquired by Harver in 2022, it became part of Harver's broader portfolio for high-volume, science-driven hiring, combining large-scale automation with innovative neuroscience assessments [38].

Features:

1. **Neuroscience Games**: Provides short, engaging games that measure traits such as memory, attention, decision-making, and risk tolerance, delivering objective behavioral data.

2. **AI-Driven Matching**: Uses machine learning models to compare candidates' profiles with success benchmarks for specific roles, emphasizing aptitude and potential over traditional CVs.
3. **Bias Mitigation**: Includes regular audits, validation studies, and scientifically supported models to reduce adverse impact and promote fairness in hiring.
4. **Integration at Scale**: Through Harver, Pymetrics is embedded into large-scale hiring workflows, enabling companies to process thousands of applicants efficiently while maintaining depth in assessment.

Limitations:

1. **Indirect Assessment Format:** The game-based approach may feel disconnected from actual job tasks, making it harder for candidates to see the relevance of the evaluation to the role they are applying for.
2. **Cultural and Contextual Differences**: Game-based assessments may not fully account for global cultural diversity, which can influence fairness and consistency across applicant pools.

Kanz

Kanz is an AI recruiting copilot and job marketplace, especially for Saudi Arabia, designed to help employers source, assess, and interview candidates faster and more efficiently via automation and broad profile reach [39].

Features:

1. **Widespread Sourcing:** Pulls from a large database (100M+ profiles) to find potential candidates.
2. **Multi-Platform Job Posting:** Post jobs to various platforms through Kanz, increasing visibility.

3. **Automated Assessment & Interview Scheduling:** Uses AI to evaluate candidates and automates scheduling of interviews, aiming to reduce turnaround time ("days, not weeks").
4. **Regional/Local Market Focus:** Tailored to the Saudi market, which could mean localization (language, regulations, culture), which few global platforms can match.

Limitations:

1. **Lack of Deep Candidate Support:** It's not clear whether they provide mock interviews, feedback on responses, or tools to help candidates improve (CV optimization, interview practice).
2. **Limited Assessment Depth:** Because the platform emphasizes speed and automation, candidate evaluations may rely on surface-level screening rather than in-depth skill or behavioral analysis.
3. **Language / Accent / Context Sensitivity:** Even though regional, AI assessments often struggle with dialects or speech variation; unless explicitly stated, it's unclear how well their system handles those.
4. **Feature Depth vs Breadth Trade-Offs:** Because they emphasize automation and speed, they may sacrifice depth in candidate evaluation (e.g. less personalized feedback, less emotional or tone evaluation) in favor of efficiency.

Kabi

Kabi is an AI-powered recruitment platform that combines ATS functionality, psychometric video assessments, and HR consulting, with a strong focus on Saudi Arabia and the wider regional market [40].

Features:

1. **AI-driven ATS (HYRDD):** Streamlines the recruitment pipeline, from posting to shortlisting and interview scheduling.

2. **Psychometric Video Assessments (INVIEWS):** Provides deeper behavioral insights into candidates beyond traditional CVs.
3. **HR Consulting Services:** Offers human capital consulting alongside technological tools.
4. **Regional and Local Focus:** Headquartered in Saudi Arabia, with regional expansion and acquisition of BLOOVO to strengthen local market presence.
5. **Efficiency and Analytics:** Emphasizes reducing time-to-hire and costs while offering analytics to support better decision-making.

Limitations:

1. **Limited Candidate Support:** Lacks clear tools for CV enhancement, mock interviews, or detailed candidate feedback.
2. **Employer-Centric Design:** The platform focuses heavily on employer tools and HR consulting, with limited features aimed at helping candidates understand their evaluation or improve their performance.
3. **Over-reliance on Assessments:** Heavy focus on video/psychometric evaluations may overshadow technical skills, experience, or multilingual ability.
4. **Language and Accent Sensitivity:** Video-based AI assessments can struggle with dialects, accents, or technical issues unless explicitly designed to handle them.
5. **Personalization Depth:** Employers may need richer, customized reports, while candidates may want clearer explanations of acceptance/rejection — areas where **Kabi** seems limited.

Tamayaz

Tamayaz is a Saudi Arabia-based AI-powered talent pool and assessment platform that connects job seekers and employers via verified candidate assessments, video and chat simulations, and auto-scoring [41].

Features:

1. **AI Auto-Scoring across Formats:** Supports scoring for video, chat simulation, and standard assessments, allowing employers to get consistent candidate evaluations.
2. **Verified Talent Pool:** Maintains a pool of pre-assessed and verified candidates, helping reduce risk and filtering effort for employers.
3. **Multi-Industry Streams:** Offers streams by industry (IT, Hospitality, Healthcare, etc.) so assessments and matching are more relevant per field.
4. **Anti-Cheating Measures:** Includes measures to maintain assessment integrity.
5. **Employer Customization:** Employers can invite candidates to take company-specific assessments/tests and use the platform's search / shortlisting tools.

Limitations:

1. **Candidate Feedback Depth:** It's not clear how detailed the feedback is for rejected candidates — whether they receive insights into their performance (strengths/weaknesses) or just scores.
2. **Mock Interview / Practice Tools:** There is no obvious mention of mock interviews or preparatory practice simulations beyond assessments.
3. **Limited Candidate Development:** While the platform scores and verifies candidates, it does not appear to offer tools for self-improvement such as practice interviews, performance feedback, or guided preparation.
4. **Language / Dialect Support:** While localized, the platform doesn't explicitly state support for multiple dialects or non-standard speech, which may affect fairness in video/chat assessments.

Table 1: 3-1 Competitive Product Analysis

Competitors / Features	HireVue	Pymetrics	Kanz	Kabi	Tamayaz	Jadeer
CV Management (Upload & AI Enhancement)	Limited – Upload for context, no AI CV editing	No	Yes	Yes	Yes	Yes
Job Search & Application	No	No	Yes	Yes	Yes	Yes
AI-Powered Interviewing	Yes	No	No	Limited, Some automated assessments.	Limited	Yes
Mock Interviews & Training	Yes	No	No	No	No	Yes
Job Posting Management	No	No	Yes	Yes	No	Yes
Application Management	Yes	Yes	Yes	Yes	Yes	Yes

In reviewing existing recruitment platforms, it is clear that each competitor brings valuable innovations yet also reveals notable gaps. **HireVue** focuses on AI-driven video interviews but offers limited transparency and minimal support for candidate preparation. **Pymetrics** emphasizes fairness and behavioral assessments but does not cover core recruitment needs such as CV management or interview simulations. **Kanz** provides regional focus and automation for job postings and scheduling but lacks detailed candidate feedback or training features. **Kabi** combines ATS functions with video assessments and consulting, yet its approach remains centered on employers with little direct support for job seekers. **Tamayaz** highlights verified talent pools and AI auto-scoring but falls short in providing mock interview practice or in-depth feedback.

Taken together, these platforms demonstrate strong progress in digitizing recruitment but leave critical user challenges unresolved. **Jadeer** addresses these gaps by focusing on both sides of the recruitment process. For job seekers, it provides AI-driven CV enhancement and mock interview practice with personalized feedback, helping them improve before applying. For employers, it offers AI-powered candidate evaluation reports with weighted scoring across multiple dimensions, including technical knowledge, psychometric traits, and vocal confidence analysis. This dual focus positions **Jadeer** as a more balanced platform that supports candidate development alongside employer decision-making.

4 System Design and Development

This section provides an overview of the **Jadeer** system, including its methodology, users, requirements elicitation process, architecture, and use case diagram, highlighting how the platform supports both companies and job seekers in the recruitment process.

4.1 Methodology

The Agile software development methodology was selected as the most suitable approach for this project due to the dynamic and evolving nature of the system requirements. Agile emphasizes flexibility, continuous improvement, and frequent delivery of working software, which allowed the team to adapt quickly to changes and incorporate feedback throughout the development lifecycle.

Figure 1: 4-1 Agile Methodology

In this project, the Scrum framework was adopted as a structured implementation of Agile. Scrum promotes collaboration and iterative progress among team members. The team followed the core components of Scrum, including roles, events, and artifacts (3×5×3).

- **Scrum Roles (3)**

The project team followed the three main Scrum roles:

- **Product Owner:** Dr. Mashael Alsaleh and Dr. Manal Albahlal, responsible for defining and prioritizing the product backlog based on project goals and stakeholder needs.
- **Scrum Master:** Dr. Mashael Alsaleh and Dr. Manal Albahlal, responsible for facilitating the Scrum process, ensuring adherence to Agile practices, and removing any obstacles faced by the team.
- **Development Team:** The student team members, responsible for designing, developing, and delivering the product increment in each sprint.

- **Scrum Events (5)**

The team actively conducted the five Scrum events throughout the project:

- **Sprint:** The project was divided into five sprints, each lasting a fixed period, where a functional increment of the system was developed.
- **Sprint Planning:** At the beginning of each sprint, the team conducted a planning meeting (with the supervisors) to select and define tasks from the product backlog.

- **Daily Scrum:** Short daily meetings were conducted to track progress, discuss challenges, and align on daily goals.
- **Sprint Review:** At the end of each sprint, the team presented the developed features to stakeholders to gather feedback and validate progress.
- **Sprint Retrospective:** After each sprint, the team evaluated their performance, identified areas for improvement, and adjusted their workflow for the next sprint.
- **Scrum Artifacts (3)**

The team utilized the three key Scrum artifacts:

- **Product Backlog:** A prioritized list of all system requirements, features, and enhancements.
- **Sprint Backlog:** A subset of tasks selected for implementation during each sprint.
- **Increment:** The working product features delivered at the end of each sprint.
- **Agile Practices in Our Project**

As a team, we applied Agile principles by maintaining continuous communication, dividing work into manageable tasks, and delivering incremental updates regularly. We adapted to requirement changes by revisiting and refining the backlog, ensuring that the system always aligned with stakeholder expectations. Collaboration and shared responsibility were key factors in maintaining productivity and quality.

- **Tools Used**

To support the Agile and Scrum process, the team utilized the following tools:

- **Jira:** Used for managing the product backlog, organizing sprints, assigning tasks, and tracking progress. It helped the team stay organized and maintain visibility over all development activities.

  Jira link: [Jira Link]

- **GitHub:** Used for version control and collaboration, allowing team members to work on the project simultaneously, track code changes, and manage contributions efficiently.

  Git-Hup link: GitHub Link

4.2 System Requirements

4.2.1 System Users

This section describes the main types and characteristics of users of the **Jadeer** platform. **Jadeer** serves two primary groups: **companies** looking to hire qualified candidates, and **individuals/job seekers** aiming to find job opportunities and improve their chances in the recruitment process in addition to the admin responsible for managing company accounts.

1) **Company users**:

   Companies using **Jadeer** vary in size, industry, and recruitment needs. Key characteristics include:

   - Users are HR professionals, recruiters, or hiring managers responsible for recruiting talent.
   - Users must be able to navigate digital platforms and understand recruitment processes.
   - Users can be from any industry or sector.
   - Users expect an efficient, unbiased, and data-driven hiring process.
   - Users require tools for job posting candidate evaluation, AI-assisted interviews, and performance reporting.

2) **Job seekers users:**

   Job seekers on **Jadeer** include individuals at different career stages who want to find suitable job opportunities and prepare effectively for interviews. Key characteristics include:

   - Users are adults, typically 18 years or older.
   - Users may be fresh graduates, early-career professionals, or experienced job seekers.
   - Users can be of any gender and nationality.

- Users should have basic digital literacy to use AI features for CV enhancement, interview practice, and job browsing.
- Users may have prior experience with job platforms or may be new to online recruitment systems.
- Users are motivated to improve their employability and seek fair recruitment processes.
- Users may have varying levels of preparation: some plan and polish their applications carefully, while others seek guidance on improving their CVs and interview skills.

3) Admin Users:

Admins in **Jadeer** are responsible for overseeing the registration and verification of company accounts. Their duties ensure that only authorized representatives can create and manage company profiles. Key characteristics include:

- Admins are platform supervisors tasked with reviewing company sign-up requests and validating the authenticity of the documents and email credentials submitted.
- Admins must be able to evaluate verification materials, such as proof of employment, official business emails, or authorization letters provided during registration.
- Admins ensure that each company account is legitimate, verifying that the user registering is officially permitted to act on behalf of the organization.
- Admins must be detail-oriented and compliant-driven, as their decisions prevent fraudulent companies from entering the system.
- Admins have the authority to approve or reject company accounts, marking them as *Verified* or *Rejected* within the platform after completing their review.

General Expectations Across Users:

- All users – except for the admin - should have a willingness to engage with AI-driven tools to optimize recruitment or job-seeking outcomes.
- Users value efficiency, accuracy, and fairness in the recruitment process.
- Users are expected to benefit from personalized recommendations, whether for job opportunities, candidate evaluations, or interview preparation.

4.2.2 Requirements Elicitation and Analysis

In this section, we will describe our approach to eliciting requirements for our project. Initially, we documented and elicited requirements using techniques such as user interviews and questionnaires. We relied primarily on these techniques to effectively gather requirements. These methods allowed us to gather valuable insights into the factors influencing the recruitment process for both companies represented by the human resources department and individuals seeking jobs, the challenges they face, and the features that could enhance their experience.

Regarding the interviews (see Appendix A: Interviews), we interviewed five stakeholders as a valuable source of information, three of them were human resources consultants and experts, and the other two were job seekers.

For HR consultants we formulated nine targeted questions focusing on HR professionals' experience with recruitment platforms, their typical hiring process, and the most time-consuming tasks they face. The questions also explored their current methods of resume screening, common reasons for applicant rejection, and the key details they look for in CVs. Additionally, we investigated how they structure interviews, the importance they place on detailed candidate assessment reports, and the features that would make a recruitment app valuable for regular use.

And for job seekers we formulated seven targeted questions focusing on how job seekers choose which jobs to apply for, the challenges they face during the application process, and the ways they usually prepare their CVs. The questions also explored their methods of preparing for interviews, their expectations from mock interview tools, the type of feedback they find most useful, and the features that would make a recruitment app valuable for regular use.

In addition, we used questionnaires (see Appendix B: Questionnaires) as another method for eliciting requirements. We distributed two separate forms one for companies and the other for individuals. The companies form composed of a set of ten carefully designed questions, combining both closed and open-ended formats, to gather valuable insights from HR professionals about their recruitment practices. We collected a total of 10 responses, which allowed us to analyze common trends and challenges in the hiring process. The questionnaire covered topics such as prior experience with recruitment platforms, the typical hiring process, time-consuming tasks, resume screening methods, common reasons for rejection, and the importance of detailed candidate reports.

As for the job seeker's form, composed of a set of 16 questions combining both closed and open-ended formats, these questions capture the perspectives of job seekers regarding their application journey. We collected a total of 30 responses, enabling us to identify common patterns, challenges, and preferences among participants. The questionnaire focused on how they select job opportunities, the biggest obstacles they face in the application process, their methods of preparing CVs and interviews, their expectations from mock interview tools, and the type of feedback they find most useful.

Overall, the combination of interviews and questionnaires enabled us to gather valuable insights from both job seekers and HR professionals, uncovering their needs, challenges, and expectations in the recruitment process. By analyzing this data and presenting it visually, we developed a comprehensive understanding of both perspectives, which guided us in shaping **Jadeer** into a smarter, more effective, and user-centered recruitment platform.

4.2.2.1 Interviews

Through the interviews, we gained valuable insights and results about job seekers' and HR professionals' experiences and preferences when using recruitment platforms in Saudi Arabia. We found that HR professionals have varying degrees of experience with recruitment software: some rely on well-known platforms such as LinkedIn, Jadara, and Job.com and value their efficiency in saving time and finding suitable candidates (from interview 2), while others still rely on manual filtering of CVs, especially in private companies, and consider the process time-consuming (from interview 1). In terms of the hiring process, HR professionals emphasized that it is often long, complex, and labor-intensive, with multiple stages including job posting, candidate screening, interviews, and final evaluation (from interviews 1 and 3). The most time-consuming tasks identified were recruiting candidates and filtering CVs, highlighting a strong need for automation and AI assistance (from interviews 1, 2, and 3).

Regarding candidate evaluation, HR professionals highlighted that personality, experience, qualifications, and compatibility with company culture are crucial factors in rejecting or selecting applicants (from interviews 1, 2, and 3). They also value detailed and organized assessment reports, especially those that measure personality, confidence, and achievements, which support objective decision-making (from interviews 2 and 3). Features that make a recruitment platform valuable include AI-driven personality analysis, efficient candidate selection, and strong performance in recommending suitable applicants (from interviews 1 and 3).

From the job seekers' perspective, participants commonly use specialized websites, personal connections, and social platforms to find job opportunities, with some relying heavily on AI tools like ChatGPT to prepare their CVs and practice interviews (from interviews 4 and 5). The main challenges they face include lack of feedback from companies, difficulty in CV preparation, and insufficient guidance for interviews (from interviews 4 and 5). Job seekers expressed interest in features that provide realistic mock interview simulations, actionable feedback on performance, and opportunities to improve CVs (from interviews 4 and 5).

Suggestions for improving recruitment platforms include providing clear and honest feedback, enhancing AI support for CV optimization and interview preparation, verifying the authenticity of job postings, and integrating all opportunities into a single accessible platform (from interviews 4 and 5). Overall, the interviews highlighted a gap between current recruitment tools and user needs, emphasizing automation, personalized recommendations, and comprehensive feedback as key factors for a successful recruitment experience.

Here is information about our participations:

Table 2: 4-1 Summarizing the participants' information

Participants	Name	Location	Gender	Age	Nationality	Occupation	Educational Level
1	Participant 1	Riyadh	Male	45	Saudi Arabia	HR	PhD
2	Participant 2	Riyadh	Male	48	Saudi Arabia	HR	Master
3	Participant 3	Jeddah	Male	51	Saudi Arabia	HR	Master
4	Participant 4	Riyadh	Female	24	Saudi Arabia	Computer science	Bachelor
5	Participant 5	Riyadh	Female	25	Syria	Software engineering	Bachelor

4.2.2.2 Questionnaires

1- Companies Survey:

We conducted a survey with **10 organizations** from different sectors (70% government, 20% private sector, 10% regulatory bodies). The findings can be summarized as follows:

- **Volume of Applications:** 40% of organizations receive more than 500 applications per job posting, 30% receive 201–500, another 30% receive 50–200, while 10% receive fewer than 50.
- **Most Time-Consuming Stage:** 60% identified *initial CV screening* as the most time-consuming stage, 30% said *candidate evaluation/reporting*, and 10% mentioned *conducting first interviews*.
- **Use of Digital Tools:** 30% currently use recruitment platforms, another 30% are planning to, 20% are unsure, and 20% do not use any tools. Reported tools included LinkedIn, HireVue, SAP SuccessFactors, and traditional recruitment systems.
- **Trust in AI Systems:** Opinions were divided: 20% expressed low trust, 30% moderate trust, 30% high trust, and 20% very high trust in delegating interviews to AI systems.
- **Concerns about AI-Generated Questions:** Main concerns were *professional tone* (50% marked as major concern), *difficulty level* (30%), *relevance to the role* (20%), and *risk of repetition* (20%).
- **Analytical Reports:** All participants (100%) found analytical reports after AI-assisted interviews useful, with 50% rating them “very useful” and 50% “somewhat useful.”
- **Most Desired Insights:** 80% wanted *strengths and weaknesses*, 70% *comparison with other applicants*, 50% *overall suitability ranking*, and 50% *communication and confidence analysis*.
- **Report Usage:** 50% would use the report as *assistant support*, 40% as *final decision support*, and 10% were unsure.
- **Suggestions for Improvement:** Included *integration with government platforms*, *benchmarking against market data*, and *general feedback requests*.

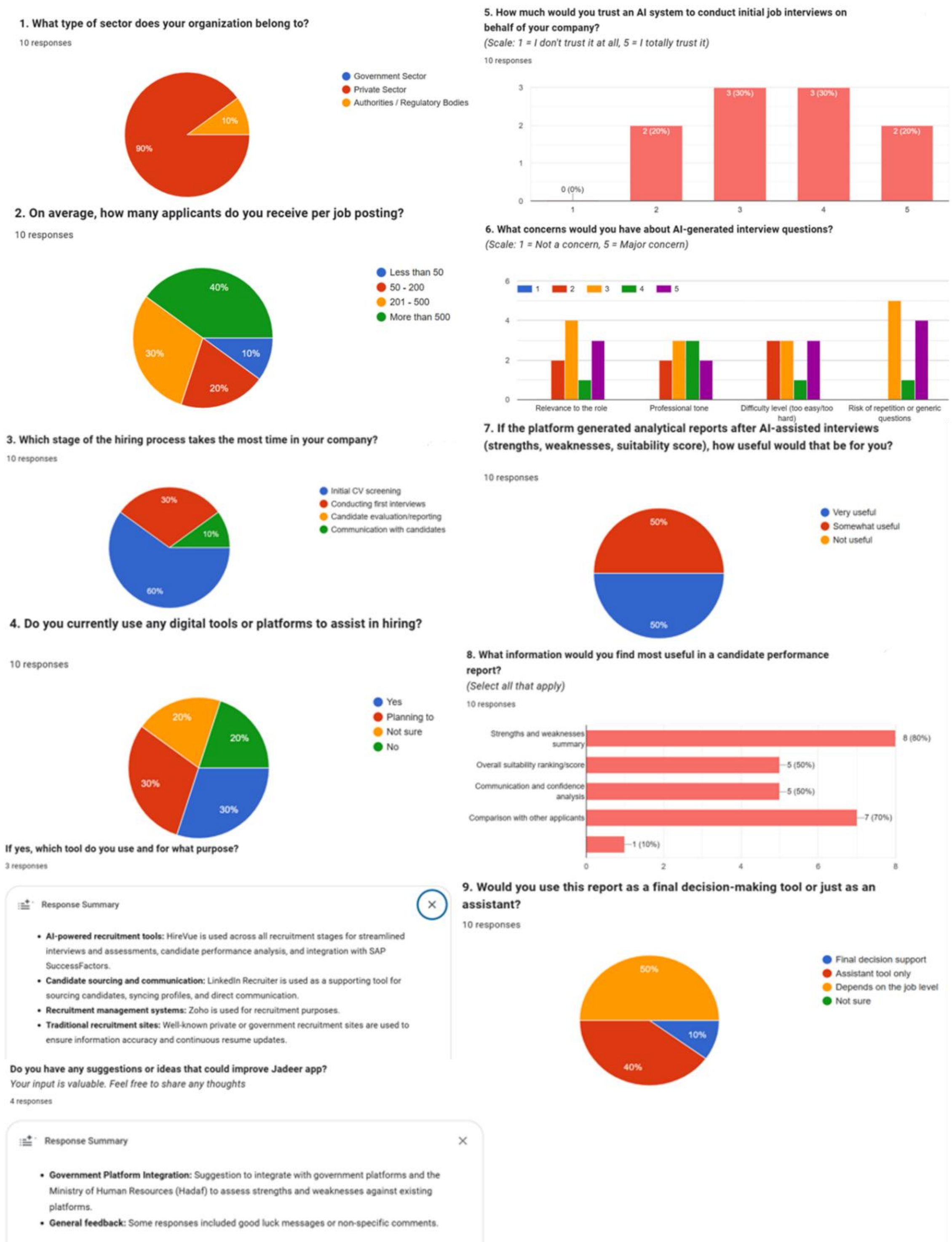

Figure 2: 4-2 Companies Survey Results

2- Individuals Survey:

A survey of **30 individuals** was conducted to understand their job-seeking experiences and expectations from AI-powered recruitment tools. The results were as follows:

- **Demographics:** 70% of respondents were *under 25*, 20% aged *25–34*, 7% aged *35–44*, and 3% aged *45–54*. Females represented 90% of participants, while males made up 10%.
- **Employment Status:** 50% were *students*, 26.7% *unemployed*, 13.3% *interns/freelancers*, and 10% *employed*.
- **Job Search Habits:** 36.7% always apply for jobs online, 26.7% often, 26.7% sometimes, and 10% rarely. None reported "never."
- **Platforms Used:** LinkedIn (76.7%), WhatsApp/Telegram groups (70%), local job boards such as Saudi Aramco/others (60%), and company career portals (36.7%). Very few used Facebook (3.3%) or Sabbar/Naukri (3.3%).
- **Challenges Faced:** The biggest obstacles were *writing/improving CVs (60%)*, *finding suitable opportunities (60%)*, *high competition (60%)*, and *preparing for interviews (46.7%)*.
- **CV Customization & Feedback:** 50% reported they do not customize their CVs for each application, 36.7% do, and 13.3% sometimes do. Regarding AI-generated CV feedback, 83.3% considered it *very important*, 10% *important*, 3.3% *moderately important*, and only 3.4% considered it unimportant. The most valued CV enhancement was *skills relevance (73.3%)*, followed by *keyword match (16.7%)* and *layout (10%)*.
- **Interview Preparation:** 63.3% practice alone, 46.7% prepare with AI tools, 43.3% watch online videos, 23.3% practice with family/friends, and 16.7% attend workshops/training sessions. Only 6.7% reported not preparing much.
- **AI Mock Interviews:** 56.7% found them "very useful," 40% "somewhat useful," and only 3.3% "not useful." Preferences for using them included *before a final/important*

interview (36.7%), *before applying to any job (33.3%)*, *regularly to improve performance (26.7%)*, and *only when underprepared (3.3%)*.

- **Comfort with AI Evaluation:** 36.7% felt very comfortable, 36.7% somewhat comfortable, 23.3% neutral, 3.3% slightly comfortable, and none reported discomfort.
- **Concerns about AI Evaluation:** Participants expressed concerns about *accuracy (53.3%)*, *data privacy/security (53.3%)*, *lack of human empathy (40%)*, and *bias/unfair judgment (23.3%)*. Only 16.7% reported no concerns.
- **Preferred Feedback After Interviews:** 46.7% valued *strengths and weaknesses*, 23.3% *confidence/tone of voice*, 23.3% *answer relevance*, and 6.7% preferred "all feedback types."
- **Suggestions for Improvement:** Key suggestions included *language support (Arabic & English)*, *guidance and advice from experienced job seekers*, *continuous reminders/notifications*, and *positive encouragement*.

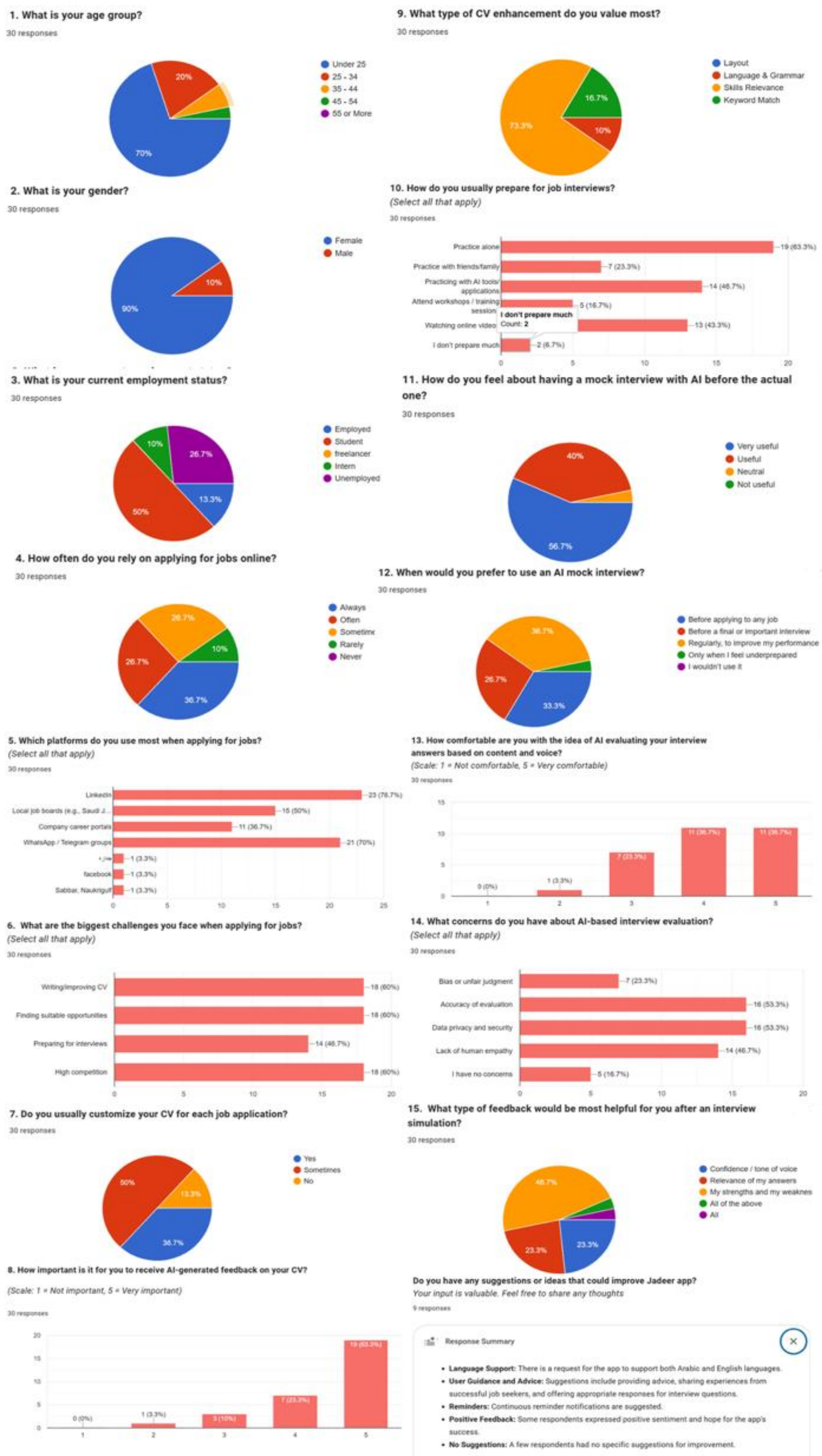

Figure 3: 4-3 individuals Survey Results

4.2.2.3 Reflections and Adjustments on Jadeer

After conducting interviews with three experts in the field of human resources, we also consulted an HR specialist at **SDAIA** and presented the project idea, which provided us with valuable insights. Based on the feedback received, we made several adjustments to the **Jadeer** application. One major enhancement was the integration of personality assessment questions during the interview process, with the results included in the final report alongside the evaluation of the candidate's CV and technical skills. This adjustment was driven by the strong emphasis from HR professionals on understanding candidates' personalities. Additionally, we introduced a feature to assist companies in drafting job descriptions once the job title is defined—an idea proposed by one of the experts we interviewed. On the candidates' side, we implemented an application status tracking feature (pending, shortlisted, or rejected) in response to the common frustration individuals expressed about not receiving feedback after interviews.

4.2.3 User Interactions

In this section, we provide the use case diagram for the **Jadeer** system. This diagram illustrates how different actors, such as the Job Seeker, Company, and Admin, interact with the system's various functionalities (use cases) like creating an account, browsing for jobs, and managing job postings.

We have two types of relationships in the diagram:

- **<include>:** This relationship is used when a use case contains the processes of another use case as a mandatory part of its normal flow of actions. For example, the "Apply for job" use case includes "Participate in AI interview" which means a user must verify their email during the account creation process.
- **<extend>:** This relationship is used for an optional behavior that can be added to a use case. For instance, the "Browse jobs" use case is extended by "Filter/Search jobs," meaning that while browsing jobs, a user has the option to filter the results.

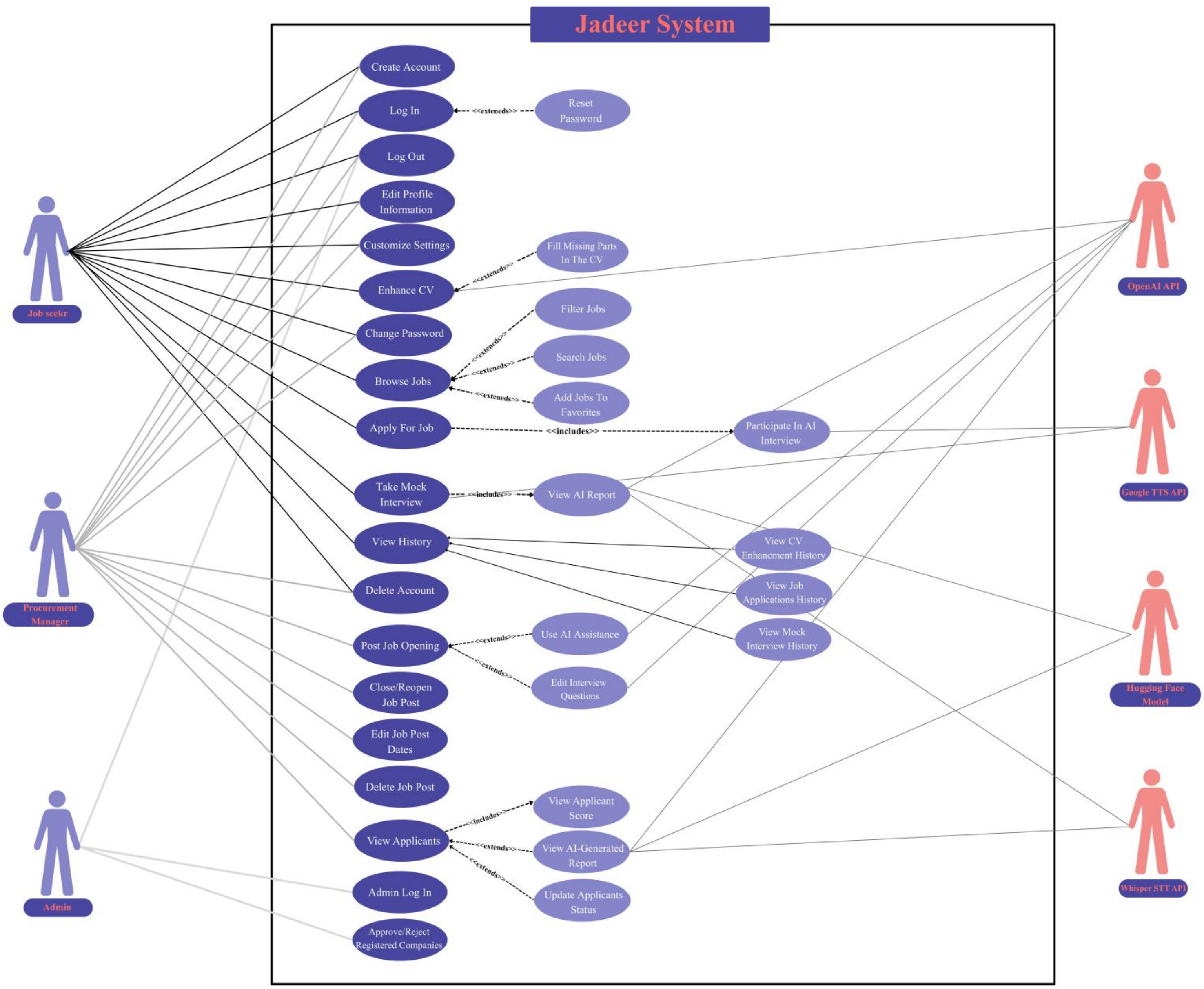

Figure 4: 4-4 Jadeer's Use Case Diagram

4.2.4 Roadmap and Product Backlog Table

This section presents the roadmap witch presents the planned development of **Jadeer** across six sprints, organized into two major releases aligned with the project timeline and the product backlog for **Jadeer** witch includes all functional user stories for job seekers, companies, and admins, as well as supporting items such as defects and technical tasks. Each story is assigned a size, type, and acceptance criteria written in an *if…then* format to ensure clarity and testability.

4.2.4.1 Product Roadmap

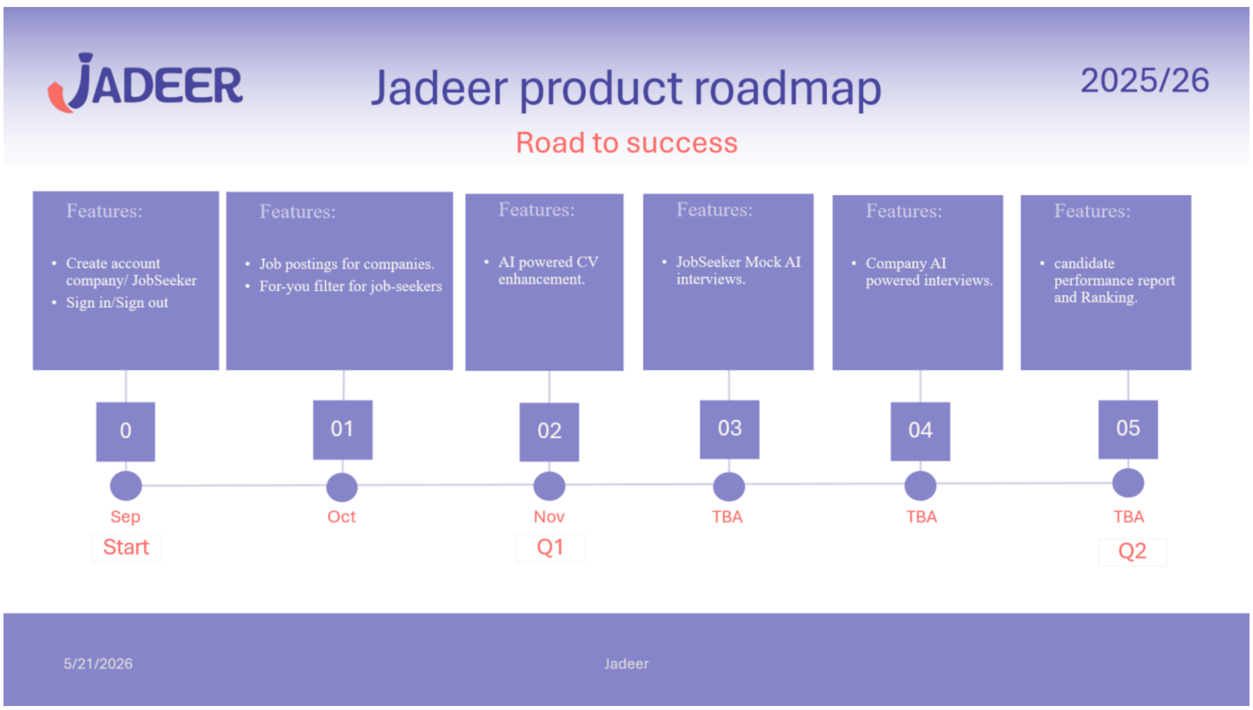

Figure 5: 4-5 Product Roadmap

4.2.4.2 Product Backlog

Table 3: 4-2 Jadeer's Product Backlog

#	**PBI** (user story)	**Size** (Story points)	**Type** (Feature, defect, technical work, knowledge acquisition)	**Status** (To do, in progress, or Done)	**Acceptance Criteria** (The conditions of satisfaction that must be met for that item to be accepted)
1	As a job seeker, I want to create an account by entering my full name, email, and password so that I can access job opportunities on **Jadeer**.	5	Feature	Done	• As a job seeker, if any required field (full name, email, or password) is missing, then I should see an error message indicating which field is required. • As a job seeker, if the full name contains numbers, special characters, or letters other than English, then I should see an error message: "Name must be in English letters only, without numbers or special characters". • As a job seeker, if the full name contains fewer than two words, then I should see an error message indicating that a full name with at least two words is required. • As a job seeker, if the email format is invalid, then I should

					see an error message "Enter a valid email address". • As a job seeker, if the email is already registered, then I should see an error message "Email already in use". • As a job seeker, if the password is shorter than 8 characters, or does not include at least one uppercase letter, or does not include at least one lowercase letter, or does not include at least one number, or does not include at least one special character, then I should see an error message explaining the password policy. • As a job seeker, if all fields are valid, then I should receive a one-time password (OTP) to verify my email. • As a job seeker, if I enter a valid OTP within 2 minutes, then my account should be activated. • As a job seeker, if the OTP expires after 2 minutes, then I

					should be required to request a new one. • As a job seeker, if I request a new OTP while the current one is still valid, then I should see a message indicating the remaining time before I can request a new code.
2	As a company, I want to create an account by providing my company name, full name, email, and password so that I can post job openings and find qualified candidates.	5	Feature	Done	• As a company, if any required field (company name, full name, email, or password) is missing, then I should see an error message identifying the missing field. • As a company, if the email format is invalid or already registered, then an error message should appear. • As a company, if the company name contains numbers or special characters, then I should see the error message: "Company name must contain only English letters and spaces, without numbers or special characters". • As a company, if the full name contains fewer than two words

					or includes non-English characters, then I should see an error message. • As a company, if the password does not meet the requirements (minimum 8 characters, at least one uppercase letter, one lowercase letter, one number, and one special character), then I should see an error message explaining the password policy. • As a company, if all fields are valid, then I should receive an OTP at my email address. • As a company, if I enter a valid OTP within 2 minutes, then I should see that my account status is Pending. • As a company, if the OTP expires after 2 minutes, then I should be required to request a new one. • As a company, after completing my registration, I should receive an email requesting the upload of an

					official company document for verification. • As a company, if the admin approves the document, then the account status changes to Verified, and a confirmation email is sent. • As a company, if the admin rejects it, then the account status changes to Rejected, and a rejection email is sent.
3	As a job seeker, I want to log in using my email and password so that I can access my account.	2	Feature	Done	• As a job seeker, if I enter a valid email and password, then I should be logged into my account successfully. • As a job seeker, if I enter an invalid email or password, then I should see an error message "Invalid credentials. Please try again". • As a job seeker, if I enter incorrect credentials 3 or 4 times, then I should see a warning message showing how many attempts I have remaining before my account is locked.

					• As a job seeker, if I exceed 5 failed login attempts in one day, then my account should be locked and I should be required to reset my password before logging in again.
4	As a company, I want to log in using my email and password so that I can access my account.	2	Feature	Done	• As a company, if I enter a valid email and password and the account status is Verified, then I should be logged into my account successfully. • As a company, if the account status is Pending, then I should see a message indicating that my account is awaiting admin approval. • As a company, if the account status is Rejected, then I should see a message indicating that my account has been rejected. • As a company, if I enter an invalid email or password, then I should see the error message: "Invalid credentials. Please try again". • As a company, if I enter incorrect credentials 3 or 4 times, then I should see a

					warning message showing how many attempts I have remaining before my account is locked. • As a company, if I exceed 5 failed login attempts in one day, then my account should be locked and I should be required to reset my password before logging in again.
5	As a job seeker, I want to reset my password using the "Forgot Password" option so that I can regain access to my account.	3	Feature	Done	• As a job seeker, if I select "Forgot Password" on the login screen, then I should be asked to enter my registered email. • As a job seeker, if I enter a registered email, then I should receive a verification code via email. • As a job seeker, if I enter an unregistered email, then I should see an error message indicating that the email was not found. • As a job seeker, if I enter an email that has not been verified yet, then I should see a message asking me to verify

					my email first before resetting my password. • As a job seeker, if I enter the verification code correctly, then I should be able to create a new password. • As a job seeker, if I enter a new password that is at least 8 characters and includes at least one uppercase letter, one lowercase letter, one number, and one special character, then my password should be updated and I should see a confirmation message.
6	As a company, I want to reset my account password using the "Forgot Password" option so that I can regain access to my account.	3	Feature	Done	• As a company, if I select "Forgot Password" on the login screen, then I should be asked to enter the registered email. • As a company, if I enter a registered email, then I should receive a verification code via email. • As a company, if I enter an unregistered email, then I should see an error message indicating that the email was not found.

					• As a company, if I enter an email that has not been verified yet, then I should see a message asking me to verify my email first before resetting my password. • As a company, if I enter the verification code correctly, then I should be able to create a new password. • As a company, if I enter a new password that is at least 8 characters and includes at least one uppercase letter, one lowercase letter, one number, and one special character, then the password should be updated and I should see a confirmation message.
7	As an admin, I want to log in using my email and password so that I can access the admin dashboard.	5	Feature	Done	• As an admin, if I enter a valid email and password, then I should be prompted to complete a one-time verification step. • As an admin, if I enter an invalid email or password, then I should see the error

					message “Invalid credentials. Please try again”. • As an admin, if I exceed the allowed number of failed login attempts, then I should be required to contact the system administrator to regain access to my account. • As an admin, if I complete the verification step successfully, then I should be logged in and redirected to the admin dashboard. • As an admin, if I enter an incorrect verification code, then I should see the error message “Invalid code. Please try again”. • As an admin, if the verification code expires, then I should be able to request a new one. • As an admin, if I am already logged in, then my session should remain active until it expires. • As an admin, if my session expires, then I should be

					redirected to the login screen and required to log in again.
8	As an admin, I want to approve or reject company registration requests so that only verified companies can access **Jadeer**.	3	Feature	Done	• As an admin, if I open the admin dashboard, then I should see a list of all registered companies with their current status (Pending, Verified, Rejected). • As an admin, if no company exist in **Jadeer**, then I should see the message: “No company found”. • As an admin, if I view a company’s details, then I should see the company name, email, date of registration, and status. • As an admin, if I approve a pending company, then its status should change to Verified and I should see that a confirmation email has been sent. • As an admin, if I reject a pending company, then its status should change to Rejected and **Jadeer** should

					automatically send a rejection email including the reason. • As an admin, if I filter by “Pending”, then only companies marked as Pending should be displayed. • As an admin, if I filter by “Verified”, then only companies marked as Verified should be displayed. • As an admin, if I filter by “Rejected”, then only companies marked as Rejected should be displayed. • As an admin, if no company match the selected filter, then I should see the message: “No companies found for this status”. • As an admin, if I clear the filter, then all companies should be displayed.
9	As a job seeker, I want to log out so that I can safely exit my account.	2	Feature	Done	• As a job seeker, if I choose to log out, then I should be taken out of my account. • As a job seeker, after logging out, if I try to access any part of

					the platform, then I should be required to log in again.
10	As a company, I want to log out so that I can safely exit my account.	2	Feature	Done	• As a company, if I choose to log out, then I should be taken out of the account. • As a company, after logging out, if I try to access any part of the platform, then I should be required to log in again.
11	As an admin, I want to log out so that I can safely exit my account.	2	Feature	Done	• As an admin, if I choose to log out, then I should be taken out of my account. • As an admin, after logging out, if I try to access any part of the platform, then I should be required to log in again.
12	As a job seeker, I want to delete my account so that my personal data and activity history are no longer available.	2	Feature	Done	• As a job seeker, if I click "Delete Account", then I should be asked to confirm the deletion. • As a job seeker, if I confirm deletion, then all my personal data and activity history should no longer be available. • As a job seeker, if I cancel the deletion request, then no changes should occur.

					• As a job seeker, if the deletion completes successfully, then I should no longer have access to my account and should see a confirmation message that my account has been deleted.
13	As a company, I want to delete my account so that my organization's data and job postings are no longer available.	2	Feature	Done	• As a company, if I click “Delete company Account”, then I should be prompted to confirm the deletion. • As a company, if I confirm deletion, then all data and content associated with the company should no longer be available. • As a company, if deletion completes successfully, then I should no longer have access to the company account and should receive a confirmation email. • As a company, if I cancel the deletion request, then no changes should occur.
14	As a job seeker, I want to customize my app settings	3	Feature	Done	• As a job seeker, if I open the Settings page, then I should see options for Appearance, Notifications, Change

	so that I can personalize my experience on **Jadeer**.				Password, My Account Details, About, Log Out, and Delete Account. • As a job seeker, if I toggle the Appearance setting, then the appearance of the app should change accordingly. • As a job seeker, if I toggle the Notifications setting, then alerts and reminders should be enabled or disabled according to my selection. • As a job seeker, if I open About, then I should see general information about the application. • As a job seeker, if I open the My Account Details page from Settings, then I should be able to view my personal information and update the fields that are allowed to be changed. • As a job seeker, if I leave any required fields empty or enter invalid input, then the update should not be completed, and I should see an error message.

					• As a job seeker, if my updates are valid, then I should see a success message confirming the changes.
15	As a company, I want to customize my app settings so that I can personalize my experience on **Jadeer**.	3	Feature	Done	• As a company, if I open the Settings page, then I should see options for Appearance, Notifications, Change Password, My Account Details, About, Log Out, and Delete Company Account. • As a company, if I toggle the Appearance setting, then the appearance of the application should change accordingly. • As a company, if I toggle the Notifications setting, then alerts and reminders should be enabled or disabled according to my selection. • As a company, if I tap My Account Details, then I should be able to view my company information and update the fields that are allowed to be changed. • As a company, if required fields are left empty or invalid

					input is entered, then the update should not be completed, and I should see an error message. • As a company, if my updates are valid, then I should see a success message confirming that the account information has been updated. • As a company, if I tap About, then I should see general information about the application.
16	As a job seeker, I want to change my password so that I can keep my account secure.	2	Feature	Done	• As a job seeker, if I want to change my password, then I must provide my current password first. • As a job seeker, if the current password I enter is incorrect, then I should see an error message. • As a job seeker, if my new password does not meet the requirements (minimum 8 characters, at least one uppercase letter, one lowercase letter, one number, and one special character), then I

					should see an error message explaining the policy. • As a job seeker, if I provide a valid current password and a valid new password, then my password should be updated and I should see a success confirmation message.
17	As a company, I want to change my password so that I can keep my account secure.	2	Feature	Done	• As a company, if I want to change my password, then I must provide my current password first. • As a company, if the current password I enter is incorrect, then I should see an error message. • As a company, if my new password does not meet the requirements (minimum 8 characters, at least one uppercase letter, one lowercase letter, one number, and one special character), then I should see an error message explaining the policy. • As a company, if I provide a valid current password and a valid new password, then my

					password should be updated and I should see a success confirmation message.
18	As a job seeker, I want to edit my profile information so that companies can better understand my background and reach me easily.	2	Feature	Done	• As a job seeker, if I upload a personal photo, then it should meet the required format and size, otherwise I should see an error message. • As a job seeker, when I enter my date of birth, then I should only be able to select a valid date. • As a job seeker, when I enter my nationality, then I should select it from a predefined list. • As a job seeker, when I enter my phone number, then it should follow a valid format, otherwise I should see an error message. • As a job seeker, if I enter my contact email, then it must follow valid email formatting. • As a job seeker, when I provide all required profile information, then I should see that my profile is complete.

					• As a job seeker, if I already have active applications, then I should be required to keep my required profile information filled.
19	As a company, I want to edit my profile information so that job seekers see accurate and up-to-date information about my company.	2	Feature	Done	• As a company, if I upload a company logo, then it should meet the required requirements, otherwise I should see an error message. • As a company, if I update the company description, then it should meet the required length, otherwise I should see a warning message. • As a company, when I enter the company location, then it should be accepted only if it is valid. • As a company, when I enter a contact email, then it should be a valid email address. • As a company, when I enter a contact phone number, then it should follow a valid format, otherwise I should see an error message.

					• As a company, when I enter a company website, then it should be a valid website address. • As a company, when all required profile information is provided, then I should see that the profile is complete. • As a company, if the update cannot be completed, then I should see an error message. • As a company, if I already have active job postings, then I should be required to keep the required profile information filled.
20	As a company, I want to create a job posting so that job seekers can view the job details and apply.	3	Feature	Done	• As a company, if I try to create a job posting, then I should be required to have a complete company profile. • As a company, if the profile is incomplete, then I should see a message asking me to complete the missing profile information. • As a company, if the profile is complete, then I should be able

					to continue to create the job posting. • As a company, if I submit the job posting with missing or invalid required information, then I should see error messages and the job posting should not be submitted. • As a company, if I submit a valid job posting, then I should be able to continue to the next step, and the job should not be visible to job seekers yet.
21	As a company, I want to generate a job description draft using AI so that I can create job postings more efficiently.	3	Feature	Done	• As a company, if I open the job creation page, then I should see an option to generate a job description automatically and information about my remaining usage. • As a company, if I provide the required job information and choose to generate a description, then a draft job description should be generated for me. • As a company, if the AI service returns a successful response, then I should see a generated

					job description that I can review and edit before continuing. • As a company, if I cannot use the automatic generation option, then I should be informed and still be able to write the job description manually. • As a company, if the description cannot be generated, then I should see an error message and still be able to write or edit the job description manually.
22	As a company, I want to edit the interview questions generated for my job posting so that I can customize them for my hiring needs before publishing the job.	3	Feature	Done	• As a company, after creating a job posting, I should be able to review the interview questions associated with that job. • As a company, I should be able to modify the text of the interview questions. • As a company, I should be able to add my own interview questions and choose their type.

					• As a company, I should be able to remove interview questions that I added myself. • As a company, when I finish editing the interview questions, I should be able to confirm them before the job is published. • As a company, if an issue prevents saving the interview questions, then I should see an error message and the job should not be published.
23	As a company, I want to edit my job posting dates so that I can keep the application timeline accurate.	2	Feature	Done	• As a company, if I click Edit on a job post, then I should be able to update the job posting dates. • As a company, if I update the job post dates with valid values, then the changes should be saved and reflected immediately. • As a company, if I enter invalid or illogical dates (e.g., end date before start date), then I should see an error message and the changes should not be saved.

					• As a company, if I cancel the edit, then no changes should be made.
24	As a company, I want to close or reopen my job posting so that I can control whether applications are being accepted.	2	Feature	Done	• As a company, if I click Close on an open job post, then the job should no longer accept new applications. • As a company, if a job is closed, then I should be able to see it as closed in my dashboard. • As a company, if I click Reopen on a closed job post and the posting dates have expired, then I should be required to enter new valid dates before the job can be reopened. • As a company, if I click Reopen on a closed job post and the posting dates are still valid, then the job should be reopened directly without requiring any date changes. • As a company, if I successfully reopen a job, then it should become visible and available to job seekers again.

25	As a company, I want to delete my job posting so that it is no longer available to job seekers.	2	Feature	Done	• As a company, if I click Delete on a job post, then I should be asked to confirm the deletion. • As a company, if I confirm deletion, then the job post should no longer be available. • As a company, if I cancel the delete action, then no changes should be made.

26	As a job seeker, I want to browse job postings so that I can explore available opportunities.	3	Feature	Done	• As a job seeker, if I open the jobs page, then I should see available job postings by default.
					• As a job seeker, if no jobs are available, then I should see a message indicating that no jobs are available.
					• As a job seeker, if I open a job posting, then I should see:
					  ○ company details: company name, logo, location, description, contact email/phone and website.
					  ○ Job details: job title, job description, and position.
					  ○ Job requirements
					• As a job seeker, if I open a job posting, then I should see the

					job application timeline clearly. • As a job seeker, if I open a job posting, then I should see available actions: ○ "Apply" (goes to CV upload/application step). ○ "Add to Favorites" (stores job for later). • As a job seeker, if I click "Apply" from the job details page, then I should be able to start the application process. • As a job seeker, if the job is Closed, then I should see that the job is closed and cannot be applied for.
27	As a job seeker, I want to add jobs to my favorites so that I can review and apply for them later.	2	Feature	Done	• As a job seeker, if I click "Add to Favorites" on a job posting, then the job should appear in my favorites list. • As a job seeker, if I click "Remove" on a job in my Favorites, then the job should be removed from my favorites list.

					• As a job seeker, if I open the Favorites page, then I should see all jobs I have added. • As a job seeker, if my Favorites is empty, then I should see a message indicating that no jobs have been added yet.
28	As a job seeker, I want to filter job postings so that I can find jobs that match my preferences.	3	Feature	Done	• As a job seeker, if I apply any filter option, then only matching job postings should be displayed. • As a job seeker, if I reset filters, then the full job list should be displayed again. • As a job seeker, if I have uploaded a CV to my profile, then personalized "For You" recommendations should be generated based on my skills and experience. • As a job seeker, if my CV is not uploaded and I enable "For You", then I should see a reminder to complete my profile. • As a job seeker, if I update my CV, then the "For You"

					recommendations should be refreshed accordingly. • As a job seeker, if no sorting option is applied, then the default view should be "Newest First", and only open jobs should be shown. • As a job seeker, if I enable personalized recommendations, then only recommended jobs should be displayed. • As a job seeker, if "Show closed jobs" is OFF, then closed jobs should be hidden in all views. • As a job seeker, if "Show closed jobs" is ON, then closed jobs should be included in the results. • As a job seeker, if I apply any filters, then I should see the total number of matching jobs. • As a job seeker, if no jobs match my filters, then I should see the message: "No jobs match your filters".

29	As a job seeker, I want to search for job postings by keyword so that I can quickly find specific opportunities.	3	Feature	Done	• As a job seeker, if I enter a keyword in the search field, then only job postings matching that keyword should be displayed. • As a job seeker, if the keyword matches the job title, company name, or job description, then those postings should appear in the results. • As a job seeker, if no job postings match my search keyword, then I should see an empty-state message: "No jobs found matching your search". • As a job seeker, if I clear the search field, then the full list of available job postings should be displayed again. • As a job seeker, if I search while a filter is active, then results should match both the keyword and the applied filter simultaneously.
30	As a job seeker, I want to view a history of my activities so that I can track my past CV	3	Feature	Done	• As a job seeker, if I open the History Page, then I should see three tabs: CV Enhancements,

	enhancements, mock interviews, and job applications.				Mock Interviews, and Job Applications. • As a job seeker, if no records exist in a tab, then I should see a message indicating that no records are available.
31	As a job seeker, I want to enhance my CV so that it is more professional and well-structured.	5	Feature	Done	• As a job seeker, if I open the CV enhancement screen, then I should see how many AI enhancement credits I have remaining for today. • As a job seeker, if I have no AI enhancement credits remaining, then I should not be able to upload a CV and should see a message that credits reset at midnight. • As a job seeker, if I upload a CV in PDF, DOC, or DOCX format and the file size does not exceed 10MB, then it should be accepted. • As a job seeker, if the uploaded CV does not meet the requirements, then I should see an error message. • As a job seeker, if I request a general CV improvement, then

					I should receive an enhanced version of my CV. • As a job seeker, if I select a specific job from Jadeer, then the CV improvement should reflect that job's details. • As a job seeker, if I provide a job title and description for a job outside Jadeer, then the CV improvement should reflect that job. • As a job seeker, if I choose "Apply for another job", then I must provide both a job title and a job description to continue. • As a job seeker, after my CV is improved, then I should be able to view and download the improved version and see the related improvement suggestions. • As a job seeker, if the improved CV cannot be downloaded, then I should see an error message. • As a job seeker, if the improvement suggestions are

					unavailable, then I should see a message indicating no improvements were recorded and still be able to download the improved CV.
32	As a job seeker, I want to fill in missing sections in my CV so that my CV is complete before enhancement.	5	Feature	Done	• As a job seeker, if I see missing sections in my CV, then I should be able to fill them using simple input fields. • As a job seeker, after filling the missing sections, I should be able to review and optionally update my existing CV sections as well. • As a job seeker, if I choose not to fill a missing section, then I should be able to skip it and continue. • As a job seeker, if I choose to skip all missing sections, then I should see a warning listing the remaining missing sections, but I should still be able to continue with CV enhancement. • As a job seeker, if I fill in one or more missing sections, then

					my CV information should reflect the changes I made. • As a job seeker, if the missing sections cannot be displayed, then I should see an error message.

33	As a job seeker, I want to view my CV enhancement history so that I can track improvements and review past feedback.	3	Feature	Done	• As a job seeker, if I open the CV Enhancements tab in the History Page, then I should see a list of all CV enhancements I performed.
					• As a job seeker, if a CV enhancement entry is displayed, then it should include:
					  ○ The date of enhancement.
					  ○ The enhanced CV version.
					  ○ The feedback provided, including comments on writing style and missing details.
					  ○ How many days remain before the entry expires.
					• As a job seeker, if a CV enhancement record has expired, then it should no longer appear in my history.

					• As a job seeker, if I delete a CV enhancement entry, then I should be asked to confirm, and if confirmed, it should be permanently removed from my history. • As a job seeker, if no CV enhancements exist, then I should see a message indicating no CV enhancements are available.
34	As a job seeker, I want to take a mock interview in my specialty so that I can practice for real job interviews in my field.	5	Feature	Done	• As a job seeker, if I open the mock interview screen, then I should see how many mock interview credits I have remaining for today. • As a job seeker, if I have no mock interview credits remaining, then I should not be able to start a mock interview and should see a message that credits reset at midnight. • As a job seeker, if I select a specialty and start a mock interview, then I should be able to begin the interview. • As a job seeker, during the mock interview, I should be

					able to answer 10 questions one by one. • As a job seeker, when a question is presented, then I should be able to read it or listen to it. • As a job seeker, after answering a question, then I should be able to proceed to the next one until the interview is completed. • As a job seeker, if I complete the mock interview, then I should see that the interview has been completed successfully and receive interview feedback. • As a job seeker, if voice tone feedback is available, then I should be able to view it as part of my interview feedback. • As a job seeker, if voice tone feedback is unavailable, then I should be informed while still being able to view the rest of the interview feedback. • As a job seeker, if I leave the mock interview before

					completing it, then the interview should end without being completed.

35	As a job seeker, I want to view my mock interview report so that I can understand my performance and identify areas for improvement.	5	Feature	Done	• As a job seeker, if I complete a mock interview, then I should be able to view a mock interview report.
					• As a job seeker, if the report is generated, then it should include:
					  ○ Strengths — areas performed well.
					  ○ Improve — areas needing improvement.
					  ○ Tips — personalized recommendations for enhancement.
					  ○ Voice Tone Analysis — insights on confidence, clarity, and communication tone.

36	As a job seeker, I want to view my mock interview history so that I can review my past practice sessions.	3	Feature	Done	• As a job seeker, if I complete a mock interview, then it should appear in my mock interview history with its date, specialty, and report. • As a job seeker, when I view my mock interview history,

					then I should see a list of my past mock interviews, including how many days remain before each entry expires. • As a job seeker, if a mock interview record has expired, then it should no longer appear in my history. • As a job seeker, if I select a past mock interview, then I should be able to view its full report. • As a job seeker, if I delete a mock interview entry, then I should be asked to confirm, and if confirmed, it should be permanently removed from my history.
37	As a job seeker, I want to apply for jobs so that I can submit my applications successfully.	2	Feature	Done	• As a job seeker, if I choose to apply for a job, then I should be required to have a complete profile. • As a job seeker, if my profile is incomplete, then I should see a message asking me to complete the missing information before applying.

					• As a job seeker, if my profile is complete, then I should be able to provide my CV as part of the application. • As a job seeker, if I provide a valid CV, then I should be able to continue with the application process. • As a job seeker, if I do not provide a valid CV, then I should see an error message and should not be able to continue with the application. • As a job seeker, if I have already applied for a job, then I should not be able to apply for the same job again, and I should see a message indicating that I have already submitted an application for this position.
38	As a job seeker, I want to complete the job interview so that my application can be considered for the position.	5	Feature	Done	• As a job seeker, if I start the interview, then I should be asked to confirm that I am ready to begin. • As a job seeker, during the interview, I should be able to

					answer the questions one by one. • As a job seeker, when a question is presented, then I should be able to read it or listen to it. • As a job seeker, after answering a question, then I should be able to proceed to the next one until the interview is completed. • As a job seeker, if I complete all interview questions, then I should see that the interview has been completed successfully. • As a job seeker, if the interview does not continue to completion, then I should see that the interview was not completed and the application cannot continue.
39	As a company, I want to view an evaluation report for each applicant so that I can make better hiring decisions.	3	Feature	Done	• As a company, if a candidate completes the required interview steps, then I should be able to view an evaluation report for that candidate.

					• As a company, if an evaluation report is available, then I should be able to access it from the candidate's application. • As a company, if a candidate has not completed the interview, then I should not see an evaluation report for that candidate. • As a company, if I open a candidate's AI report, then it should contain: ○ The CV the candidate attached with the application. ○ Checklist of job requirements met / not met. ○ Psychometric analysis (confidence, communication, personality traits). ○ Voice tone analysis (stress, confidence, hesitation, clarity). ○ Technical evaluation (domain-specific knowledge). ○ Full list of interview questions and the candidate's answers.
40	As a company, I want to view applications for a job	3	Feature	Done	• As a company, if I view a job posting, then I should be able

	posting so that I can evaluate candidates.				to see a list of candidates who applied for that job. • As a company, if I select a status filter (Pending, Shortlisted, Rejected), then only applications with that status should be displayed. • As a company, if no applications match the selected filter, then I should see a message indicating no results. • As a company, if applications exist, then I should be able to view each candidate’s application information and evaluation report. • As a company, if I select a candidate’s application, then I should be able to view the candidate’s profile information and submitted documents. • As a company, if an evaluation report is available for a candidate, then I should be able to access it from the application. • As a company, if I open a candidate’s application, then I

					should see a Download Report action. • As a company, if I click Download Report, then the report should be downloaded as PDF. • As a company, if the report is not yet generated, then the Download Report action should be disabled with a tooltip/message: "Report not available yet". • As a company, if a download error occurs, then I should see a clear error message and be able to retry. • As a company, if I open a candidate's application, then I should be able to view their profile information including: ○ Full name, nationality, date of birth, and contact details (phone/email). ○ Personal photo. ○ Uploaded CV (view and download).
41	As a company, I want to view an overall evaluation	5	Feature	Done	• As a company, if a candidate completes the required

	score for each applicant so that I can objectively compare and rank candidates.				evaluation steps, then I should be able to view an overall evaluation score for that candidate. • As a company, if a candidate completes the AI interview, then **Jadeer** should calculate a final score out of 100 using weighted scoring, where each category (CV analysis, Job requirements match, Psychometric analysis, Voice tone analysis, Technical evaluation) is scored from 0 to 100 and combined using the following weights: 30% CV analysis, 20% Job requirements match, 20% Psychometric analysis, 10% Voice tone analysis, and 20% Technical evaluation. The final score should be rounded to the nearest whole number. • As a company, if multiple candidates apply for the same job, then I should be able to see candidates ordered based on their evaluation scores.

					• As a company, if two or more candidates have the same final score, then they should be ranked equally. • As a company, if I open a candidate’s score breakdown, then I should see each category’s partial score clearly listed alongside its contribution to the final score.
42	As a company, I want to update the status of a candidate’s application so that I can reflect my hiring decision.	3	Feature	Done	• As a company, if I open a candidate’s application, then I should be able to change the status to Shortlisted or Rejected. • As a company, if I view the list of applicants for a job, then I should be able to select multiple candidates at once. • As a company, if I click "Select All", then all visible candidates should be selected at once. • As a company, if all candidates are selected and I click "Deselect All", then all selections should be cleared. • As a company, if a candidate's status is Shortlisted or

					Rejected, then I should see a badge showing how many days remain before the application record expires (180-day window from the status update date). • As a company, if I change the status for one or multiple selected candidates, then the new status should be saved and reflected immediately. • As a company, if I confirm a bulk status change, then all selected candidates should update accordingly. • As a company, if any status update fails for specific candidates, then I should receive feedback identifying which candidates failed with the option to retry only those.
43	As a job seeker, I want to view my job application history so that I can track the status of each application.	3	Feature	Done	• As a job seeker, if I have applied for one or more jobs, then I should be able to view a list of my job applications. • As a job seeker, if the report is sent to the company, then the

					job application status should set to "Pending". • As a job seeker, if I apply to multiple jobs, then all applications should appear in my Reports section, showing: o The job title and company name. o The current application state (Pending, Shortlisted, Rejected). o The application/interview date. • As a job seeker, if my application status is Rejected, then I should see how many days remain before the application record is removed from my history. • As a job seeker, if no applications exist, then an empty-state message should appear: "No job applications found".

4.2.4.3 Non-functional Requirements

This section lists the non-functional requirements of **Jadeer**. These requirements describe critical operational qualities such as performance, reliability, security, and scalability. Each is expressed as a user story with measurable acceptance criteria to ensure **Jadeer** meets quality standards.

Table 4: 4-3 Non-functional requirements

#	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
1	As a user, I want the **Jadeer** to respond within 5 seconds so that I can complete my tasks quickly.	3	Feature	• As a job seeker, if I navigate between home, jobs, profile, or reports, then each page should load in ≤ 5 seconds under normal load.
2	As a user, I want **Jadeer** to be available 99% of the time so that I can use the app.	3	Feature	• As a recruiter, if **Jadeer** availability is monitored at 1-minute intervals over a month, then uptime should be ≥ 99%. • As a recruiter, if an outage occurs, then it should be restored within ≤ 30 minutes.
3	As a user, I want login attempts to be limited to 5 failed tries in	2	Feature	• As a user, if login fails ≥ 5 times in one day, then login

	one day so that my account remains secure.			should be temporarily blocked. • As a user, if account lock occurs, then I should be required to reset my password before attempting to log in again.
4	As a job seeker, I want my interview recordings stored for no more than 3 days so that my privacy is protected.	2	Feature	• As a job seeker, if an interview recording is saved, then it should be encrypted and automatically deleted after 3 days.
5	As a user, I want at least 90% of tasks to be completed successfully on the first attempt by test users so that **Jadeer** is user-friendly.	3	Feature	• As a user, if 10 test users attempt core tasks (upload CV, apply for job, view job details), then ≥ 9 users should complete them successfully on the first attempt without assistance.

4.3 System Design

4.3.1 Architectural Diagram

This section details the **Jadeer** system's architecture, which follows a client-server model. The system is split into two primary components: the client and the server. The client operates on a user's mobile device, handling the user interface and data transmission. The server is responsible for business logic, data storage, and intelligent processing. The server receives requests from the client and interacts with a database and various **APIs** to process them and enhance the system's functionality.

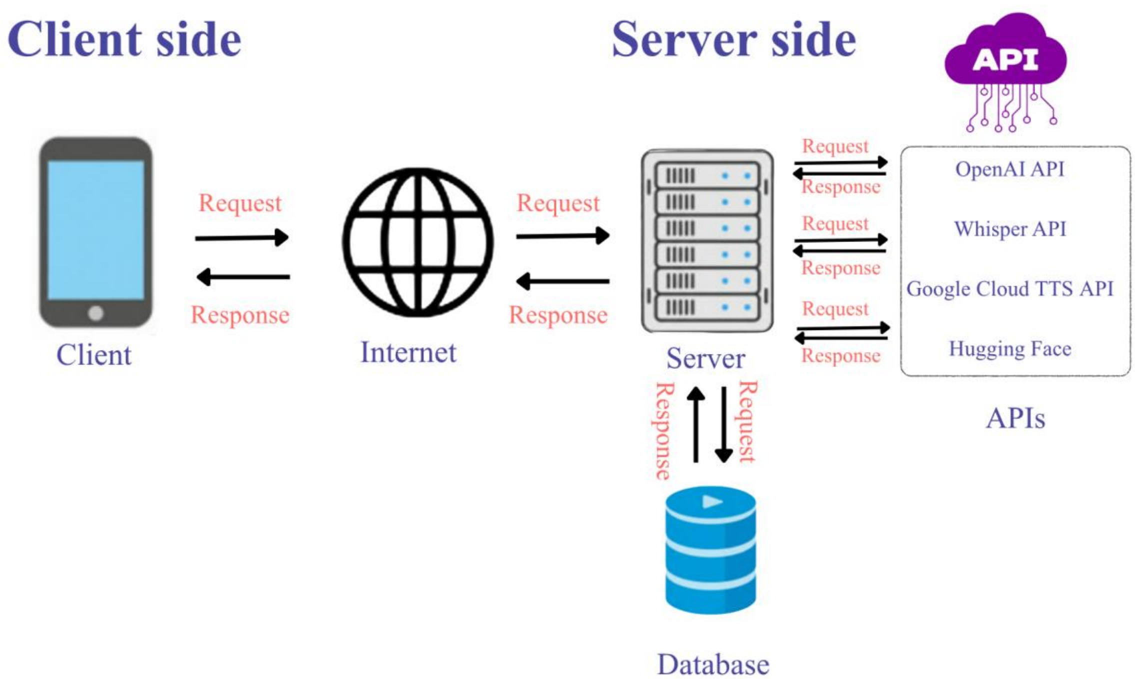

Figure 6: 4-6 Jadeer-Client-Server-Architecture

The architecture of **Jadeer** follows the client–server model, where the system is divided into two main components: the client and the server. The client runs on the user's mobile device and manages the user interface, input mechanisms, and lightweight logic for rendering and transmitting data. It is also responsible for initiating communication with the server over the Internet and displaying the responses it receives.

On the server side, the system handles business logic, data storage, and intelligent processing. The server receives requests from the client and processes them by interacting with the database, which securely stores structured information such as user profiles, system logs, and application data, ensuring efficient retrieval. It also interacts with the external API systems to power its intelligent features; it utilizes the Whisper API to process and transcribe user voice inputs into text, the OpenAI GPT API to analyze CVs and generate tailored smart career recommendations, the Hugging Face API to support advanced language processing and specialized models, and the Google TTS API to synthesize text into clear audio responses, enhancing the system's functionality and responsiveness.

When a request is received from the client, the server coordinates between the database and the API systems as needed, processes the request, and sends the appropriate response back to the user's device. This response may include data, confirmation of an action, or feedback from the system.

The client–server model was chosen for **Jadeer** due to its scalability, flexibility in handling multiple concurrent requests, and centralized nature, which makes updates and maintenance easier. Moreover, integrating APIs capabilities on the server side provides advanced analytical functions that improve user experience and support decision-making within the application.

4.3.2 Class Diagram

The class diagram shows a high-level structure of **Jadeer** 's application, illustrating the interactions between key roles and main components. It presents an overview of the main attributes and methods associated with each class, highlighting their relationships. The diagram serves as a visual representation of the system's architecture and aids in understanding the overall design and functionality of **Jadeer**.

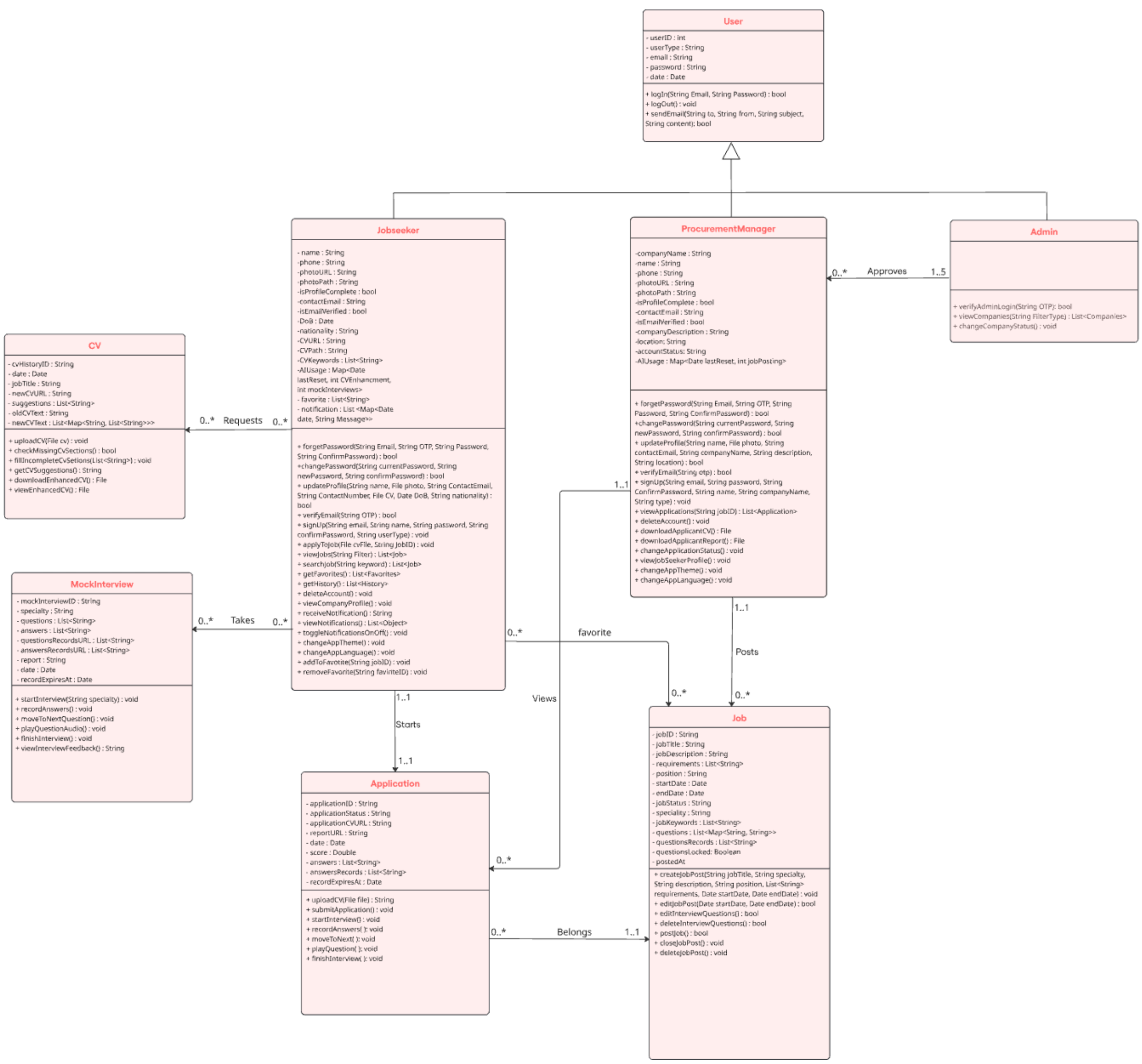

Figure 7: 4-7 Class Diagram

4.3.3 Component Level Design

4.3.3.1 Pseudocode

1) Pseudocode For CV-Driven "For You" Jobs

Step 1 – Check CV Completeness

Start Personalized Jobs Flow

Check if User.CV is uploaded and readable

IF CV is missing OR cannot be analyzed THEN

  Display message: "Upload your CV to see personalized matches"

  Stop personalized matching

ELSE

  Continue to matching logic

END IF

Step 2 – Extract Skills From CV

When CV is uploaded or updated:

Call backend Cloud Function

Extract keywords (skills, tools, technologies)

Save extracted list to User.CVKeywords

IF extraction fails OR User.CVKeywords is empty THEN

  Display message: "We couldn't analyze your CV. Try uploading a clearer version."

  Stop personalized matching

ELSE

  Continue

END IF

Step 3 – Enable "For You" Mode (Reset Filters)

IF User toggles ForYou = ON THEN

  Set Filter.Specialty = "All"

  Set Sort = "Newest first"

```
    Set ShowClosed = OFF
    Clear SearchQuery
    Load jobs list
END IF
```

Step 4 – Match User to Jobs

```
Fetch active jobs where Job.Status != "Closed"

For each Job:
    Combine Job.Specialty + Job.Keywords
    Compare with User.CVKeywords
    Count matches → MatchCount

    IF MatchCount ≥ 2 THEN
        Mark Job as Recommended
    END IF
END FOR

After finishing all jobs:

IF RecommendedJobs is empty THEN
    Display: "No strong matches found yet – check All Jobs for more opportunities."
ELSE
    Sort RecommendedJobs by StartDate descending
    Display RecommendedJobs
END IF
```

Step 5 – Using Filters While "For You" Mode is ON

```
IF User changes any filter (Specialty, Search, ShowClosed, Sort) THEN
    Apply filters only to RecommendedJobs
```

Refresh displayed job list

END IF

2) Pseudocode For Job Posting Feature

Step 1 – Check company Profile

Start Job Posting Process

Check company profile completeness

IF profile is incomplete THEN

Display message: "Please complete your company profile before posting a job"

Stop process

ELSE

Display job posting form

END IF

Step 2 – Fill Job Information

Display job posting form

company provides:

- JobTitle
- (required)
- Description (required)
- Specialty (required)
- Requirements (at least one required)
- Position (optional unless using AI Generator)
- StartDate (required)
- EndDate
- (required)

Optional: AI Job Description Generator

IF company selects AI Generator THEN
Request:
JobTitle (required)

Specialty (required)
Position (required)
Generate AI draft based on provided fields

IF AI draft generation fails THEN
Display message: "AI draft failed. Please create manually."
ELSE
Insert AI draft into Description field for review/editing
END IF
END IF

Validate fields
IF any required field is missing THEN
Display inline validation errors
ELSE
Navigate to Questions Page and pass JobData
END IF

Step 3 – Interview Questions Generation
On Questions Page:
Receive JobData

Automatically initiate AI question generation using:

- JobTitle
- Specialty
- Position (if provided)

- Requirements
- Description

Prepare request body:

- JobID
- Title
- Position
- Specialty
- Requirements
- Description
- Mix object (technical vs psychometric ratio)
- Difficulty object (easy/medium/hard ratio)

Send request to backend AI generation service

Display loading state: "Generating interview questions…"

Validate request:

IF JobID is missing OR Title is missing THEN

  Return error: "Missing required fields (JobID/Title)."

END IF

Check AI configuration:

IF AI_KEY is missing or invalid THEN

  Return error: "Invalid or missing AI key."

END IF

Map Specialty to knowledge profile:

- Technical categories
- Psychometric/behavioral dimensions

Build AI prompt using:

- Title
- Position
- Specialty
- Requirements
- Description
- Technical categories
- Behavioral dimensions

Target result:

TotalQuestions = 10

- 5 Technical
- 5 Psychometric

Call AI model

```
IF AI request fails OR returns invalid data THEN
  Mark AI generation as failed
ELSE
  Extract QuestionsList

  IF QuestionsList is empty OR invalid THEN
    Mark AI generation as failed
  ELSE
    Sanitize QuestionsList:
     - Remove duplicates
     - Remove empty strings
     - Ensure QuestionType in ["technical", "psychometric"]
     - Ensure Difficulty in ["easy", "medium", "hard"]
```

```
        - For psychometric: assign trait from Big Five or null
        - Ensure list contains exactly 10 questions

      Return QuestionsList to frontend
   END IF
END IF

IF backend returns valid QuestionsList THEN
   Display QuestionsList
   Allow company to:
      - Edit any question
      - Add up to 3 custom questions
      - Delete only custom questions
ELSE
   Display message: "AI generation failed. Please try again or add questions manually."
   Stop AI step, allow manual question entry
END IF

When company selects Done:
Display confirmation: "After publishing, questions will be locked."

IF confirmed THEN
   Create JobPost with:
      JobID
      Title
      Description
      Specialty
      Requirements
      Position
      StartDate
```

EndDate
Status = "Open"
QuestionsList
QuestionsLocked = true

Display message: "Job created successfully."
Add JobPost to company dashboard
END IF

Step 4 – Manage Job Posts
In company Dashboard show options:

- Edit Date
- Close Job
- Delete Job

Edit Date

IF company selects Edit Date THEN
Allow editing StartDate & EndDate only
Save changes
END IF

Close Job

IF company selects Close THEN
Set Status = "Closed"
Display message to job seekers: "This job is no longer accepting applications."
END IF

Delete Job

```
IF company selects Delete THEN
    Display confirmation
    IF confirmed THEN
        Delete JobPost and all related:
          - applications
          - interviews
          - reports
        Remove JobPost from job seekers' AppliedJobs lists
    END IF
END IF
```

4.3.3.2 Flowchart

1) Flowchart For CV Enhancement

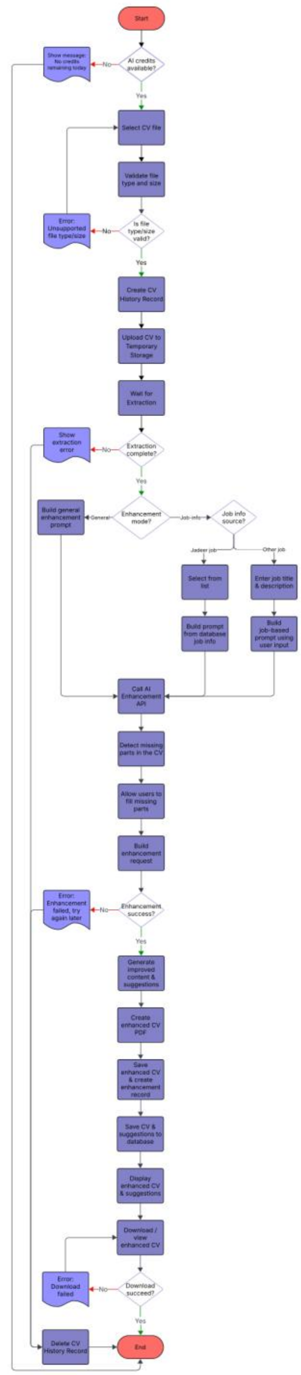

Figure 8: 4-8 CV enhancement flowchart

2) Flowchart For Job Interview

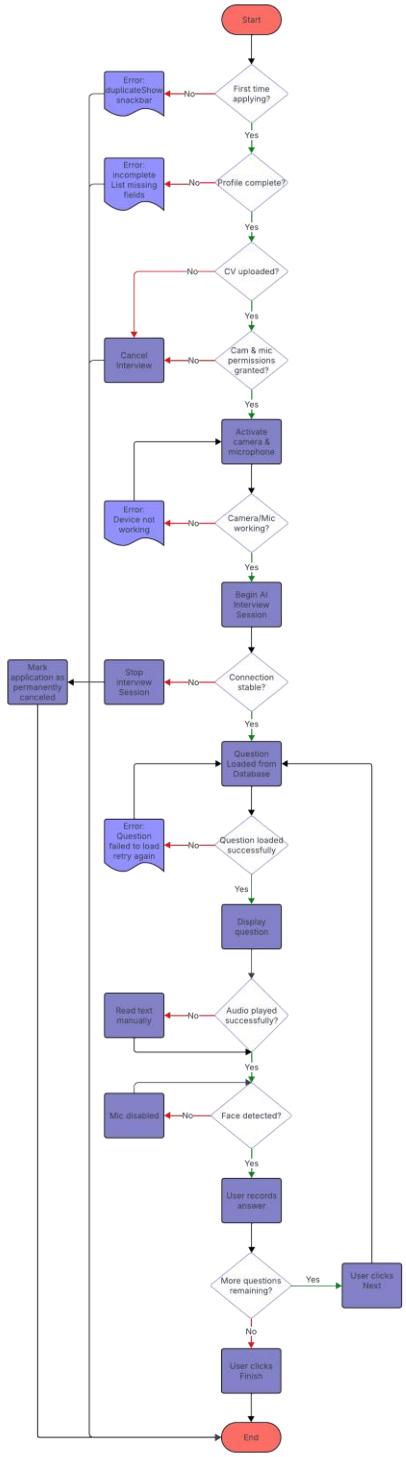

Figure 9: 4-9 Job Interview flowchart

4.4 Data Design

4.4.1 Data Models

- ER Diagram

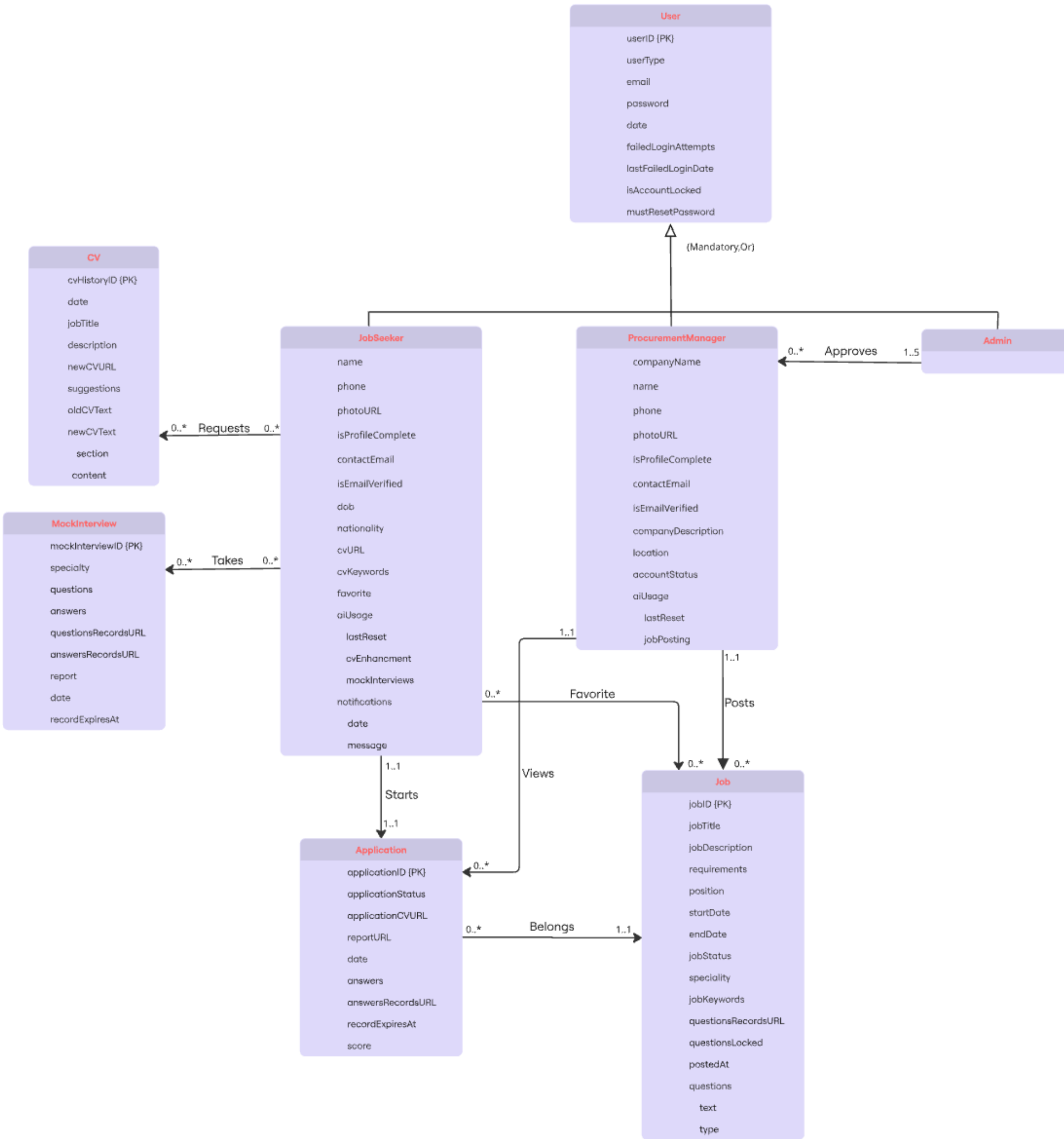

Figure 10: 4-10 ER Diagram

The Entity–Relationship (ER) diagram above provides a comprehensive visualization of the core entities within **Jadeer** application and the interactions that govern its functionality. The diagram follows UML conventions by representing each entity using singular, capitalized names, while attributes are written in lowercase. This consistent naming ensures clarity and supports industry-standard modeling practices.

The ERD illustrates how different system actors— HR departments**, job seekers, and admins**—interact with platform elements such as job postings, applications, interviews, enhancements, and AI-generated reports. Relationships are depicted through connecting lines with clearly marked multiplicities, indicating the logical cardinality between entities. In alignment with UML notation, foreign keys are intentionally excluded from the entity boxes and are implied through relationship connectors.

The model also highlights optional and mandatory associations. For example, a *Job Seeker* may submit multiple applications, while each application must be linked to exactly one job. Similarly, HR departments can manage several postings, whereas a job post belongs to a single HR department. These multiplicities provide precision when translating the conceptual model into a relational database schema.

To interpret this diagram, consider the following scenarios:

1. **Job Posting Lifecycle:**
   A HR department can publish multiple *Job Posts*, each containing attributes such as requirements, responsibilities, and skills. A job post may receive zero or many *Applications*, but each application must be tied to a single job.
2. **Job Seeker Interactions:**
   A *Job Seeker* can apply to multiple jobs and may save several job posts as favorites. Additionally, job seekers can participate in interviews scheduled based on their applications. Each interview belongs to exactly one job seeker and one job post.
3. **AI-Generated CV Enhancement:**
   Job seekers may request *CV Enhancements* or *Mock Interviews* powered by AI. These

interactions form optional relationships, meaning a job seeker may use none, some, or all of these services.

- Non-Relational Model Tree

Figure 11: 4-11 Non-Relational Model Tree

- Non-Relational Model Relation Approach

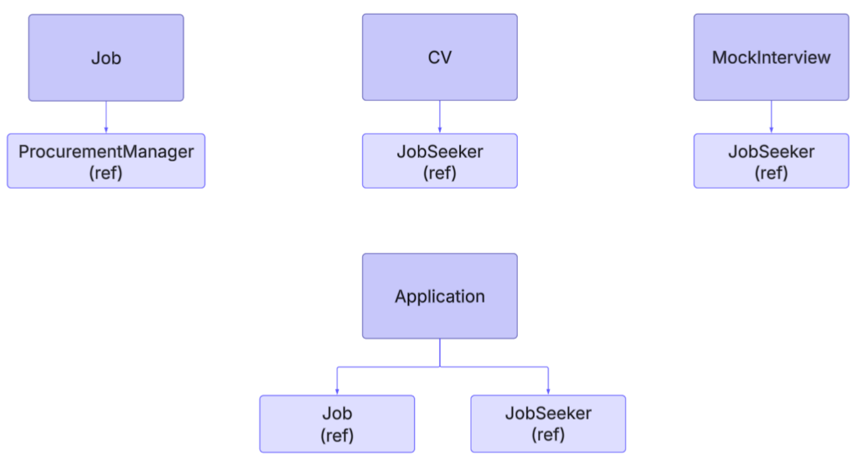

Figure 12: 4-12 Non-Relational Model Relation Approach

In the context of database design, modeling the database structure is essential for understanding its organization and the relationships between collections. Figure 10 illustrates the hierarchical tree structure of collections, documents, and their attributes, where indentation reflects different levels within the non-relational model. Additionally, Figure 11 demonstrates how these collections interact through references.

The following points describe the reference-based relationships in our Firestore non-relational data model:

- The **"userID"** attribute in the **"Job"** collection is a reference to the **"HRDepartment"** collection, indicating which HR department created the job posting.
- The **" userID "** attribute in the **"CV"** collection is a reference from the **"JobSeeker"** collection, indicating the user who submitted a CV for enhancement.
- The **" userID"** attribute in the **"MockInterview"** collection is a reference from the **"JobSeeker"** collection, indicating the user who requested a mock interview simulation.
- The **"jobID"** attribute in the **"Application"** collection is a reference from the **"Job"** collection, identifying the job the user applied for.
- The **"userID"** attribute in the **"Application"** collection is a reference from the **"JobSeeker"** collection, identifying the applicant submitting the application.
- The **"jobID"** attribute in the **"Interview"** collection is a reference from the **"Job"** collection, indicating the job associated with its corresponding interview.
- The **"userID"** attribute in the **"Interview"** collection is a reference from the **"JobSeeker"** collection, identifying the candidate being interviewed.

4.4.2 Dataset Preparation

In this section, we outline the dataset preparation process for **Jadeer's** speech-based confidence assessment model used in the interview simulations. The speech-based confidence model was developed using the First Impressions V2 dataset, which contains short video samples with continuous annotations related to personality traits and interview perception. Since the objective of the model is to estimate interview-related confidence, the interview annotation was selected as the target variable. This value is represented as a continuous score between 0 and 1, making the task suitable for regression.

The original annotation file was provided in a serialized .pkl format. Therefore, it was first converted into a structured tabular format, where each row represented one video sample and its corresponding annotation values. After matching the available video files with their labels, a total of 5,998 matched samples were used in the pipeline.

Since the dataset was originally provided as videos, the audio track was extracted from each video file using FFmpeg. All extracted audio files were converted into a standardized format: WAV, mono-channel, and 16 kHz sampling rate. This standardization was necessary to ensure compatibility with the speech feature extraction models and to maintain consistency across all samples.

To reduce the risk of identity leakage, a speaker-independent splitting strategy was applied. Instead of randomly splitting individual samples, the dataset was divided using speaker-based grouping, ensuring that samples from the same speaker did not appear in more than one split. This prevents the model from memorizing speaker-specific vocal characteristics and encourages it to learn more general speech patterns related to interview performance.

The final dataset was divided into three subsets: 4,803 samples for training, 592 samples for validation, and 603 samples for testing. The validation set was used during training to monitor generalization and select the best model checkpoint, while the test set was held out for final evaluation on unseen data.

4.5 Interface Design

The interface of **Jadeer** was designed following established principles of Human–Computer Interaction (HCI) to ensure that the application provides an intuitive, accessible, and efficient user experience for both job seekers and Companies. In designing the user interface, we adopted Shneiderman's Eight Golden Rules [42] as the primary UX framework to guide layout decisions, interaction patterns, and component behavior.

This section presents the application's sitemap and outlines the UX principles applied during the design phase.

- **Sitemap**

The following diagram shows the navigation structure of **Jadeer**. It highlights the main screens and flows for all three roles in the system — Job Seekers, Companies, and Admins. The sitemap covers the core processes such as creating accounts, browsing and applying for jobs, posting and managing job listings, and enhancing CVs.

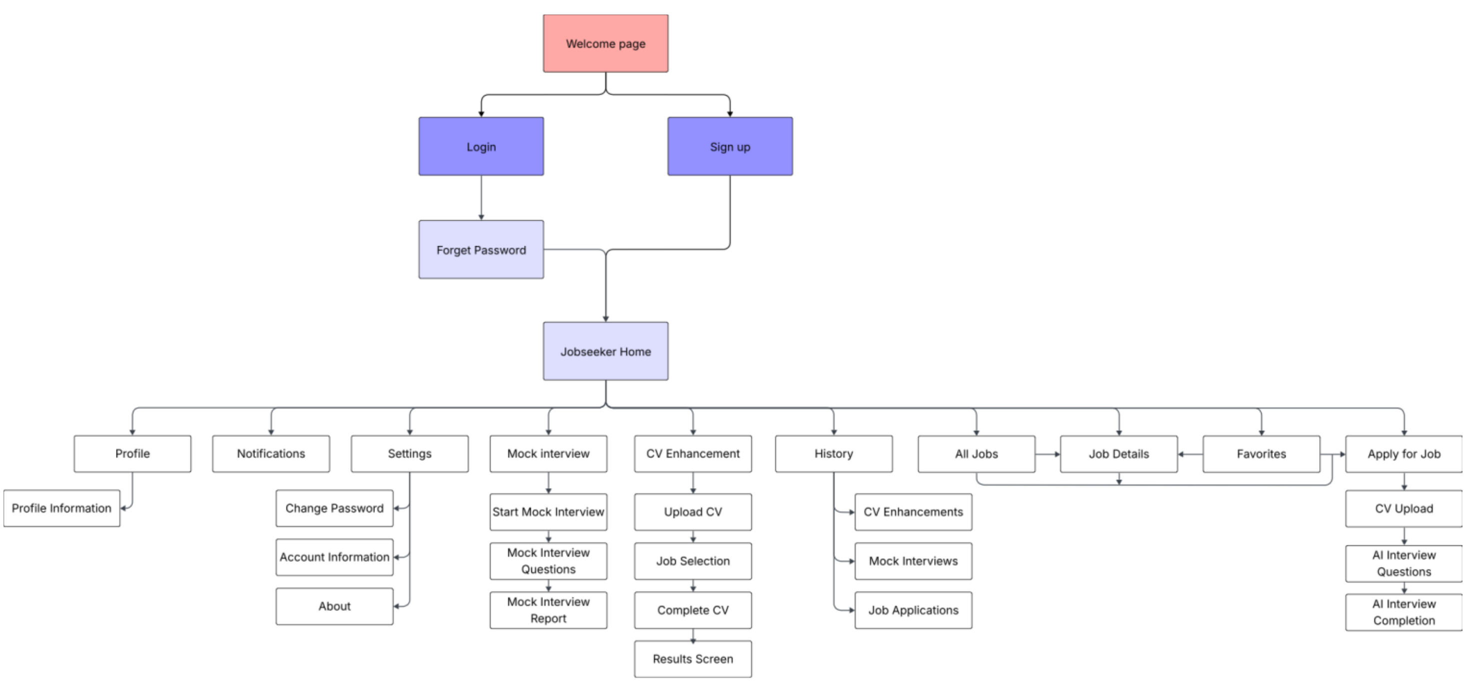

Figure 13: 4-13 Job Seeker Sitemap

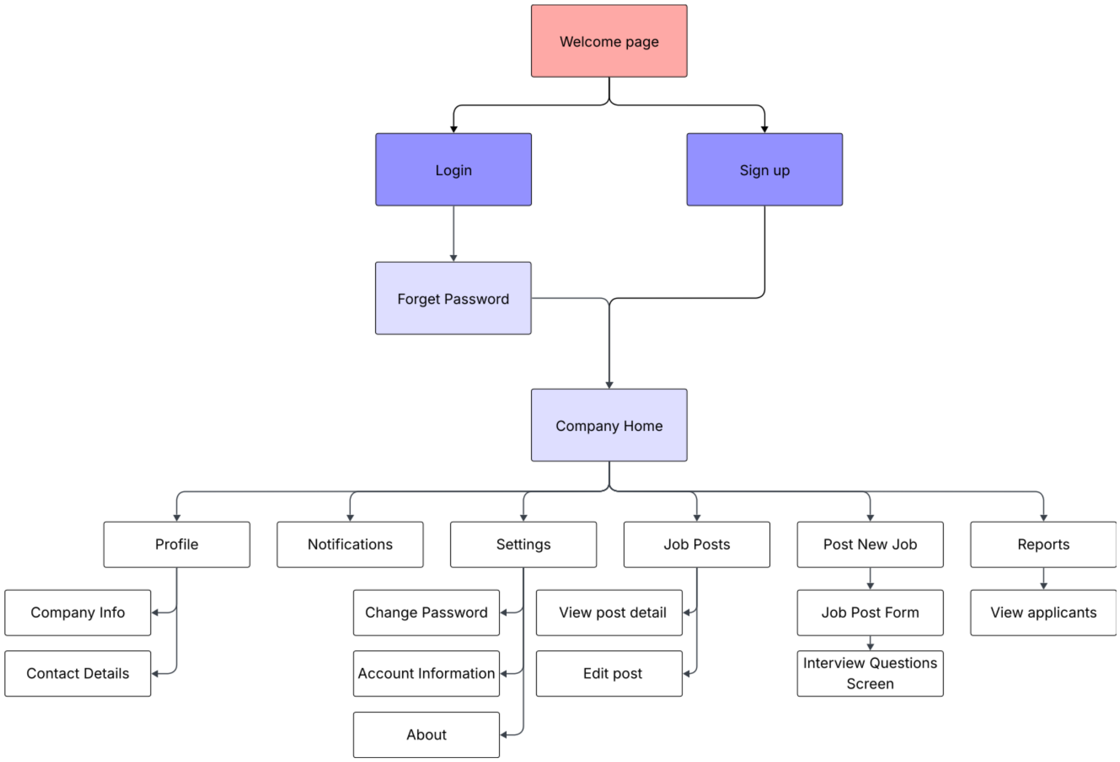

Figure 14: 4-14 Company Sitemap

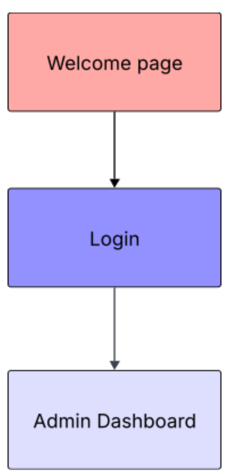

Figure 15: 4-15 Admin Sitemap

- **UX Guidelines**

During the design of **Jadeer**, we incorporated Shneiderman's Eight Golden Rules [42] to ensure clarity, reduce user effort, and enhance task efficiency. The following guidelines summarize how each rule was applied within the application:

1. **Strive for Consistency**

   Consistency across colors, icons, typography, and layouts was maintained throughout all screens. This ensures predictable interaction patterns and reduces users' learning time. In **Jadeer**, consistent visual elements are used in navigation bars, job cards, input forms, and buttons for both job seekers and Companies.

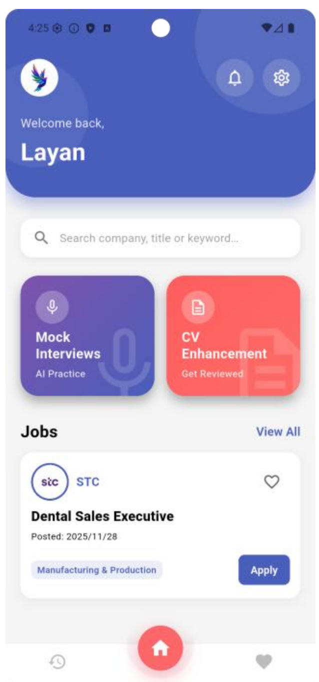

Figure 16: 4-16 Job Seeker Home Page in Jadeer

2. **Cater to Universal Usability**

The interface was designed to accommodate users with varying levels of digital experience. Clear wording, simple layouts, and step-by-step flows (such as job posting and CV enhancement) ensure accessibility for different user groups, including fresh graduates and HR professionals.

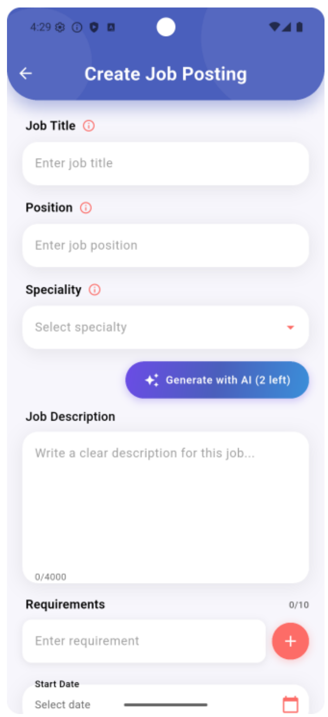

Figure 17: 4-17 Create Job Posting Page in Jadeer

3. Design Dialogs to Yield Closure

Each major action ends with a clear sense of completion. For instance, once a job is successfully posted, users receive a confirmation message. After the AI processes a CV or generates interview questions, the results are displayed clearly, signaling the end of the process.

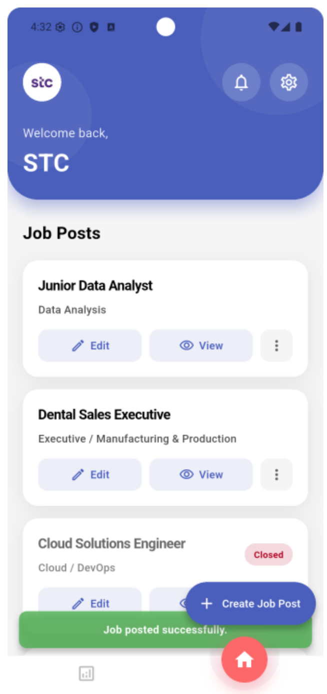

Figure 18: 4-18 Confirmation Message After Posting a Job

4. Prevent Errors and Offer Simple Error Handling

Validation checks are applied across all input fields to reduce user mistakes. Invalid entries such as incorrect email formats, missing job details, or weak passwords trigger descriptive error messages. Additionally, users are prevented from submitting incomplete forms, reducing the likelihood of downstream errors.

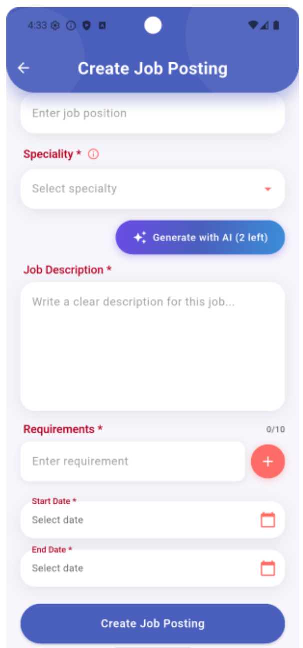

Figure 19: 4-19 Error Prevention Through Required Fields

5. **Permit Easy Reversal of Actions**

The system supports safe exploration by providing options to cancel or reverse certain actions. Examples include confirmation prompts before deleting job postings or AI-generated question sets, as well as the ability to edit posted job information.

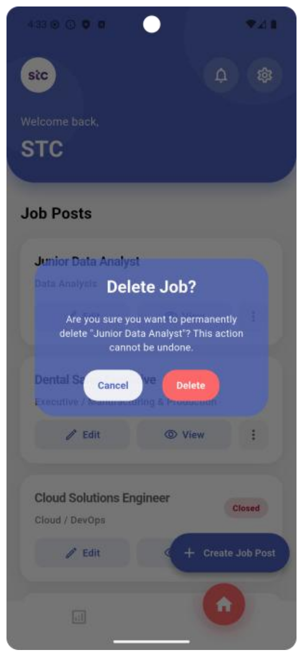

Figure 20: 4-20 Delete Confirmation Dialog

6. **Support Internal Locus of Control**

The interface ensures that users feel in control of their actions. Navigation is straightforward, buttons are clearly labeled, and system behavior aligns with user expectations. This is particularly important in multi-step processes such as job posting and profile completion.

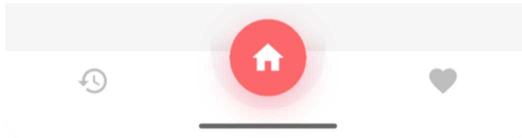

Figure 21: 4-21 User-Controlled Bottom Navigation

7. **Reduce Short-Term Memory Load**

   The design minimizes cognitive effort by presenting all necessary information directly on the screen. Clear labels, placeholders, and categorized question groups help users understand requirements without needing to recall information. Job seekers can easily compare job details, and Companies can manage postings without navigating through multiple screens.

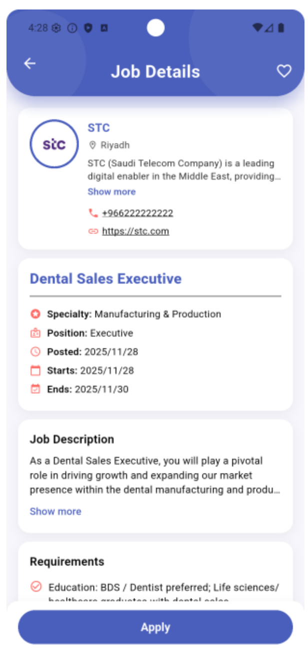

Figure 22: 4-22 Single-Page Job Details Layout

4.6 System Implementation

This section describes the implementation of the **Jadeer** system, covering the hardware and software components used, integration with third-party services, and custom features developed. We explain the technical architecture, API connections, and key implementation decisions. Additionally, we discuss the challenges encountered during development and the consultations that guided the process. Finally, we provide a link to the project repository.

4.6.1 Hardware and Software Components

The **Jadeer** application was developed using Visual Studio Code as the primary IDE with Flutter and Dart extensions. Android Studio was used for Android device emulation and testing. The development team used personal laptops with specifications including AMD Ryzen 7 6800HS processors (3.20 GHz), 16GB RAM, and 952GB storage. Git and GitHub were used for version control and team collaboration.

The frontend was built using Flutter framework (SDK version >=3.5.3 <4.0.0) with Dart programming language for cross-platform mobile development. Firebase was chosen as the backend platform, providing Authentication for user management, Cloud Firestore as a NoSQL database, Cloud Storage for file management, and Cloud Functions for serverless backend processing. Firebase was integrated by creating a Firebase project in the Firebase Console, downloading the configuration files, and placing them in the Flutter project directory.

4.6.2 Out-of-the-Box Components

The following third-party packages and services were integrated into the application:

4.6.2.1 Flutter Packages:

- **firebase_core, firebase_auth, cloud_firestore, firebase_storage, cloud_functions:** Firebase SDK packages for backend services integration
- **provider:** State management solution for managing application state across widgets
- **file_picker:** File selection functionality for CV uploads

- **image_picker:** Image selection for profile photos and company logos
- **http:** HTTP client for API communication with backend services
- **url_launcher:** Opening external URLs, email clients, and phone dialer
- **shared_preferences:** Local data storage for user preferences
- **intl:** Internationalization and localization support
- **iconsax_flutter, dotted_border, awesome_dialog, dropdown_button2, share_plus:** UI component libraries for enhanced user interface

4.6.2.2 Cloud Functions Packages:

- **openai (v6.3.0):** Official OpenAI Node.js SDK for GPT-4 API integration
- **pdf-parse, textract:** Libraries for extracting text from PDF and DOCX files
- **pdfkit:** PDF generation library for creating enhanced CV documents
- **nodemailer:** Email service for sending OTP codes and notifications
- **mammoth:** DOCX file processing

4.6.2.3 Pre-trained AI Model:

- **OpenAI GPT-4o:** Large language model used for CV enhancement and content generation

4.6.3 API Connections and Third-Party Integration

4.6.3.1 Firebase Backend Services:

The application connects to Firebase services for all backend operations:

- **Firebase Authentication:** Handles user registration and login using email/password authentication. Integrated using firebase_auth package with built-in security features including password hashing and account management
- **Cloud Firestore:** NoSQL database storing user profiles, job postings, applications, favorites, and CV history. Main collections include: Users, Jobs, Applications, Favourite, CVHistory. Provides real-time data synchronization across devices

- **Firebase Storage:** Stores user-uploaded files including CVs (PDF/DOCX), profile photos, and company logos. Files organized in folders by user ID and document type (cv/, NewCV/, temp_cv_extraction/)
- **Cloud Functions:** Serverless backend functions for CV text extraction, keyword extraction, and OpenAI API integration

4.6.3.2 OpenAI API Integration:

The CV enhancement feature integrates OpenAI's GPT-4o model through Firebase Cloud Functions (Generation 2). The implementation works as follows:

4.6.3.3 Prompt Design and AI Output Control

Jadeer uses prompt templates to control the behavior and output of the Large Language Model across multiple AI-powered features. These prompts are implemented through the OpenAI API using a system message and a user message pattern. The system message defines the model's role and general behavior, while the user message provides the feature-specific input data, task instructions, constraints, and required output format.

Most text-based prompts in **Jadeer** require structured JSON output to make the responses easier to validate, store, and display inside the application. This is especially important where the output must be processed by the system rather than displayed as free text only. Full prompt templates are provided in 9.5 Appendix E.

4.6.3.4 CV Enhancement Prompt

- **Purpose:** This prompt rewrites and restructures a candidate's CV to improve grammar, formatting, ATS compatibility, and relevance to a target job when one is provided. it handles multiple CV sections and must preserve the accuracy of the candidate's original information.
- **Inputs:** The prompt receives the full extracted CV text, an optional target job title, an optional job description, and any additional section content that the user adds through the

interface, such as missing sections they choose to fill in. The prompt follows conditional rules: if no target job is provided, the model is instructed to preserve the CV content and improve its quality; if a target job is provided, the model is instructed to tailor the CV toward the role while only using information that already exists in the provided input.

- **Output format:** The model returns a structured JSON object with two top-level fields: enhanced_cv and improvements. The enhanced_cv field contains an array of section objects, where each object includes a section name and structured content. The improvements field contains a short list of plain-text improvements summarizing the main improvements made to the CV.
- **Role in the system:** The parsed enhanced_cv array is used to reconstruct and display the enhanced CV in the application's CV builder view. The improvements array is displayed as a summary panel alongside the result. The most important constraint in this prompt is data integrity: the model is explicitly forbidden from inventing any information that is not present in the original CV or the user-provided additional content.

4.6.3.5 CV Missing Sections Detection

- **Purpose:** This prompt analyzes the extracted CV text and identifies which common CV sections are missing or empty. Its output is used to prompt the user to provide missing information before continuing the CV enhancement process.
- **Inputs:** The prompt receives the full extracted CV text. The target sections are listed directly in the prompt, along with detection rules that instruct the model to recognize section content based on meaning rather than relying only on explicit section headers.
- **Output format:** The model returns a minimal JSON object containing a missingSections array. Only section names from the predefined list are accepted as valid values. If no sections are missing, the array is returned empty. Since this task is closer to classification than content generation, a lower temperature is used to reduce variation in the output.
- **Role in the system:** The returned list controls the CV enhancement flow. If missing sections are detected, the interface presents the user with a form to supply the missing content before the CV enhancement prompt is called. This two-step design helps ensure

that the enhancement prompt receives more complete input and reduces the need for the model to make up CV information.

4.6.3.6 CV Skills Suggestion Generation

- **Purpose:** This prompt only works if a specific job was selected for CV enhancement, it generates a list of 15–20 role-relevant skills based on a target job. It is used to suggest suitable technical and soft skills when a user is tailoring their CV to a specific job and their skills section is missing, sparse, or incomplete.
- **Inputs:** The prompt receives the target job title and job description. Based on these inputs, the model generates a concise list of skills that are commonly expected for the role, covering both technical and soft skills.
- **Output format:** The model returns a JSON object containing a skills array. Each skill is limited to a short phrase of one to three words to ensure compatibility with the CV skills field and selectable skill chips in the user interface. A moderate temperature setting is used to allow useful variation while keeping the suggestions relevant to the job context.
- **Role in the system:** This prompt supports the CV enhancement workflow when a target job is available. The returned skills are displayed to the user as selectable suggestions, allowing them to add relevant skills before the enhanced CV is generated. This helps provide the CV enhancement prompt with richer input while still keeping the final decision under the user's control.

4.6.3.7 Job Description Generation Prompt

- **Purpose:** This prompt generates a complete, professionally written job description from minimal structured input. It is used during job-posting creation to assist employers who may not have a pre-written description.
- **Inputs:** The prompt receives three fields: job title, position level, and specialty/field. The model generates a general job description based on these structured inputs.
- **Output format:** The model returns plain text. The output is structured into three parts: a Role Overview, Key Responsibilities, and a Closing Statement. The prompt specifies the

expected length for each part, including 5–6 sentences for the overview, 8–12 bullet points for responsibilities, and 2–3 sentences for the closing statement. It also prevents markdown formatting, headers, and meta-commentary so the result can be displayed cleanly in the job posting form.

- **Role in the system:** The generated description is pre-populated into the job description field of the posting form, where the employer can review and edit it before publishing. A key constraint in the prompt prevents the model from inventing a company name, ensuring that the output remains accurate until the employer completes the form.

4.6.3.8 Job Interview Questions Generation Prompt

- **Purpose:** This prompt generates a set of ten interview questions tailored to a specific job posting. The questions are split equally between technical and psychometric dimensions, calibrated to reflect the seniority level and domain of the role.
- **Inputs:** The prompt receives five runtime variables: job title, position/seniority level, specialty, a list of must-have requirements, and a job description excerpt. From these inputs, the system also computes derived values, including the applicable technical categories and psychometric work-style themes, which are injected into the prompt to guide category coverage.
- **Output format:** The model returns a JSON object containing a questions array. Each element carries five fields: text (the question string), type ("technical" or "psychometric"), category (a domain or trait label), difficulty ("easy", "medium", or "hard"), and trait (one of the Big Five personality dimensions, or null for technical questions). The difficulty distribution is specified explicitly in the prompt and is validated programmatically after parsing.
- **Role in the system:** The output is stored in Firestore and displayed to the interviewer before the session begins. The type and category fields drive the interview session UI, which groups and sequences questions. The trait field feeds into the psychometric scoring component of the interview report.

4.6.3.9 Job Interview Performance Report Prompt

- **Purpose:** This prompt analyzes the transcribed audio responses from a completed job interview and generates a structured candidate evaluation report intended for the hiring manager.
- **Inputs:** The prompt is assembled from four sources: the job context, including the job title, specialty, company name, requirements list, and job description excerpt; the interview question texts, each labeled as either technical or psychometric; the corresponding transcriptions produced by the Whisper API; and the job requirements checklist, which guides the model in evaluating requirement coverage. If one or more answers cannot be transcribed, a transcription failure note is included in the prompt.
- **Output format:** The model returns a JSON object containing four independent GPT-generated scores: cvAnalysisScore, jobRequirementsMatchScore, psychometricScore, and technicalScore, each on a 0–100 scale. The output also includes an overallSummary, strengths, weaknesses, recommendations, a requirementsChecklist, a psychometricAnalysis object, and a questionsAndAnswers array. The requirementsChecklist contains each posted requirement with a met/not met status. The psychometricAnalysis includes confidence and communication sub-scores, along with personality traits and work style descriptions. The shared scoring rubric is embedded in the prompt, where 80–100 indicates Excellent, 60–79 indicates Good, 40–59 indicates Average, and 0–39 indicates Below Expectations. All feedback is required to be written in third person.
- **Voice tone scoring:** The voiceToneScore used in the final grade is not generated by this GPT prompt. It is computed separately using a Speech Emotion Recognition model hosted on Hugging Face. The model processes each audio recording and returns a confidence assessment per question. The system then averages these results into a single 0–100 voiceToneScore. If the SER model is unavailable for a recording, that question contributes a score of zero instead of being excluded, so the voice tone score defaults to 0 when all analyses fail.

- **Role in the system:** After the GPT response is parsed, the five scores are combined into a single finalScore using a fixed weighted formula: CV Analysis 30%, Technical 20%, Job Requirements Match 20%, Psychometric 20%, and Voice Tone 10%. The finalScore and sub-scores are passed to a PDF generation function, and the resulting report file is uploaded to Firebase Storage. The Firestore record for the application stores both the structured JSON data and the public PDF URL.

4.6.3.10 Mock Interview Questions Generation Prompt

- **Purpose:** This prompt generates ten practice questions for a specialty-based mock interview. Unlike the job interview questions generation prompt, the questions are not tied to a specific job posting. Instead, they are broader and designed to help job seekers practice interview scenarios independently based on their selected specialty.
- **Inputs:** The prompt receives the candidate's selected specialty and its parent specialty bucket. These inputs are used to determine the relevant domain categories and behavioral work-style themes, which are included in the prompt to ensure balanced question coverage.
- **Output format:** The model returns a JSON object containing a questions array. The output schema is similar to the job interview questions generation prompt. Each question includes fields such as text, type, category, difficulty, and trait. The difficulty distribution is fixed at three easy, five medium, and two hard questions.
- **Role in the system:** The generated questions are stored and presented to the candidate during the mock interview session. Because the output follows the same structured format used in the real interview module, the system can reuse the same front-end rendering, text-to-speech, audio recording, and answer submission workflow.

4.6.3.11 Mock Interview Performance Report Prompt

- **Purpose:** This prompt generates feedback for a candidate who has completed a mock interview session. The feedback is written in second person and addressed directly to the candidate, designed to feel encouraging and coaching-oriented rather than evaluative.

- **Inputs:** The prompt receives the candidate's specialty, the set of transcribed question-and-answer pairs from the completed mock session produced by the Whisper API, and an overall voice confidence score computed from the Speech Emotion Recognition model. This voice score is injected into the prompt as an "Acoustic Tone Reading" parameter before the GPT call is made.
- **Output format:** The model returns a JSON object containing overallScore, overallSummary, strengths, weaknesses, advice, and mockVoiceComment. The overallScore is an integer from 0 to 100, based on the scoring rubric embedded in the prompt: 80–100 indicates Exceptional, 60–79 indicates Good, 40–59 indicates Average, and 0–39 indicates Below Expectations. The strengths, weaknesses, and advice fields each contain three items. The mockVoiceComment is a dynamically generated coaching comment where the model interprets the candidate's vocal delivery based on the injected voice tone analysis score. All feedback is written in second person using "you" and "your."
- **Voice tone analysis**: The voice confidence dcoring pipeline runs before the GPT call. The system first checks that the Hugging Face Space is in a running state, then sends each recorded audio answer to the model for analysis. The results are averaged into a single overallVoiceConfidenceScore which is injected into the GPT prompt. After the GPT response is parsed, three voice-related fields are saved to Firestore: VoiceToneAnalysis containing the GPT-generated voice comment, VoiceToneRawResults containing the raw SER output per question, and VoiceConfidenceScore containing the computed numerical score.
- **Role in the system:** The parsed GPT report is written to the Report field in the Firestore mock interview document. The mockVoiceComment is stored in VoiceToneAnalysis. The front-end voice tab displays per-question scores from VoiceToneRawResults, the overall GPT voice comment from VoiceToneAnalysis, and the average confidence score from VoiceConfidenceScore.

4.6.3.12 Integration Architecture:

- **API Access:** OpenAI Node.js SDK (v6.3.0) used in Cloud Functions with API key stored securely in environment variables
- **Model:** gpt-4o (latest GPT-4 optimized model)
- **Endpoint:** openai.chat.completions.create() method from OpenAI SDK
- **Response Format:** Structured JSON object with CV sections (PersonalInformation, Summary, Experience, Education, Skills, Certifications, Languages) and improvement suggestions
- **Configuration:** Temperature: 0.7, Max tokens: 4000, Response format: JSON mode

4.6.3.13 Implementation Flow:

- User uploads CV file (PDF/DOCX) through Flutter app using file_picker package
- File uploaded to Firebase Storage with metadata (cvHistoryId, userId)
- Cloud Function (extractCVKeywords) automatically triggered by storage upload, extracts text using pdf-parse or textract libraries
- Extracted text saved to Firestore CVHistory collection, keywords saved to Users collection
- Flutter app listens to Firestore updates in real-time and detects when extraction is complete
- User optionally selects target job (from **Jadeer** jobs or enters custom job details)
- Flutter app calls detectMissingSections Cloud Function with cvHistoryId
- Cloud Function uses OpenAI GPT-4o-mini to analyze OldCVText and identify completely missing or empty sections (PersonalInformation, Summary, Experience, Education, Skills, Certifications, Languages)
- If job information is provided, Cloud Function also generates suggested skills relevant to the target position
- If missing sections are detected, the user navigates to CVNextStepsScreen where they can fill in missing information section by section with guided forms
- Filled information stored in additionalSections object in Flutter app state
- If no missing sections or after user completes filling, Flutter app calls enhanceCV Cloud Function with cvHistoryId, job information, and additionalSections (if any)

- Cloud Function constructs detailed prompt merging original CV text with user-filled additional sections, sends to OpenAI GPT-4o API
- GPT-4o returns structured JSON with enhanced CV sections and improvement suggestions
- Cloud Function automatically generates professional PDF using PDFKit library
- PDF uploaded to Firebase Storage and URL saved to Firestore
- User navigates to CVReadyScreen (PublishScreen) to view enhanced CV and suggestions, can download PDF

4.6.4 Customizations Applied

1. CV-Job Keyword Matching Algorithm

A custom keyword-based matching algorithm was developed to filter and personalize job recommendations based on CV content. The algorithm implementation includes:

Keyword Extraction (Cloud Function):

When a CV is uploaded, a Cloud Function automatically extracts keywords using a custom algorithm that:

- Tokenizes CV text using regex pattern to extract words (minimum 3 characters)
- Filters out common English stopwords using a curated stopwords set (articles, prepositions, common verbs, pronouns)
- Includes technology-specific terms allowlist (programming languages, frameworks, tools) to preserve important short keywords like 'C++', 'API', 'ML', 'UI/UX'
- Calculates keyword frequency and selects top 20 most relevant keywords
- Stores extracted keywords in Firestore Users collection (CVKeywords field)

Job Matching (Flutter App):

When user activates 'For You' filter in job browsing:

- Retrieves user's CV keywords from Firestore
- For each job posting, extracts keywords from job's specialty and keywords fields
- Tokenizes job keywords using regex pattern (splits on special characters, converts to lowercase)

- Compares user's CV keywords with job's tokenized keywords
- Counts matching keywords and returns true if at least 2 keywords match
- Displays only matching jobs when 'For You' filter is active

2. Multi-Criteria Job Filtering System

Custom filtering system allowing users to combine multiple filters:

- Text search across job title, position, specialty, company name, and keywords
- Specialty filtering from 30+ predefined categories (Technology, Engineering, Business, Healthcare, etc.)
- Sort order toggle (newest first or oldest first based on StartDate)
- Show/hide closed jobs filter
- 'For You' personalized filter using CV keyword matching

3. Favorites System

Users can save jobs for later viewing. Implementation includes composite document IDs (UserID_JobID format) in Firestore Favourite collection, synchronization using Firestore snapshots, and optimistic UI updates with error handling.

4. CV Enhancement Workflow with Job Targeting

Multi-step CV enhancement process with three job selection options: select from **Jadeer** job postings (with wishlist/all jobs toggle), enter custom job details for external applications, or enhance CV for general use without job targeting. System provides tailored suggestions based on selected option.

5. Profile Completeness Tracking

System monitors profile completion status and displays contextual reminder messages when users attempt to use features requiring complete profiles (like 'For You' recommendations without uploaded CV), with direct navigation to profile completion screen.

4.6.5 Model Implementation (Vocal Confidence Scoring Model)

The proposed model, referred to as **Vocal Confidence Scoring Model** (architecturally named JadeerNet), was designed as a hybrid regression model that combines deep speech embeddings with handcrafted acoustic features. This hybrid approach was selected to benefit from both high-level contextual speech representations and interpretable acoustic measurements.

- **First Feature Stream:** Uses a pretrained WavLM-Base model to extract deep speech embeddings from each audio sample. The audio is loaded at 16 kHz and passed into WavLM, which produces contextual hidden-state representations. Global average pooling is then applied across the temporal dimension to convert the variable-length speech representation into a fixed-length 768-dimensional embedding for each audio file.
- **Second Feature Stream:** Uses openSMILE with the eGeMAPS v02 feature set to extract handcrafted acoustic features. This produces 88 acoustic functional features related to vocal properties such as pitch, jitter, shimmer, loudness, and spectral characteristics. These features provide interpretable information about voice stability, expressiveness, and other acoustic patterns that may contribute to perceived confidence.

Before model training, the acoustic features were normalized using Z-score normalization through StandardScaler. This step was applied because WavLM embeddings and openSMILE features exist on different numerical scales. Normalization helps prevent one feature type from dominating the learning process during training.

The **Vocal Confidence Scoring Model** follows a late-fusion dual-branch architecture. The WavLM branch processes the 768-dimensional deep embeddings through a dense layer with batch normalization, Leaky ReLU activation, and dropout. The openSMILE branch processes the 88-dimensional acoustic features through a separate dense layer with the same regularization approach. The outputs of the two branches are then concatenated into a unified feature vector and passed through a regression head.

The final regression head outputs a single value between 0 and 1 using a sigmoid activation function, matching the scale of the dataset's interview annotation. The model was trained using Mean Squared Error (MSE) as the loss function. The Adam optimizer was used with a learning rate of 0.001, and the model was trained for 50 epochs with a batch size of 32. During training, the best checkpoint was saved based on the lowest validation loss.

4.6.5.1 Architectural Benchmarking and Baselines

To validate the architectural viability of **JadeerNet**, the hybrid late-fusion design was benchmarked against baseline frameworks established in recent speech evaluation literature. Specifically, the system references state-of-the-art approaches in speech trait prediction—such as the multi-modal fusion networks utilizing handcrafted acoustic parameters proposed by Schuller et al. [43], and recent benchmarks evaluating self-supervised speech embeddings like WavLM for downstream regression tasks [44].

Typical baseline implementations in these standard benchmarks utilize either flat feature concatenation or independent feature branches, resulting in reported Mean Absolute Errors (MAE) fluctuating between **0.12 and 0.14** on continuous emotion and traits evaluation scales. The integration of specialized dense normalization layers and localized dropouts within JadeerNet's customized dual-branch architecture was specifically engineered to reduce this variance and tighten the prediction error boundary below traditional baseline boundaries.

4.6.5.2 Technical Configuration and Deployment

To support mobile execution without introducing computational lag on the client side, the fine-tuned PyTorch model pipeline for the **Vocal Confidence Scoring Model** is compiled and deployed as an independent containerized microservice hosted on the cloud via **Hugging Face Spaces**. The serverless Firebase Cloud Functions interact with this microservice using HTTPS REST API communication. When an interview audio track is sent, the hosting environment extracts the combined features, executes runtime inference, and returns structured JSON confidence weights back to the system.

4.6.6 Implementation Challenges

The development team encountered several technical and research-oriented challenges during the system's implementation phase:

- **OpenAI API Structural Reliability:** Managing LLM outputs to guarantee exact, parsable JSON matching our application schemas required rigorous refinement of prompt architectures and shifting to strict JSON response configurations.

- **API Cost and Rate Management:** Optimizing cloud invocation budgets and reducing duplicate API network loops by storing static generation outcomes inside Firestore records to control expenses for multiple users.
- **Heterogeneous Document Processing:** Resolving library character extraction failures when parsing specific font structures from legacy .pdf or .docx layouts, fixed by deploying stable dependencies and embedding fallback exception catches.
- **Vocal Dataset Scarcity and Realism:** Finding an appropriate, naturally recorded dataset that mimics actual professional interview setups was a major challenge. Most open-source speech emotion repositories contain over-acted, synthetic, or highly dramatic expressions (such as shouting or extreme anger) recorded in studio environments. These do not represent the subtle, natural variations of confidence and stress found in real-world corporate recruitment sessions, requiring intensive filtering to establish a realistic evaluation baseline.
- **Baseline Selection and Benchmark Discrepancies:** Establishing a definitive scientific baseline for comparison presented a significant hurdle due to extreme variance in existing literature. Published papers using advanced frameworks—such as combining pretrained WavLM deep models with openSMILE eGeMAPS acoustic features—exhibit huge discrepancies in training strategies, custom hyperparameter setups, data splits, and diverse downstream regression head architectures. This lack of a unified benchmark standardized for interview confidence scoring forced the team to perform extensive empirical hyperparameter tuning from scratch to stabilize our hybrid late-fusion model.

4.6.7 Consultations Received

The team received regular guidance from the project Supervisor throughout the development process. Her guidance included feedback on system architecture decisions, AI integration approaches, best practices for mobile development, and ensuring alignment with academic project requirements. Specific feedback received included a software specialist Professor recommendation to separate the machine learning model into its own class in the class diagram, as the team would be responsible for training the model, ensuring proper architectural separation of concerns.

The team conducted consultations with HR professionals from a well-known companies like **SDAIA** to validate feature requirements and gather insights on recruitment challenges. Key feedback received:

- Project success depends on quality of outcomes and accurate candidate evaluation
- Medium and small companies would benefit more from the system than larger enterprises
- Suggested integration of psychometric tests (DISC, 16 Personalities) as personality assessment is crucial for company-candidate fit
- HR professionals prioritize personality tests and soft skills evaluation over voice tone analysis
- Many companies use personality test links during hiring process to filter candidates based on results
- Recommended conducting field studies and interviewing companies across different sizes and sectors to better understand their specific needs
- Overall positive feedback on project concept and potential market impact

4.6.8 Project Repository

Repository link: https://github.com/WS888-CODER/gp_2025_11

5 System Testing

5.1 Experimental Results

The **Vocal Confidence Scoring Model** was evaluated through three practical testing phases to verify its mathematical accuracy, its immediate software behavior, and its ability to generalize to real-world interviews:

5.1.1 Testing on the Dataset Test Set

In this phase, the model was evaluated using the held-out test split from the core dataset, which contains **603 unseen audio records** to measure foundational statistical accuracy. The results are summarized in the Table 5:

Table 5: 5-1 Model Performance Metrics on the Dataset Test Set

Evaluation Dataset Split	Mean Squared Error (MSE)	Mean Absolute Error (MAE)	Accuracy
Test Set	0.0195	0.1102	0.8898

The model achieved a Mean Squared Error (MSE) of 0.0195 and a Mean Absolute Error (MAE) of 0.1102. Since the confidence score ranges from 0 to 1, an MAE of 0.11 means that the model's predictions differ from the actual human ratings by only 11% on average, resulting in an overall baseline accuracy score of **88.98%**. This proves the model is mathematically stable when processing new voices.

When compared to established architectural benchmarks in speech trait evaluation—such as standard hybrid WavLM and openSMILE fusion baselines that typically yield an MAE fluctuating between 0.12 and 0.14—the **Vocal Confidence Scoring Model** successfully out-performed traditional setups by achieving a lower error boundary of 0.1102. This statistical improvement proves that our custom dual-branch regularization, specific dropout configurations, and dense normalization head successfully minimize prediction variance, rendering the model highly stable and precise when processing entirely new voices.

5.1.2 Preliminary Testing (4 Live Custom Scenarios)

Before expanding the evaluation, a preliminary test was conducted directly within the model pipeline using four custom recorded voice tracks. This step was crucial to verify that the integrated backend layers—such as the Voice Activity Detection (VAD) and automated pause-counting algorithms—react correctly to distinct vocal behaviors. The operational decisions and scoring outputs for these four test cases are summarized in the Table 6:

Table 6: 5-2 Preliminary Model Test Results

Test Case / Scenario	Tested Vocal Behavior	Resulting Score	Final System Decision & Action
Test 1: Normal Pace *(The Calm Intro)*	Steady delivery with stable speech, clear pronunciation, and no major pauses.	61.47%	Good Presence. Accepted as a normal, valid interview response.
Test 2: High Hesitation *(The Anxious Case)*	Speech filled with frequent verbal fillers ("umms", "ahhs") and long gaps.	26.95%	High Vocal Anxiety. Gaps flagged; automated hesitation penalty applied.
Test 3: Non-Human Input *(The Noise Case)*	Background environment noise or complete silence with no human speech signals.	N/A	REJECTED. Audio blocked via automated Silero Voice Activity Detection (VAD).

Test 4: High Confidence *(The Professional Peak)*	Fluid, clear, articulate, and well-paced speech delivery.	95.56%	Excellent Professional Flow. Candidate marked as a highly confident communicator.

5.1.3 Extended Real-World Testing (100 Records)

To expand the evaluation phase beyond manual checks, a large-scale blind test was conducted using a randomized batch of 100 actual human voice records taken from the **CREMA-D dataset**. This dataset was specifically selected because it contains a wide variety of unconstrained real-world voices with diverse lexical phrasing (12 distinct conversational sentences) and high acoustic variation, making it ideal for testing behavioral robustness. This phase was designed to stress-test the model's behavioral logic and observe how the insight generator reacts to completely unseen human speakers. The overall distribution of the final system decisions across the 100 test samples is summarized in the Table 7:

Table 7: 5-3 Distribution of System Decisions Across 100 Real-World Test Records

#	Final System Decision & Action	Test Count	Primary Behavioral Patterns Observed
1	Excellent Professional Flow. Steady, fluent, and highly confident vocal delivery.	3	High score (> = 85%), clear articulation, and zero speech gaps.
2	Good Presence. Stable professional presence with minor speech artifacts.	11	Balanced score (70% - 84.9%), normal pacing with safe vocal limits.

3	Significant Silent Pauses. Gaps flagged; automated hesitation penalty applied.	0	Multiple structural silent gaps exceeding the 1.0-second threshold.
4	High Vocal Anxiety. Vocal instability and tremors detected in speech.	85	Lower regression score caused by high local jitter and shimmer features.
5	Frequent Verbal Fillers. Monotone delivery or repetitive vocal fillers flagged.	0	Low pitch variation and flat standard deviation metrics.
6	Low Vocal Volume. Weak audio signal; may affect professional presence.	0	Overall loudness levels falling below the required system threshold.
7	REJECTED. Audio blocked due to data corruption or zero human voice signals.	1	Isolated automatically by the integrated Silero VAD layer (Sample 100).
Total Evaluated Records: 100			

The statistical breakdown shows that the model successfully processed and classified all 100 records from the CREMA-D dataset. A significant majority (85%) triggered the **High Vocal Anxiety** decision. This high density is normal and directly corresponds to the acoustic nature of this dataset split, which is inherently filled with intense emotional stress, vocal tremors, and high jitter and shimmer values. The model accurately caught these physical micro-variations and mapped them to lower confidence brackets.

Simultaneously, the system isolated 11% as **Good Presence** and 3% as **Excellent Professional Flow**, where the combination of WavLM deep embeddings and openSMILE features reflected strong, stable signal patterns.

Crucially, **Sample 100** was automatically **REJECTED** by the system, proving that the Silero Voice Activity Detection (VAD) layer works reliably as a protective filter to block corrupt data or empty audio frames before they reach the scoring backend. This extended test proves that the pipeline is behaviorally sound, reliable, and ready for deployment.

5.2 User Acceptance Testing

To evaluate user acceptance of the application, we assembled a team of 20 participants: 10 of them were HR consultants representing companies, and the other 10 were individuals actively seeking jobs from within our network. We had them test the application and then complete a survey to measure their level of satisfaction.

The following are the results of the test.

5.2.1 Demographics of Participants

The participants in the application testing were asked about their age, gender, education level, and experience with mobile applications. The following figures displays their answers.

As for the Companies, these were their responses:

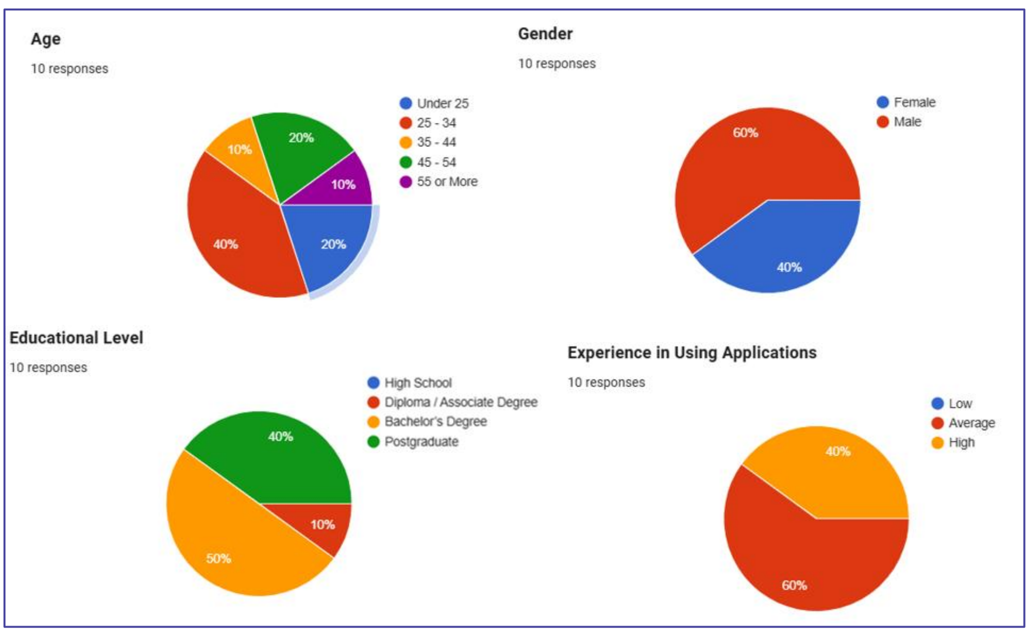

Figure 23: 5-1 Companies Demographics

For the job seekers, these were their responses:

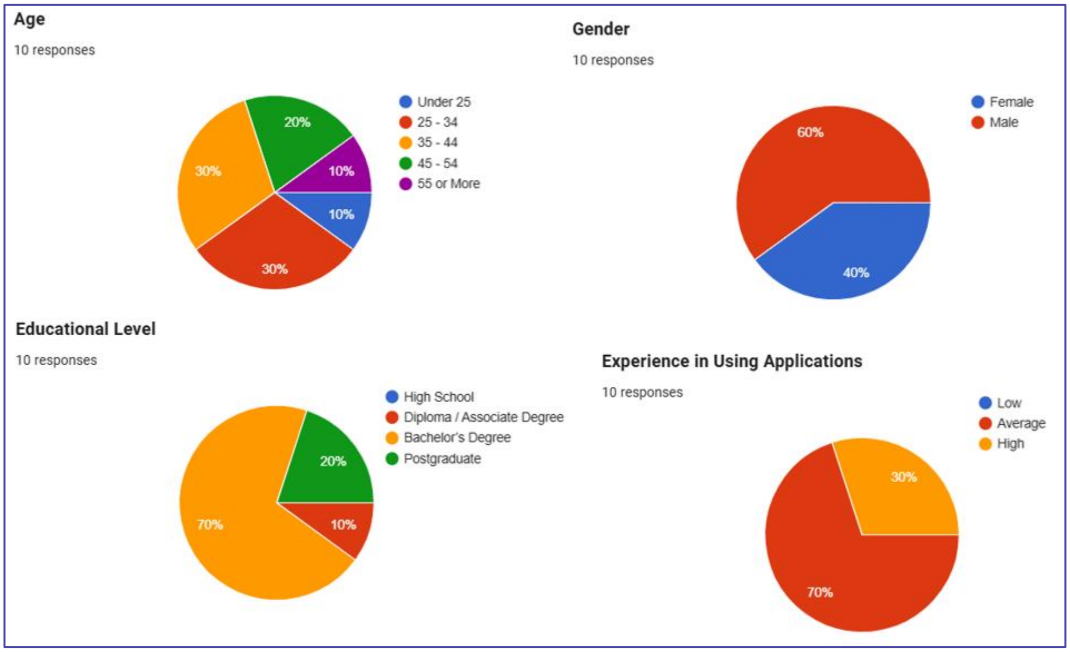

Figure 24: 5-2 Job Seeker Demographics

5.2.2 Questionnaire/Interview Results

Since our application targets two user groups — **Companies** and **job seekers** — we conducted user acceptance testing on both segments using questionnaires. A total of **20 participants** were involved: **10** Companies and **10 job-seeker**. The survey results are summarized in the following two tables (for more details see Appendix D).

Companies' acceptance testing survey results:

Table 8: 5-4 Companies UAT Results

Question	Number of respondents (Percentage of responses)				
	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I needed previous experience to use the app.			2 (20%)	5 (50%)	3 (30%)
2. I found the application easy to use.			2 (20%)	4 (40%)	4 (40%)
3. I found the navigation between pages smooth and clear.			1 (10%)	3 (30%)	6 (60%)
4. I found the color palette and visual design comfortable.			1 (10%)	4 (40%)	5 (50%)
5. I found the interface design clear and intuitive.			1 (10%)	4 (40%)	5 (50%)
6. Some icons or labels were unclear to me.			4 (40%)	1 (10%)	5 (50%)
7. The app responded quickly and did not lag.				5 (50%)	5 (50%)

8. Pages and features loaded without noticeable delays.			2 (20%)	5 (50%)	3 (30%)
9. I was able to create a job post correctly, easily, and clearly.		1 (10%)	1 (10%)	6 (60%)	2 (20%)
10. I found the AI-generated job descriptions useful and relevant.		1 (10%)	2 (20%)	5 (50%)	2 (20%)
11. The AI-generated interview questions were relevant to the job post.		1 (10%)	2 (20%)	3 (30%)	4 (40%)
12. The difficulty level and clarity of the questions were appropriate.		1 (10%)	1 (10%)	3 (30%)	5 (50%)
13. I was able to view, edit, or add to the AI-generated questions easily.				5 (50%)	5 (50%)
14. I was able to edit my profile correctly, easily, and clearly.			1 (10%)	4 (40%)	5 (50%)

15. I was able to reset my password without any difficulty.			2 (20%)	7 (70%)	1 (10%)
16. The pop-up messages I got from the app were helpful and clear.			2 (20%)	4 (40%)	4 (40%)
17. I think I will use this application frequently.				6 (60%)	4 (40%)
18. I will recommend this application to others.				6 (60%)	4 (40%)
19. I was able to view candidate applications for my job postings clearly and easily.		1 (10%)	3 (30%)	4 (40%)	2 (20%)
20. The AI-generated candidate evaluation report was comprehensive and useful for hiring decisions.		1 (10%)	1 (10%)	6 (60%)	2 (20%)
21. The candidate's overall score and sub-scores (technical,			1 (10%)	7 (70%)	2 (20%)

psychometric, voice tone, CV, job requirements) were clear and easy to interpret.					
22. The candidate ranking based on evaluation scores helped me compare applicants objectively			1 (10%)	7 (70%)	2 (20%)
23. I was able to download the candidate evaluation report as a PDF without difficulty.			1 (10%)	5 (50%)	4 (40%)

Job seekers acceptance testing survey results:

Table 9: 5-5 Job Seeker UAT Results

Question	Number of respondents (Percentage of responses)				
	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I needed previous experience to use the app.		2 (20%)	1 (10%)	5 (50%)	2 (20%)
2. I found the application easy to use.			3 (30%)	6 (60%)	1 (10%)

3. I found the navigation between pages smooth and clear.			4 (40%)	4 (40%)	2 (20%)
4. I found the color palette and visual design comfortable.		1 (10%)	3 (30%)	4 (40%)	2 (20%)
5. I found the interface design clear and intuitive.			3 (30%)	4 (40%)	3 (30%)
6. Some icons or labels were unclear to me.			4 (40%)	4 (40%)	2 (20%)
7. The app responded quickly and did not lag.			3 (30%)	6 (60%)	1 (10%)
8. Pages and features loaded without noticeable delays.			5 (50%)	4 (40%)	1 (10%)
9. I was able to upload my CV to my profile correctly, easily, and clearly.			4 (40%)	5 (50%)	1 (10%)
10. I found the AI CV enhancement helpful and easy to understand.			2 (20%)	5 (50%)	3 (30%)
11. The AI suggestions I received after			3 (30%)	6 (60%)	1 (10%)

enhancing my CV were clear, relevant, and helpful.					
12. The system correctly identified missing or incomplete information in my CV and helped me understand what to add.			2 (20%)	7 (70%)	1 (10%)
13. I liked having the option to choose jobs so the AI could tailor the CV enhancement toward those roles.		1 (10%)	1 (10%)	5 (50%)	3 (30%)
14. The enhanced CV I received (PDF) was clear, well-formatted, and improved compared to my original one.		1 (10%)	2 (20%)	4 (40%)	3 (30%)
15. I was able to browse and search for job posts easily and clearly.			1 (10%)	6 (60%)	3 (30%)
16. The job recommendations I			3 (30%)	4 (40%)	3 (30%)

received based on my CV were relevant and useful.					
17. I was able to save or favorite job posts correctly and easily.			3 (30%)	6 (60%)	1 (10%)
18. I was able to find my previous CV enhancements on the History page easily.			2 (20%)	7 (70%)	1 (10%)
19. I was able to edit my profile correctly, easily, and clearly.		1 (10%)	1 (10%)	5 (50%)	3 (30%)
20. I was able to reset my password without any difficulty.			3 (30%)	7 (70%)	
21. The pop-up messages I got from the app were helpful and clear.		1 (10%)	2 (20%)	6 (60%)	1 (10%)
22. I think I will use this application frequently.		1 (10%)	1 (10%)	3 (30%)	5 (50%)

23. I will recommend this application to others.			1 (10%)	7 (70%)	2 (20%)
24. I was able to apply for a job correctly, easily, and clearly.			1 (10%)	7 (70%)	2 (20%)
25. The interview questions presented during the AI interview were relevant to the job I applied for.		1 (10%)	2 (20%)	3 (30%)	4 (40%)
26. I was able to start and complete the mock interview without difficulty.		1 (10%)	2 (20%)	5 (50%)	
27. The mock interview questions were relevant to my selected specialty.			2 (20%)	7 (70%)	1 (10%)
28. The mock interview report I received (strengths, weaknesses, and advice) was clear, accurate, and helpful.			4 (40%)	4 (40%)	2 (20%)
29. The voice tone analysis included in my mock interview report			2 (20%)	6 (60%)	

provided useful insight into my communication style.					
30. I was able to find my previous mock interviews and job applications on the History page easily.			2 (20%)	5 (50%)	3 (30%)

5.3 Quality Attributes (NFR testing)

Table 10: 5-6 Non-functional requirements testing

PBI (user story)	Quality attribute	Measure	Results
As a user, I want the **Jadeer** to respond within 5 seconds so that I can complete my tasks quickly.	Performance: How responsive the system is when users navigate between pages?	Measure page load time (in seconds) for all main pages: Home, Job Search, History (CV, Mock Interview, Job Application), Favorites, Profile, Settings, Notifications. Use manual timing from navigating to page to fully loaded page.	**Test Results (5 attempts per page):** • Home: 0.22 sec avg • Job Search: 0.40 sec avg • History (CV): 0.51 sec avg • History (Mock Interview): 0.59 sec avg • History (Job Application): 0.46 sec avg • Favorites: 0.38 sec avg • Profile: 0.49 sec avg • Settings: 0.29 sec avg

			• Notifications: 0.43 sec avg **Conclusion:** All pages load in < 1 second (under 5-second requirement).
As a user, I want **Jadeer** to be available 99% of the time so that I can use the app.	Availability: How consistently is the system accessible and available to users?	Compute the percentage of application availability. **Jadeer** relies on Firebase for authentication and data storage.	**Jadeer** is built on Firebase, which provides 99.95% uptime according to Google Cloud's SLA documentation. This exceeds the 99% availability requirement [45].
As a user, I want login attempts to be limited to 5 failed tries in one day so that my account remains secure.	Security: How well are user accounts protected from unauthorized access attempts?	Test login failure handling by attempting invalid logins 5+ times. Verify account lock, error messaging, and unlock mechanism.	The system locked the account after exactly 5 failed login attempts. A dialog displayed: "Account Locked - You have entered an incorrect password 5 times today." Users must reset their password via OTP verification to regain access.
As a job seeker, I don’t want my interview recordings stored after the system is done with them so that my privacy is protected.	Data Protection: How is user data protected and managed throughout its lifecycle?	Verify interview recordings are automatically deleted from Firebase Storage after the system is done with processing, by uploading a test file and confirming deletion.	Interview recordings uploaded to Firebase Storage are automatically deleted after the system is done with processing them. Auto-delete lifecycle rule is configured and functioning as intended.

As a user, I want at least 90% of tasks to be completed successfully on the first attempt by test users so that **Jadeer** is user-friendly.	Usability: How easily can users complete core tasks without assistance on their first attempt?	Have 8 test users attempt 3 core tasks without guidance: Sign up, Apply to job, CV enhancement. Calculate success rate: (Successful completions / 24 total attempts) × 100. Target: ≥ 90%.	8 test users completed 3 tasks each (24 total attempts). Successful completions: 22 / 24 = 91.7%. Two minor usability issues identified: User feedback indicated need for clarification on CV enhancement sections and their purpose, and questions about interview flow and post-application process. Overall, system demonstrated strong usability with minimal friction.

5.4 AI Output Validation

To ensure that the AI-generated interview reports are reliable and accurate, we conducted a structured validation process using a three-layer approach: consistency testing, LLM-as-judge evaluation, and human review.

5.4.1 Test Setup

We selected a Cyber Security Analyst position and created two test scenarios with distinct candidate profiles. The first scenario represented a strong candidate with over two years of SOC experience, hands-on SIEM proficiency with Splunk, practical knowledge of NIST CSF and ISO 27001 frameworks, and real vulnerability discovery experience. The second scenario represented a weak candidate with only academic exposure to cybersecurity concepts and no practical experience. Both candidates were asked the same five interview questions covering SIEM tools, incident response, security frameworks, vulnerability management, and professional development.

Each scenario was processed five times through our GPT-4 report generation function in independent sessions to ensure no memory carryover between runs.

5.4.2 Results

The strong candidate received scores of 88, 90, 88, 92, and 88, with an average of 89.2 out of 100 and a range of only 4 points. The weak candidate received scores of 38, 34, 32, 42, and 32, with an average of 35.6 out of 100 and a range of 10 points. The average gap between the two scenarios was 53.6 points, confirming that the AI clearly differentiates between strong and weak candidates.

For the LLM-as-judge evaluation, a separate AI prompt was used to rate each generated report on five criteria: strengths accuracy, weakness relevance, advice usefulness, score appropriateness, and consistency with job requirements. Each criterion was rated on a scale of 1 to 5, giving a total out of 25. The strong candidate reports averaged 20.8 out of 25, and the weak candidate reports averaged 22 out of 25.

Finally, team members manually reviewed the AI-generated reports and confirmed that the identified strengths, weaknesses, and recommendations were relevant and accurate for both scenarios.

5.4.3 Conclusion

This three-layer validation process confirms that the AI report generation system produces consistent, differentiated, and relevant interview evaluations. The system reliably assigns higher scores to qualified candidates and lower scores to underqualified ones, while providing accurate and actionable feedback in both cases.

5.5 Voice Tone Analysis Validation

To validate the reliability of the Speech Emotion Recognition (SER) model integrated into the application, we conducted a structured evaluation to ensure that the model produces consistent and meaningful vocal confidence assessments for interview recordings.

5.5.1 System Overview

The voice tone analysis pipeline operates in two stages. First, each recorded interview answer is sent to a custom SER model hosted on Hugging Face (Jadeer Hybrid Vocal Confidence Scoring Engine). The model analyzes the audio signal and returns an Assessment Score (0–100) along with a textual insight describing the candidate's vocal delivery, such as "Steady, fluent, and highly confident vocal delivery" or "Vocal instability and tremors suggest nervousness." The per-question scores are then averaged to produce an Overall Voice Confidence Score.

In the second stage, the overall voice confidence score is passed to GPT-4 as part of the interview analysis prompt. GPT-4 uses this score to generate a personalized voice feedback comment that addresses the candidate's pacing, conviction, and delivery without revealing the exact numerical score. This two-stage approach combines objective acoustic analysis with contextual AI-generated feedback.

5.5.2 Integration in Reports

For mock interviews, the voice analysis is presented to the job seeker in a dedicated Voice tab containing an informational message explaining the analysis methodology, per-question scores with color-coded progress bars and insight text, an overall voice feedback comment generated by GPT-4, and the average voice confidence score displayed at the bottom of the tab.

For job interview reports, the per-question voice analysis results are included in the PDF report generated for the company. Each question displays its confidence score out of 100, a visual progress bar, and the model's insight. The average voice confidence score is presented at the end of the section. The voice tone score contributes 10% to the candidate's weighted final score in the company evaluation.

5.5.3 Weighted Score Formula

The final candidate evaluation score is calculated using the following weighted formula as defined in the product backlog:

Final Score = (CV Analysis × 30%) + (Job Requirements Match × 20%) + (Psychometric Analysis × 20%) + (Voice Tone Analysis × 10%) + (Technical Evaluation × 20%)

Each component is scored from 0 to 100, and the final score is rounded to the nearest whole number. The voice tone score is derived from the average of all per-question SER assessment scores.

5.5.4 Conclusion

The voice tone analysis system provides an objective, AI-driven assessment of a candidate's vocal confidence during interviews. By combining a specialized SER model for acoustic analysis with GPT-4 for contextual feedback, the system offers both quantitative scores and qualitative insights. The 10% weight in the final score ensures that vocal delivery is considered as a supplementary factor alongside technical knowledge, psychometric traits, and job requirement alignment, without disproportionately affecting the overall evaluation.

5.6 Discussion

Based on the results of the system evaluation, which comprised User Acceptance Testing (UAT) conducted with 20 participants — 10 job seekers and 10 HR representatives — alongside objective Non-Functional Requirements (NFR) testing, the overall outcomes were highly favorable and indicate that **Jadeer** successfully met its core design and development objectives.

The UAT results demonstrated strong user satisfaction across both groups. Job seekers consistently rated the AI-powered CV enhancement features at 4–5, with the system's ability to identify missing CV information receiving 70% at rating 4. The mock interview module was well received, with question relevance to selected specialties rated 4 by 70% of job seekers. HR representatives rated AI-generated job descriptions and interview questions at 4–5, and the ability to edit AI-generated content received uniformly high scores. Most significantly, both groups expressed strong intention to use the application frequently and willingness to recommend it to others, confirming genuine product-market fit. The NFR testing further validated the system's technical soundness: all pages loaded under 1 second (well within the 5-second requirement), the system achieved 99.95% availability through Firebase's infrastructure, the account lockout security mechanism functioned exactly as designed, interview recordings were automatically deleted after processing, and the first-attempt task success rate reached 91.7%, exceeding the 90% usability target.

However, the evaluation also surfaced several recurring areas of concern. Icon and label clarity drew consistently mixed responses from both user groups, with a notable portion of participants rating it at 3. Feature discoverability — particularly around the History page and the mock interview flow — generated neutral responses, suggesting that while the features are functionally sound, their navigation entry points are not sufficiently prominent. The PDF output quality for enhanced CVs, while generally positive, showed occasional inconsistency. Additionally, pop-up messages were rated as adequate but not particularly informative or actionable.

Reflecting on the development process, the choice to build **Jadeer** on Flutter and Firebase proved effective in delivering a cross-platform, responsive application with minimal latency. The integration of the OpenAI API for CV enhancement, job description generation, interview question creation, and candidate evaluation was validated by users as genuinely useful and contextually relevant. However, the UI/UX design process would have benefited from earlier-stage usability testing — particularly for icon design and navigation structure — rather than relying solely on post-development evaluation. Incorporating iterative prototyping with representative users earlier in the process would have likely surfaced the clarity and discoverability issues before implementation.

To improve the system, the following enhancements are recommended:

- Icon and Label Clarity: Conduct a targeted UI audit to identify unclear visual elements, add descriptive text labels or tooltips, and retest with users.
- Feature Discoverability: Redesign navigation to surface the History page and mock interview entry points more prominently, and introduce an optional onboarding walkthrough for first-time users.
- PDF Output Consistency: Implement automated quality checks and a preview mode before final export to ensure consistent rendering across all CV formats.
- Mock Interview Report Depth: Enrich feedback reports with more specific, role-targeted insights and actionable improvement guidance.
- Localization: Add Arabic language support and RTL interface adaptation to better serve the Saudi market target audience.

In conclusion, the system evaluation yielded strongly positive results, validating both the technical implementation and the user experience design of **Jadeer**. The identified improvement areas are targeted and actionable, positioning the platform as a viable foundation for real-world deployment beyond the academic context of this graduation project.

6 Conclusions and Future Work

This section reflects on the journey of developing **Jadeer**, starting from identifying the recruitment challenges in Saudi Arabia to designing, scaling, and implementing an end-to-end AI-powered solution that addresses these gaps across multiple development phases. The report began by introducing the problem of inefficient and biased hiring processes, followed by an in-depth background study on recruitment technologies and their limitations. We explored existing platforms through a literature review, identifying gaps such as lack of candidate feedback, limited fairness, and weak personalization. These insights guided the comprehensive design of **Jadeer** as a comprehensive recruitment ecosystem serving both companies and job seekers, expanding its capabilities from core initial prototypes to a fully integrated, production-ready application.

Moving forward, we defined system requirements through interviews and surveys, ensuring that **Jadeer** meets real user needs. The design phase translated these requirements into a structured architecture, incorporating advanced technologies such as Large Language Models, and Speech-to-Text systems. Implementation focused on delivering a comprehensive suite of features across all development cycles: AI-powered CV enhancement, interactive speech-based and AI-generated mock interviews for candidates with automated performance feedback, alongside AI-driven evaluation, customized job posting generation, and candidate ranking for Companies. Finally, extensive user acceptance testing validated the usability and effectiveness of **Jadeer's** entire ecosystem, confirming its potential to streamline recruitment and improve fairness. This conclusion summarizes the project's comprehensive impact, limitations, and contributions, and outlines future directions for enhancement.

6.1 Global and Local impact

Jadeer significantly impacts recruitment both locally and globally. On a local level, it supports Saudi Vision 2030 by promoting workforce empowerment, national talent readiness, and labor market efficiency. Companies benefit from automated CV screening, AI-driven mock and automated interviews, and structured candidate evaluation reports, reducing hiring delays and operational costs. Job seekers gain intelligent tools for CV optimization and realistic voice-interactive interview practice, improving their chances of securing suitable roles. On a global scale, **Jadeer** demonstrates how advanced AI orchestration can transform recruitment processes by reducing systemic bias and serving as a model for future HR technologies worldwide.

6.2 Problems and Challenges Encountered During Software Development

During the development of **Jadeer**, the team faced several challenges that required adaptability and problem-solving. One of the initial difficulties was reaching stakeholders to gather accurate requirements. This was resolved by leveraging the Nawafith [46] platform to schedule consultations with HR experts, ensuring that the system design aligned with real-world recruitment needs.

Another challenge involved technical issues with OTP (One-Time Password) functionality, which initially failed to work within the Kingdom. After multiple attempts and adjustments, the team identified and implemented a workaround to ensure successful verification.

Database design also posed significant challenges, particularly in determining the necessary tables and attributes to support **Jadeer's** comprehensive recruitment functionality and expanding schema for new features. This required multiple iterations and refinements throughout the stages of the project.

Additionally, defining the methods for the class diagram and synchronizing the real-time AI API communication with backend servers was a complex task. The team sought guidance from the supervisor and external experts, including consultations with specialists, to finalize the appropriate methods, manage asynchronous requests, and ensure accurate representation of system behavior. These challenges collectively strengthened the team's ability to adapt, acquire new knowledge, and refine the system design effectively.

6.3 Limitations of the system

Despite its innovative features, **Jadeer** has certain limitations in its current release. The application is focused on its localized cross-platform release, which may require further optimization across diverse device models. Additionally, **Jadeer** supports English language exclusively for its core processing, which may limit usability for non-English speakers. The system focuses primarily on comprehensive recruitment-related functionalities and does not include broader HR modules such as payroll or attendance management. Furthermore, advanced features like real-time multi-language support and fully independent offline processing are planned for future versions but are not yet fully extended.

6.4 The main contribution of the project

The main contribution of **Jadeer** lies in its ability to bridge the gap between employers and job seekers through a unified, fully realized AI-powered platform. Unlike traditional Applicant Tracking Systems that rely on keyword filtering, **Jadeer** offers a holistic approach by combining CV optimization, AI-generated job postings, and unbiased speech-integrated interview evaluations. By leveraging technologies such as LLMs, and speech analysis, **Jadeer** ensures fairer recruitment decisions and provides actionable feedback to candidates, ultimately improving hiring quality and candidate preparedness across the entire job application lifecycle.

6.5 Future work

Jadeer aims to enhance the user experience in its upcoming journey by introducing several new features. Future plans include expanding language support to incorporate multiple languages and adding Arabic localization to increase adoption within the Saudi market. The application will also focus on expanding accessibility for a wider audience by continuously optimizing its performance across different platforms and operating systems. Additionally, **Jadeer** intends to integrate live real-time video interviews for more interactive assessments and develop advanced analytics dashboards to help Companies track long-term recruitment metrics effectively. Further improvements will include refining AI models for deeper psychometric analysis and personalized job recommendations. These enhancements demonstrate a strong commitment to optimizing the platform and meeting the evolving needs of both employers and job seekers.

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9 Appendix

9.1 Appendix A (Interviews):

Since our application **Jadeer** has two types of users, we conduct interviews with both types to gather requirements more accurately. Let us begin with the first type of users, which are the Companies. We interviewed employees and consultants in the field of **Human Resources**, and after introducing them to the idea of our application **Jadeer**, we asked them the following questions:

Companies' Interview Questions:

1) Do you have prior experience using recruitment software or platforms? If yes, which tools have you used, and what did you like or dislike about them?
2) Could you describe your typical hiring process?
3) What are the most time-consuming tasks in your recruitment process?
4) Can you walk me through your current process for screening resumes? What tools or methods do you rely on, and what would you consider the main strengths and weaknesses of this approach?
5) What are the most common factors or reasons that lead to the rejection of applicants during the recruitment process?
6) What details of a candidate's CV are most important for you in the first review?
7) How do you usually structure your interviews — for example, what kinds of questions do you start with, and how do you move on from there?
8) How do you evaluate the importance of receiving detailed and objective assessment reports about applicants, and what aspects of such reports matter to you the most?
9) What feature would make a recruitment app valuable enough for you to use it regularly?

Table 11: 9-1 Interview's Transcription for first Interviewee

Interview (1)	
Interviewee: Participant 1	**Interviewer:** Walaa Saif Aleslam
Location/Medium: Online via Nawafith app	**Appointment Date:** 15 Sep 2025 **Start Time:** 12:00 PM **End Time:** 01:00 PM
Objectives: • Understand prior experience with recruitment software and user preferences. • Explore the typical hiring process followed by HR professionals. • Identify the most time-consuming tasks in recruitment. • Assess current resume screening methods, tools, and their strengths and weaknesses. • Understand common reasons for applicant rejection. • Determine which CV details are most important during initial review. • Explore how interviews are structured and conducted. • Evaluate the perceived importance of detailed and objective candidate assessment reports. • Identify features that make the recruitment platform valuable for regular use.	**Reminders:** The interviewee constantly emphasizes the importance of quality of application output.
Agenda: Introduction: Background in project: Overview of interview: Topic to be covered: Permission to record: Question 1:	**Approximate Time:** 12 min 15 min 4 min 2 min 1 min 3 min

Question 2:	5 min
Question 3:	4 min
Question 4:	2 min
Question 5:	2 min
Question 6:	2 min
Question 7:	1 min
Question 8:	2 min
Question 9:	2 min
Summary of major points:	1 min
Questions from interviewee:	1 min
Closing:	1 min
General Observations: The interviewee showed interest and curiosity about the project and advised focusing on measuring the personal side of the candidate.	
Topic not covered: Due to the lack of time and the interviewee's limited appointments, there was not enough time for me to ask him about the names of companies he would recommend me to meet with, because he advised me to focus on medium and small companies.	
Interviewee: Khalid Al-Kwelet	**Date:** 15 Sep 2025
Questions:	**Answers and Notes:**
Q1: Do you have prior experience using recruitment software or platforms? If yes, which tools have you used, and what did you like or dislike about them?	Yes, I have experience with recruitment programs, for example, LinkedIn. What I like about it is that it gives me the ability to post jobs for free, and it allows me to search for candidates based on their city of residence. It also suggests people suitable for the job. **Observations:** To persuade businesses to embrace **Jadeer**, we need to perform better than current applications.
Q2: Could you describe your typical hiring process?	First, we post the available position and explain the job description and employee responsibilities in detail. Then, we see which candidates meet the requirements. After that,

	we send them to the decision-maker. Then, an interview is organized with them, whether in person or online. The interview is attended by a committee consisting of the head of the department for the available position, a human resources employee, and an employee from a neutral department. After that, each applicant is evaluated by this committee. Finally, we offer the position to the appropriate person, and the appropriate salary is determined. **Observations:** The recruitment process is complex, long, time-consuming and labor-intensive.
Q3: What are the most time-consuming tasks in your recruitment process?	We suffer from all the stages of the hiring cycle that I mentioned previously, and they waste our time, from announcing the job until we hire the right person. **Observations:** There is a real need to automate the process.
Q4: Can you walk me through your current process for screening resumes? What tools or methods do you rely on, and what would you consider the main strengths and weaknesses of this approach?	I work in the private sector. We do not have a specific tool that we use to filter CVs, and we rely more on the manual filtering method. **Observations:** Human resources employees suffer from the filtering process, especially since it is done manually in the private sector, which causes a delay in selecting the right person.
Q5: What are the most common factors or reasons that lead to the rejection of applicants during the recruitment process?	The main reasons we reject job applicants can be summed up in three reasons: First, the skills may be overqualified or underqualified than required. Second, the salary may be higher or lower than the company's budget. Third, the personality, if it does not suit the work environment and system, will result in his rejection.

	Observations: We should focus on these things when suggesting jobs to people and also when training them for interviews.
Q6: What details of a candidate's CV are most important for you in the first review?	The most important detail in any CV is the applicant's suitability for the job. This varies from one job to another, but the most important thing is to ensure that he is able to carry out all the job responsibilities. **Observations:** The candidate should concentrate on emphasizing his abilities in a way that aligns with the job posting.
Q7: How do you usually structure your interviews — for example, what kinds of questions do you start with, and how do you move on from there?	Each role has its own questions that show us the person's skills in the field. **Observations:** The importance of asking appropriate questions in each interview according to the field
Q8: How do you evaluate the importance of receiving detailed and objective assessment reports about applicants, and what aspects of such reports matter to you the most?	From every job description, we get an assessment sheet, which becomes like a checklist. Every interviewer fills them out, and the final report is made up of them. **Observations:** The importance of arranging the final report in the form of a checklist
Q9: What feature would make a recruitment app valuable enough for you to use it regularly?	Focus on analyzing the applicant's personality, not just his skills. Also, the efficiency of the artificial intelligence in your application and its ability to select suitable candidates are the two most important points I can advise you on. **Observations:** Performance is what distinguishes us

Table 12: 9-2 Interview's Transcription for second Interviewee

Interview (2)	
Interviewee: Participant 2	**Interviewer:** Walaa Saif Aleslam
Location/Medium: Online via Nawafith app	**Appointment Date:** 15 Sep 2025 **Start Time:** 02:30 PM **End Time:** 03:25 PM
Objectives: • Understand prior experience with recruitment software and user preferences. • Explore the typical hiring process followed by HR professionals. • Identify the most time-consuming tasks in recruitment. • Assess current resume screening methods, tools, and their strengths and weaknesses. • Understand common reasons for applicant rejection. • Determine which CV details are most important during initial review. • Explore how interviews are structured and conducted. • Evaluate the perceived importance of detailed and objective candidate assessment reports. • Identify features that make the recruitment platform valuable for regular use.	**Reminders:** The importance of providing solutions to individuals that qualify them to obtain a job.
Agenda: Introduction: Background in project: Overview of interview: Topic to be covered: Permission to record:	**Approximate Time:** 12 min 15 min 4 min 2 min 1 min

Question 1:	3 min
Question 2:	5 min
Question 3:	4 min
Question 4:	2 min
Question 5:	2 min
Question 6:	2 min
Question 7:	1 min
Question 8:	2 min
Question 9:	2 min
Summary of major points:	1 min
Questions from interviewee:	1 min
Closing:	1 min
General Observations: The interviewee believes that our Company side is not new and that we need to be excellent at implementing it if we want to be truly helpful and attract corporations' attention. Despite their frustration with the drawn-out and difficult hiring process, I also observed that not all HR staff members are happy with the integration of AI into their jobs. This could be due to a lack of confidence in technology or a reluctance to let AI take their place.	
Topic not covered: I didn't have time to question him about why he wasn't amenable to AI doing job interviews.	
Interviewee: Mohammed Al-Zahrani	**Date:** 15 Sep 2025
Questions:	**Answers and Notes:**
Q1: Do you have prior experience using recruitment software or platforms? If yes, which tools have you used, and what did you like or dislike about them?	Yes, I use more than 150 recruitment platforms, the most prominent of which are Jadara, LinkedIn, and Job.com. Their advantages are that they all facilitate the process and save energy. **Observations:** HR professionals are already accustomed to recruitment platforms.
Q2: Could you describe your typical hiring process?	There is no specific process. Each company has its own method of hiring. **Observations:**

	The hiring process varies between companies.
Q3: What are the most time-consuming tasks in your recruitment process?	Recruiting candidates is the most difficult part of the process. **Observations:** A broad audience of job seekers must be built.
Q4: Can you walk me through your current process for screening resumes? What tools or methods do you rely on, and what would you consider the main strengths and weaknesses of this approach?	I use Jadara and ATS system to bring me the best candidates **Observations:** The importance of researching the systems currently used
Q5: What are the most common factors or reasons that lead to the rejection of applicants during the recruitment process?	The most frequent reason I turn away job applicants is that their personalities or methods don't fit the workplace. **Observations:** Focus on character analysis
Q6: What details of a candidate's CV are most important for you in the first review?	Experience, qualifications and skills **Observations:** How to build a CV
Q7: How do you usually structure your interviews — for example, what kinds of questions do you start with, and how do you move on from there?	In order to better grasp his or her personality, I first ask them to tell me about themselves. The remainder is dependent on the work and specialty. **Observations:** Againe, focus on character analysis
Q8: How do you evaluate the importance of receiving detailed and objective assessment reports about applicants, and what aspects of such reports matter to you the most?	The most important point in the report is measuring and analyzing the applicant's personality. We will know the rest during the trial period. **Observations:** Againe, focus on character analysis

Q9: What feature would make a recruitment app valuable enough for you to use it regularly?	Although I represent companies and HR staff for your app, I am very interested in your individual side and the features you offer them. **Observations:** Focus on the features of individuals.

Table 13: 9-3 Interview's Transcription for third Interviewee

Interview (3)	
Interviewee: Participant 3	**Interviewer:** Walaa Saif Aleslam
Location/Medium: Online via Nawafith app	**Appointment Date:** 15 Sep 2025 **Start Time:** 08:00 PM **End Time:** 09:00 PM
Objectives: • Understand prior experience with recruitment software and user preferences. • Explore the typical hiring process followed by HR professionals. • Identify the most time-consuming tasks in recruitment. • Assess current resume screening methods, tools, and their strengths and weaknesses. • Understand common reasons for applicant rejection. • Determine which CV details are most important during initial review. • Explore how interviews are structured and conducted. • Evaluate the perceived importance of detailed and objective candidate assessment reports. • Identify features that make the recruitment platform valuable for regular use.	**Reminders:** He very much welcomes the idea of automating the recruitment process and having artificial intelligence choose the most suitable candidates, but he is strict about the quality of the results.

Agenda:	**Approximate Time:**
Introduction:	12 min
Background in project:	15 min
Overview of interview:	4 min
Topic to be covered:	2 min
Permission to record:	1 min
Question 1:	3 min
Question 2:	5 min
Question 3:	4 min
Question 4:	2 min
Question 5:	2 min
Question 6:	2 min
Question 7:	1 min
Question 8:	2 min
Question 9:	2 min
Summary of major points:	1 min
Questions from interviewee:	1 min
Closing:	1 min
General Observations: The idea is not new to human resources employees, yet they believe there is still plenty of scope for work on it and developing applications to serve employees and facilitate the recruitment process.	
Topic not covered: There was not enough time to ask him about the most important sources of data collection.	
Interviewee: AbuBaker Balbid	**Date:** 15 Sep 2025
Questions:	**Answers and Notes:**
Q1: Do you have prior experience using recruitment software or platforms? If yes, which tools have you used, and what did you like or dislike about them?	I use LinkedIn, Monster and many others. The advantage of LinkedIn is that because people are keen on it, it is always up-to-date and provides their latest data and gives an impression of the person from a personal perspective other than the qualifications mentioned in the CV.

	Observations: The importance of getting an initial impression of individuals before hiring them for companies
Q2: Could you describe your typical hiring process?	The process begins with the recruitment plan approved by the administration according to the budget. After it is translated into job vacancies, the job is announced after writing an accurate description for it. Candidates are searched for according to the required specifications, either through the published post or through acquaintances to save time. Although the matter is not completely fair, it shortens the path. **Observations:** Letting AI choose job candidates could ensure fair hiring.
Q3: What are the most time-consuming tasks in your recruitment process?	The CV screening process takes time, so we often hire people we know who are qualified. **Observations:** Providing a way to select the right employee without having to filter the entire CVs helps HR staff.
Q4: Can you walk me through your current process for screening resumes? What tools or methods do you rely on, and what would you consider the main strengths and weaknesses of this approach?	I don't rely on the ATS system very often and typically filter manually because it doesn't always filter appropriately. For example, it instantly deletes a CV that isn't written in the required format without reading its contents, even though the applicant may be qualified. **Observations:** Providing a way to evaluate CVs without filtering is important.

Q5: What are the most common factors or reasons that lead to the rejection of applicants during the recruitment process?	Good question, the first reason is that he does not have sufficient experience and qualifications, and the second reason is his compatibility with the system in terms of values and personality. **Observations:** Understanding the reasons for rejection is the first step to addressing the problem.
Q6: What details of a candidate's CV are most important for you in the first review?	Ease of reading, organization, language accuracy and experience. **Observations:** Important details to improve CVs
Q7: How do you usually structure your interviews — for example, what kinds of questions do you start with, and how do you move on from there?	The applicant must prove to me that he is ready to contribute to the place and the system in which he is coming to work and that he is familiar with the details of the company and the job. My questions will be focused on understanding and extracting these points from the applicant. **Observations:** Ask the right questions
Q8: How do you evaluate the importance of receiving detailed and objective assessment reports about applicants, and what aspects of such reports matter to you the most?	Two important things: the degree of a person's self-confidence, whether from the tone of voice or body language, and secondly, his achievements and the way he talks about them. These reports are important, but the way they are written is more important. **Observations:** Provide organized and comprehensive reports.
Q9: What feature would make a recruitment app valuable enough for you to use it regularly?	Having strong candidates in the application and evaluation process ensures that I get the right employee. **Observations:** System efficiency

Then, we conducted interviews with the second type of users, who are **job seekers**, and we asked them the following questions:

Individuals Interview Questions:

1) How do you usually choose the jobs you apply for (job sites, connections, LinkedIn)?
2) Can you describe your biggest challenge when applying for jobs?
3) How do you usually prepare your CV, and what kind of help would you like to receive?
4) How do you usually prepare for interviews?
5) What features would you expect from mock interviews or interview practice tools, and if you tried them what improvements would you expect from them?
6) What type of feedback would be most useful for you after an interview?
7) What feature would make a recruitment app valuable enough for you to use it regularly?

Table 14: 9-4 Interview's Transcription for fourth Interviewee

Interview (4)	
Interviewee: Participant 4	**Interviewer:** Walaa Saif Aleslam
Location/Medium: Online via Zoom	**Appointment Date:** 16 Sep 2025 **Start Time:** 08:00 PM **End Time:** 08:35 PM
Objectives: • Understand how job seekers choose which jobs to apply for. • Identify the main challenges faced during job applications. • Explore users' CV preparation process and support needs. • Assess how users prepare for interviews. • Explore expectations for mock interview tools and possible improvements. • Determine the most useful feedback after interviews.	**Reminders:** The interviewee relies on ChatGPT to prepare for interviews and improve her CV, and she suffers from companies not providing any feedback after applying for the job or interviews.

• Identify features that make the app valuable for regular use.	
Agenda: Introduction: Background in project: Overview of interview: Topic to be covered: Permission to record: Question 1: Question 2: Question 3: Question 4: Question 5: Question 6: Question 7: Summary of major points: Questions from interviewee: Closing:	**Approximate Time:** 3 min 3 min 1 min 2 min 1 min 2 min 5 min 3 min 3 min 5 min 2 min 2 min 1 min 1 min 1 min
General Observations: Job seekers rely heavily on artificial intelligence, specifically ChatGPT, to prepare for interviews and create resumes.	
Topic not covered: Due to time constraints, the interviewee did not give me any tips or additional features that she would like to see in the application.	
Interviewee: Nourah Al-Manee	**Date:** 16 Sep 2025
Questions:	**Answers and Notes:**
Q1: How do you usually choose the jobs you apply for (job sites, connections, LinkedIn)?	I usually search for jobs on graduate development websites and from my acquaintances and friends. **Observations:**

	She doesn't use LinkedIn much or other popular sites.
Q2: Can you describe your biggest challenge when applying for jobs?	The most challenging and difficult thing I face is that companies do not give me any feedback after applying for jobs, and this makes me confused. **Observations:** The importance of providing feedback in **Jadeer** app for interviewee
Q3: How do you usually prepare your CV, and what kind of help would you like to receive?	Depending on the role I applied for, I conduct research on the most important skills that the candidate should have, then I focus on them in my CV. **Observations:** She has the intelligence to prepare the CV to catch the company's attention.
Q4: How do you usually prepare for interviews?	I use ChatGPT to conduct a mock interview for me and get a thorough understanding of the questions that could be asked during the interview. **Observations:** Convincing users to use AI in mock interviews can be easy.
Q5: What features would you expect from mock interviews or interview practice tools, and if you tried them what improvements would you expect from them?	It should be able to answer questions quickly and teach you how to present yourself in a good way. It also should focus more on the role and give you an honest report, not just praise me. **Observations:** Job seekers are already familiar with mock interviews
Q6: What type of feedback would be most useful for you after an interview?	Of course, the honest feedback that really taught me what I need to improve on. **Observations:** Accurate and clear results are the most important thing.

Q7: What feature would make a recruitment app valuable enough for you to use it regularly?	It is important that the opportunities offered are real and that I receive feedback after applying. **Observations:** Business identity verification is important.

Table 15: 9-5 Interview's Transcription for fifth Interviewee

Interview (5)	
Interviewee: Participant 5	**Interviewer:** Walaa Saif Aleslam
Location/Medium: Online via Zoom	**Appointment Date:** 16 Sep 2025 **Start Time:** 05:25 PM **End Time:** 06:00 PM
Objectives: • Understand how job seekers choose which jobs to apply for. • Identify the main challenges faced during job applications. • Explore users' CV preparation process and support needs. • Assess how users prepare for interviews. • Explore expectations for mock interview tools and possible improvements. • Determine the most useful feedback after interviews. • Identify features that make the app valuable for regular use.	**Reminders:** The interviewee uses many websites to apply and search for jobs and suffers from companies not providing any feedback after applying for the job or interviews.
Agenda: Introduction: Background in project: Overview of interview:	**Approximate Time:** 3 min 3 min 1 min

Topic to be covered: Permission to record: Question 1: Question 2: Question 3: Question 4: Question 5: Question 6: Question 7: Summary of major points: Questions from interviewee: Closing:	2 min 1 min 2 min 5 min 3 min 3 min 5 min 2 min 2 min 1 min 1 min 1 min
General Observations: Job seekers have had many unsuccessful experiences in searching for jobs and also in creating CVs and they believe that most job advertisements are misleading.	
Topic not covered: Due to time constraints, the interviewee did not give me any tips or additional features that she would like to see in the application.	
Interviewee: Lubabah Al-Zawadi	**Date:** 16 Sep 2025
Questions:	**Answers and Notes:**
Q1: How do you usually choose the jobs you apply for (job sites, connections, LinkedIn)?	I search for opportunities through a site called NAKURI, LinkedIn, and also a freelancing site. I also hear about opportunities through people by attending events related to my field. I also go to companies and ask them. **Observations:** Providing a single platform that brings together all opportunities will greatly help job seekers and save them effort.
Q2: Can you describe your biggest challenge when applying for jobs?	The most challenging and difficult thing I face is that companies do not give me any feedback after applying for jobs, and this makes me confused.

	Observations: The importance of providing feedback in **Jadeer** app for interviewee
Q3: How do you usually prepare your CV, and what kind of help would you like to receive?	I usually present it the same, without changing it, because I apply for jobs without a specific job title. **Observations:** Job seeker struggling with CV creation and preparation
Q4: How do you usually prepare for interviews?	I focus on the skills that I told them I know because they will definitely ask me about them, so I practice the way I answer on myself. **Observations:** Providing a way to practice for the interview is very good.
Q5: What features would you expect from mock interviews or interview practice tools, and if you tried them what improvements would you expect from them?	I haven't tried it, but I expect it to be a virtual environment with a complete simulation that makes me feel the atmosphere and live the full experience and the same pressure, and at the same time it might be helpful. **Observations:** Job seekers are already familiar with mock interviews
Q6: What type of feedback would be most useful for you after an interview?	I want it to give me feedback about my self-confidence, the tone of my voice, did I answer correctly? And can I be accepted? **Observations:** Accurate and clear results are the most important thing.
Q7: What feature would make a recruitment app valuable enough for you to use it regularly?	The CV improvement alone is enough to make me use it. **Observations:** Job seeker struggling with CV creation and preparation

9.2 Appendix B (Questionnaires):

Also, for the survey, we collected information from the two types of **Jadeer** users. Here are the questions we presented in the Company **survey**, which was directed to Human Resources employees:

Link: Companies Survey Sheet

Companies' Questionnaire Questions:

1) What is your name?
2) What is your organization's name?
3) What type of sector does your organization belong to?
4) On average, how many applicants do you receive per job posting?
5) Which stage of the hiring process takes the most time in your company?
6) Do you currently use any digital tools or platforms to assist in hiring? If yes, which tool do you use and for what purpose?
7) How much would you trust an AI system to conduct initial job interviews on behalf of your company?
8) What concerns would you have about AI-generated interview questions?
9) If the platform generated analytical reports after AI-assisted interviews (strengths, weaknesses, suitability score), how useful would that be for you?
10) What information would you find most useful in a candidate performance report?
11) Would you use this report as a final decision-making tool or just as an assistant?
12) Do you have any suggestions or ideas that could improve **Jadeer** app?
13) If you are interested in trying **Jadeer** app in the future, please leave your email so we can stay in touch.

As for the job seeker, here are the questions we presented to them in the survey:

Link: Job seeker Survey Sheet

Job seeker Questionnaire Questions:

1) What is your age group?
2) What is your gender?
3) What is your current employment status?
4) How often do you rely on applying for jobs online?
5) Which platforms do you use most when applying for jobs?
6) What are the biggest challenges you face when applying for jobs?
7) Do you usually customize your CV for each job application?
8) How important is it for you to receive AI-generated feedback on your CV?
9) What type of CV enhancement do you value most?
10) How do you usually prepare for job interviews?
11) How do you feel about having a mock interview with AI before the actual one?
12) When would you prefer to use an AI mock interview?
13) How comfortable are you with the idea of AI evaluating your interview answers based on content and voice?
14) What concerns do you have about AI-based interview evaluation?
15) What type of feedback would be most helpful for you after an interview simulation?
16) Do you have any suggestions or ideas that could improve **Jadeer** app?

9.3 Appendix C (Important links):

- Git-Hup link: GitHub Link
- Jira link: Jira Link

9.4 Appendix D (UAT):

Link: Companies UAT Survey Sheet

Company's Questionnaire Questions:

1) I needed previous experience to use the app.
2) I found the application easy to use.
3) I found the navigation between pages smooth and clear.
4) I found the color palette and visual design comfortable.
5) I found the interface design clear and intuitive.
6) Some icons or labels were unclear to me.
7) The app responded quickly and did not lag.
8) Pages and features loaded without noticeable delays.
9) I was able to create a job post correctly, easily, and clearly.
10) I found the AI-generated job descriptions useful and relevant.
11) The AI-generated interview questions were relevant to the job post.
12) The difficulty level and clarity of the questions were appropriate.
13) I was able to view, edit, or add to the AI-generated questions easily.
14) I was able to edit my profile correctly, easily, and clearly.
15) I was able to reset my password without any difficulty.
16) The pop-up messages I got from the app were helpful and clear.
17) I think I will use this application frequently.
18) I will recommend this application to others.
19) I was able to view candidate applications for my job postings clearly and easily.
20) The AI-generated candidate evaluation report was comprehensive and useful for hiring decisions.
21) The candidate's overall score and sub-scores (technical, psychometric, voice tone, CV, job requirements) were clear and easy to interpret.

22) The requirements checklist in the candidate report accurately reflected whether the candidate met the job requirements.
23) I was able to download the candidate evaluation report as a PDF without difficulty.

Link: Job seeker UAT Survey Sheet

Job seeker Questionnaire Questions:

1) I needed previous experience to use the app.
2) I found the application easy to use.
3) I found the navigation between pages smooth and clear.
4) I found the color palette and visual design comfortable.
5) I found the interface design clear and intuitive.
6) Some icons or labels were unclear to me.
7) The app responded quickly and did not lag.
8) Pages and features loaded without noticeable delays.
9) I was able to upload my CV to my profile correctly, easily, and clearly.
10) I found the AI CV enhancement helpful and easy to understand.
11) The AI suggestions I received after enhancing my CV were clear, relevant, and helpful.
12) The system correctly identified missing or incomplete information in my CV and helped me understand what to add.
13) I liked having the option to choose jobs so the AI could tailor the CV enhancement toward those roles.
14) The enhanced CV I received (PDF) was clear, well-formatted, and improved compared to my original one.
15) I was able to browse and search for job posts easily and clearly.
16) The job recommendations I received based on my CV were relevant and useful.
17) I was able to save or favorite job posts correctly and easily.
18) I was able to find my previous CV enhancements on the History page easily.
19) I was able to edit my profile correctly, easily, and clearly.

20) I was able to reset my password without any difficulty.
21) The pop-up messages I got from the app were helpful and clear.
22) I think I will use this application frequently.
23) I will recommend this application to others.
24) I was able to apply for a job correctly, easily, and clearly.
25) The interview questions presented during the AI interview were relevant to the job I applied for.
26) I was able to start and complete the mock interview without difficulty.
27) The mock interview questions were relevant to my selected specialty.
28) The mock interview report I received (strengths, weaknesses, and advice) was clear, accurate, and helpful.
29) The voice tone analysis included in my mock interview report provided useful insight into my communication style.
30) I was able to find my previous mock interviews and job applications on the History page easily.

9.5 Appendix E (AI Prompt Templates Used in Jadeer):

This appendix presents the AI prompt templates used in **Jadeer's** main AI-powered features. The prompts are documented using summarized templates rather than full raw code prompts to keep the appendix readable while still showing the input variables, model instructions, and expected output structures.

E.1 Job Interview Questions Generation Prompt

Model:
gpt-4o-mini
System Prompt:
You are an expert interviewer that outputs strict JSON.
User Prompt Template:

Job Context:
- Title: {jobTitle}
- Position/Seniority: {positionLevel}
- Specialty: {specialty}
- Requirements: {requirementsList}

- Job Description: {jobDescription}

Generate exactly 10 interview questions (5 technical, 5 psychometric).
Difficulty: easy {easyPct}%, medium {mediumPct}%, hard {hardPct}%.
Technical categories: {techCategories}
Psychometric dimensions: {psychometricDimensions}
Use the Big Five model for psychometric questions.

Return ONLY a JSON object:
{ "questions": [{ "text", "type", "category", "difficulty", "trait" }] }
- type: "technical" | "psychometric"
- difficulty: "easy" | "medium" | "hard"
- trait: Big Five value or null (null for technical questions)

Expected Output:
```
{
 "questions": [
   { "text": "...", "type": "technical", "category": "System Design",
     "difficulty": "hard", "trait": null },
   { "text": "...", "type": "psychometric", "category": "Motivation",
     "difficulty": "easy", "trait": "Openness" }
 ]
}
```

E.2 Mock Interview Questions Generation Prompt

Model:
gpt-4o-mini
System Prompt:
You output strict JSON only.
User Prompt Template:
Mock Interview Context:
- Specialty: {selectedSpecialty}
- Specialty bucket: {bucketSpecialty}

Generate exactly 10 questions (5 domain, 5 psychometric).
Difficulty: easy 3, medium 5, hard 2.
Domain categories: {domainCategories}
Work-style themes: {behaviouralThemes}
Use the Big Five model for psychometric questions.

Return ONLY a JSON object:
{ "questions": [{ "text", "type", "category", "difficulty", "trait" }] }
- type: "domain" | "psychometric"
- difficulty: "easy" | "medium" | "hard"

- trait: Big Five value or null (null for domain questions)

Expected Output:

```
{
 "questions": [
   { "text": "...", "type": "domain", "category": "Data Analysis",
     "difficulty": "medium", "trait": null },
   { "text": "...", "type": "psychometric", "category": "Teamwork",
     "difficulty": "easy", "trait": "Agreeableness" }
 ]
}
```

E.3 Job Interview Performance Report Prompt

Model:

gpt-4

System Prompt:

You are an expert recruiter writing a candidate evaluation report for a hiring manager.

Always use third person ('the candidate', 'they'). Never use second person ('you').

Always respond with valid JSON only, no markdown.

User Prompt Template:

You are an expert recruiter writing a candidate evaluation report for a hiring manager.

Always refer to the candidate in third person ("the candidate", "they").

NEVER use second person ("you"). This report is viewed by the company, not the candidate.

Position: {jobTitle}

Specialty: {specialty}

Company: {companyName}

Job Requirements:

- {requirement1}
- {requirement2}

Job Description: {jobDescriptionExcerpt}

{transcriptionFailureNote}

Based on the following interview transcripts, provide a comprehensive evaluation in VALID JSON format with NO markdown code blocks.

Note: Questions marked [technical] assess domain knowledge. Questions marked [psychometric]

assess personality traits and work style.

Interview Transcripts:

[technical] Q1: {question1} / A1: {answer1}
[psychometric] Q2: {question2} / A2: {answer2}
...
For each of the following job requirements, evaluate whether the candidate demonstrated meeting it based on their interview answers. Return a "requirementsChecklist" array:
- "{requirement1}"
- "{requirement2}"
Return JSON with:
- cvAnalysisScore (0–100)
- jobRequirementsMatchScore (0–100)
- psychometricScore (0–100)
- technicalScore (0–100)
- overallSummary, strengths[], weaknesses[], advice[]
- requirementsChecklist[{ requirement, met: bool }]
- psychometricAnalysis { confidence, communication, personalityTraits, workStyle }
- questionsAndAnswers[{ question, answer }]
SCORING GUIDELINES (0-100):
- 80-100: Excellent — strong knowledge, clear communication, strong role fit
- 60-79: Good — adequate knowledge, room for improvement
- 40-59: Average — partial knowledge, needs development
- 0-39: Below expectations — unclear answers, significant gaps
CRITICAL: Write ALL feedback in third person. NEVER use second person ("you").

Expected Output:

```
{
  "cvAnalysisScore": 78,
  "jobRequirementsMatchScore": 65,
  "psychometricScore": 82,
  "technicalScore": 71,
  "overallSummary": "The candidate demonstrated...",
  "strengths": ["..."],
  "weaknesses": ["..."],
  "advice": ["..."],
  "requirementsChecklist": [
```

```
    { "requirement": "3+ years backend", "met": true }
  ],
  "psychometricAnalysis": {
    "confidence": 80, "communication": 85,
    "personalityTraits": "...", "workStyle": "..."
  },
  "questionsAndAnswers": [
    { "question": "...", "answer": "..." }
  ]}
```

E.4 Mock Interview Performance Report Prompt

Model:
gpt-4
System Prompt:
You are a friendly mock interview coach. Always respond with valid JSON only, no markdown.
User Prompt Template:
You are an expert interview coach giving personal mock interview feedback to a jobseeker practicing for a {specialty} position. IMPORTANT: Address the user directly using "you" and "your".

Specialty: {specialty}
{transcriptionFailureNote}
Interview Transcripts:
Q1: {question1} / A1: {answer1}
Q2: {question2} / A2: {answer2}
...
Acoustic Tone Reading Parameters:
The microphone analysis registered an overall vocal confidence metric of {overallVoiceConfidenceScore} out of 100.

Respond with a JSON object containing:
{
"overallScore": <number between 0-100>,
"overallSummary": "<2-3 sentences summarizing performance, using 'you' language.>",
"strengths": ["<strength 1>", "<strength 2>", "<strength 3>"],
"weaknesses": ["<weakness 1>", "<weakness 2>", "<weakness 3>"],
"advice": ["<recommendation 1>", "<recommendation 2>", "<recommendation 3>"],
"mockVoiceComment": "<Based on the voice score, generate a friendly comment offering voice tone advice. Do NOT mention the exact score number or percentages. Analyze their pacing, conviction, or hesitation, and provide constructive advice tailored to a jobseeker trying to sound confident in a {specialty} interview. Speak to the user as 'you'.>"

}
SCORING GUIDELINES (0-100 scale):
- 80-100: Exceptional — strong knowledge, clear communication, confident delivery
- 60-79: Good — adequate knowledge and communication, room for improvement
- 40-59: Average — partial knowledge, needs development in key areas
- 0-39: Below expectations — unclear answers, significant gaps

CRITICAL: NEVER use references like "the candidate". Always use "you" and "your".
Expected Output:
{
"overallScore": 72,
"overallSummary": "You showed a solid understanding of core concepts and communicated your ideas clearly throughout the session. Your
answers were well-structured, though some responses could benefit from more concrete examples.",
"strengths": ["You clearly explained your approach to...", "...", "..."],
"weaknesses": ["You could improve the depth of...", "...", "..."],
"advice": ["You should practise the STAR method...", "...", "..."],
"mockVoiceComment": "Your delivery showed genuine enthusiasm and conviction in most of your answers. You could work on slowing your
pace slightly during technical explanations to let your points land with more confidence."
}

E.5 Job Description Generation Prompt

Model:

gpt-4o-mini
System Prompt:
You are an expert HR professional who writes compelling, detailed job descriptions
that attract top talent.

User Prompt Template:
Job Title: "{jobTitle}"
Position Level: "{positionLevel}"
Speciality/Field: "{specialty}"

Write a 300–500 word job description including:
1. Role Overview (5–6 sentences)
2. Key Responsibilities (8–12 bullet points)
3. Closing Statement (2–3 sentences)

Do NOT mention any company names. Output plain text only, no markdown.

Expected Output:

Plain prose, ~300–500 words. Three sections: Role Overview, Key Responsibilities (bullet points), and a Closing Statement. No markdown, no company names.

E.6 CV Enhancement Prompt

Model: gpt-4o
response_format:
json_object

System Prompt:
You are a professional CV enhancement assistant. Always return valid JSON. CRITICAL: Never invent or create fake data — only use information that exists in the provided CV.

User Prompt Template:
Target Job: {jobTitle} — {jobDescription}

Original CV Text:
{originalCVText}

{additionalSectionsText}

Enhance the CV for ATS and the target job. Only use existing information — do not invent data.
Return JSON with:
- enhanced_cv: array of section objects, each with "section" name and typed "content"
  Sections: PersonalInformation, Summary, Experience, Education, Skills, Projects, Awards, Publications, Patents, Research, Certifications, Internships, VolunteerWork, Courses, Training, Workshops, Conferences, Achievements, ProfessionalMemberships, Portfolio, Languages, ExtracurricularActivities, Interests
- suggestions: string[] (max 5, describing changes made)

Expected Output:
```
{
 "enhanced_cv": [
   { "section": "PersonalInformation",
    "content": { "full_name": "...", "email": "...", "phone": "...",
             "location": "...", "links": [] } },
   { "section": "Summary", "content": "..." },
   { "section": "Experience",
    "content": [{ "title": "...", "company": "...", "years": "...", "description": "..." }] },
   { "section": "Skills", "content": ["...", "..."] }
 ],
 "suggestions": [
   "Tailored experience descriptions to match the target role.",
   "Removed unrelated sections to focus the CV."
```

]
}

E.7 CV Missing Sections Detection Prompt

Model:
gpt-4o-mini
response_format:
json_object

System Prompt:
You are a CV analysis assistant. Always return valid JSON.

User Prompt Template:
Analyze the following CV and identify which of these 7 sections are completely missing or empty:
PersonalInformation, Summary, Experience, Education, Skills, Certifications, Languages.

CV Text:
{originalCVText}

Return JSON: { "missingSections": ["SectionName", ...] }
Use exact section names. Return an empty array if nothing is missing.

Expected Output:
{ "missingSections": ["Certifications", "Languages"] }

E.8 Skills Suggestion Generation Prompt

Model: gpt-4o-mini
response_format:
json_object

System Prompt:
You are a career expert assistant. Always return valid JSON.

User Prompt Template:
Job Title: {jobTitle}
Job Description: {jobDescription}

Suggest 15–20 relevant skills (technical and soft) for this role.
Each skill must be 1–3 words.

Return JSON: { "skills": ["Skill 1", "Skill 2", ...] }

Expected Output:

{ "skills": ["Python", "Data Analysis", "SQL", "Machine Learning", "Communication", "Problem Solving", "..."] }

E.9 Voice Feedback Comment Generation Prompt

Model: gpt-4

System Prompt: You are a friendly mock interview coach. Always respond with valid JSON only, no markdown.

User Prompt Template: Acoustic Tone Reading Parameters: The microphone analysis registered an overall vocal confidence metric of {overallVoiceConfidenceScore} out of 100.

Based on the voice score of {overallVoiceConfidenceScore}/100, generate a friendly comment offering voice tone advice. Do NOT mention the exact score number, percentages, or basic instructions like "take a deep breath". Instead, analyze their pacing, conviction, or hesitation, and provide constructive advice tailored to a jobseeker trying to sound confident in a {specialty} interview. Speak to the user as "you".

Expected Output:

"Your delivery showed genuine enthusiasm and conviction in most of your answers. You could work on slowing your pace slightly during technical explanations to let your points land with more confidence."